COMP2711 / DISCRETE MATHEMATICAL TOOLS

Counting

Past Paper Solutions

178	4	2013-25
verified exercises	topic groups	source papers

English solutions aligned with the supplied official answers.

Each answer has two parts: the original Official Answer and the existing Supplementary Explanation. Source PDF pages are retained for verification.

Generated 2 October 2026

All verified, in-scope questions linked to Counting in the knowledge base.

Each question is included once, including cross-topic applications.

READING GUIDE

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Basic Counting
56 exercises — C001-C056

Permutations and Combinations
93 exercises — C057-C149

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16 exercises — C150-C165

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How to use this booklet

Questions are grouped by their first Counting subtopic, then ordered by difficulty and year (newest first). Each question appears once as C001-C178. Difficulty ranges from 1 (introductory) to 4 (challenging).

Official Answer reproduces the answer in the source paper, with its physical PDF page reference.

Supplementary Explanation retains the existing independently authored reasoning, including the stated result and derivation. The two parts remain separate throughout the booklet.

Source references use physical PDF pages. File links work with the local source collection at its existing location.

The answer provenance shown for each exercise retains its knowledge-base classification; attaching an official quotation does not reclassify a derived correction as official.

C001

Basic Counting

2025 Fall Midterm | Question 4(d) | Source page 13 | Difficulty 1 | 2 marks
Q-2025F-MID-P04-d | Answer provenance: official

> (d) [2 point] We call a 5-digit number palindrome if it is of form $\overline{abcba}$. For example 11111 and 12421 are both palindromes. However, 21013 is not. How many 5-digit palindrome numbers exist? Note that the leftmost digit can not be 0.

Official Answer

Source: physical PDF page 13 (COMP2711_F2025_Midterm.pdf).

> (d) $9 \times 10 \times 10$

Supplementary Explanation

Answer. 900.

Derivation/proof. The first digit a has nine choices, one through nine. The second digit b and middle digit c each have ten choices, zero through nine. The fourth digit must equal b, and the fifth must equal a, so these positions add no freedom. Each triple a,b,c determines exactly one five-digit palindrome and conversely. Therefore $9 \cdot 10 \cdot 10 = 900$.

C002

Basic Counting

2022 Final Part 1 | Question 3(a) | Source page 2 | Difficulty 1 | 2 marks
Q-2022-FINAL1-P03-a | Answer provenance: official

> Consider a web system for online shopping that will generate an ID for each order. The order ID is a sequence of 7 characters where each character could be either one of the 26 upper case English letters (A to Z) or one of the 10 digits (0 to 9).
>
> How many possible order IDs are there?

Official Answer

Source: physical PDF page 2 (2022-final-part-1.pdf).

> (a) 36^7

Supplementary Explanation

Answer. 36^7.

Derivation/proof. The available alphabet has 26+10=36 distinct characters. With no repetition restriction each of the seven positions independently has 36 choices. The product rule therefore gives 36·36·36·36·36·36·36=36^7. A leading zero is a valid character because these are IDs, not seven-digit integers.

There are $26 + 10 = 36$ symbols; a leading zero is also valid.

C003

Basic Counting

2019 Final | Question 8(a) | Source page 10 | Difficulty 1 | 2 marks
Q-2019-FINAL-P08-a | Answer provenance: official

> How many strings of six lowercase letters from the English alphabet contain
>
> (a) the letter a?

Official Answer

Source: physical PDF page 10 (2019-final.pdf).

The only reasonable way to do this is by subtracting from the number of strings with no restrictions the number of strings that do not contain the letter a. The answer is $26^6 - 25^6$.

Supplementary Explanation

Answer. $26^6 - 25^6$.

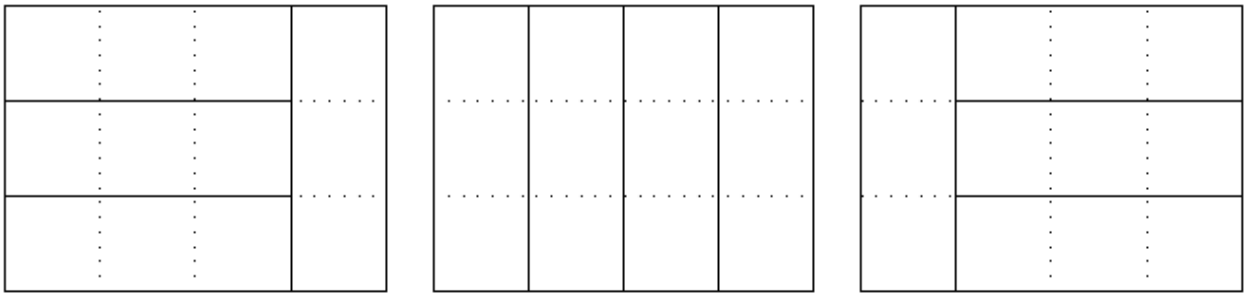
Derivation/proof. There are 26^6 unrestricted length-six strings. A string with no a has 25 choices independently at each position, giving 25^6. The complement of no a is at least one a, so subtracting counts every desired string exactly once, including those with several occurrences of a.

The requested letter needs at least one occurrence; subtracting strings with none avoids having to split into every possible number of occurrences.

C004

Basic Counting

2018 Final | Question 2(a) | Source page 4 | Difficulty 1
Q-2018-FINAL-P02-a-E1 | Answer provenance: official

> Let E_n be the number of ways in which you can tile a 3-by-n board with 1-by-3 pieces. For example, when $n = 4$ the board is 3-by-4 and $E_4 = 3$. The 3 ways are shown as follows:
>
> 
>
> (a) Calculate E_1, E_2, and E_3. [Atomic request: E_1.]

Official Answer

Source: physical PDF page 4 (2018-final.pdf).

1, 1, 2.

Supplementary Explanation

Answer. $E_1 = 1$.

Derivation/proof. The board has three cells in one column. A horizontal piece of length three cannot fit in width one, so the only possible placement is a single vertical piece covering all three cells. This placement completes the board and is unique.

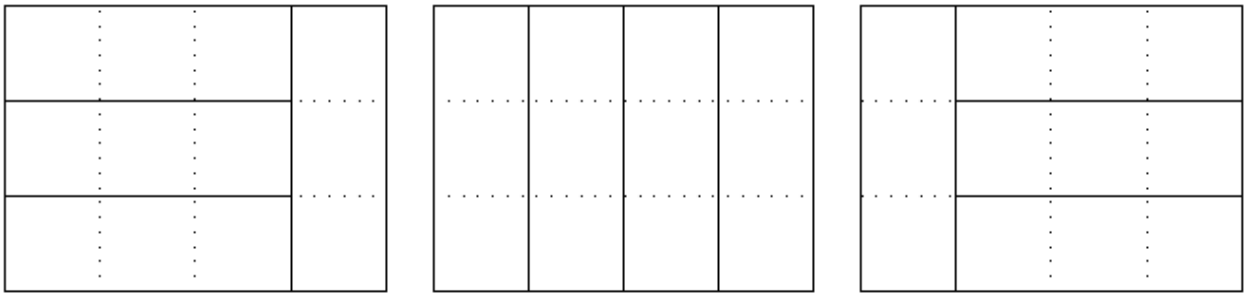
The three printed values are respectively E_1, E_2, E_3; this atomic question asks for the first.

A 1×3 tile covers the entire column in the one-column board. This is the sole tiling and gives the first initial value.

C005

Basic Counting

2018 Final | Question 2(a) | Source page 4 | Difficulty 1
Q-2018-FINAL-P02-a-E2 | Answer provenance: official

> Let E_n be the number of ways in which you can tile a 3-by-n board with 1-by-3 pieces. For example, when $n = 4$ the board is 3-by-4 and $E_4 = 3$. The 3 ways are shown as follows:
>
> 
>
> (a) Calculate E_1, E_2, and E_3. [Atomic request: E_2.]

Official Answer

Source: physical PDF page 4 (2018-final.pdf).

1, 1, 2.

Supplementary Explanation

Answer. $E_2 = 1$.

Derivation/proof. A horizontal piece requires three columns but this board has only two. Thus both columns must be filled vertically, one piece per column. Those two placements cover every cell and there is no choice about their positions, so only one tiling exists.

The three printed values are respectively E_1, E_2, E_3; this atomic question asks for the second.

In a two-column board, each of the two columns must be covered vertically: a horizontal tile would need three columns. These two tiles cover its six cells uniquely.

C006

Basic Counting

2018 Midterm | Question 8(a) | Source page 10 | Difficulty 1
Q-2018-MID-P08-a | Answer provenance: official

> You are given a suitcase that has a combination lock. It consists of five dials, each of which can be rotated to take a value from 0–9. The suitcase only opens if you guess the correct 5-digit solution.
>
> Briefly justify your answers.
>
> (a) How many different solutions are there?

Official Answer

Source: physical PDF page 10 (2018-midterm.pdf).

All 10 digits are possible for each of the 5 dials, hence 10^5.

Supplementary Explanation

Answer. $10^5 = 100000$.

Derivation/proof. Each of the five labeled dial positions has ten choices. With repetition allowed and no leading-zero restriction on a lock, the product rule gives $10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 = 10^5$. Different ordered digit lists are different settings.

A leading zero is permitted because these are lock settings rather than three-digit positive integers; therefore the first dial has ten choices as well.

C007

Basic Counting

2017 Midterm | Question 10(a) | Source page 12 | Difficulty 1
Q-2017-MID-P10-a | Answer provenance: official

> Assume you have 50 candies and four boxes, labeled A, B, C, and D.
>
> (a) How many ways can you put the candies in the boxes, assuming that the candies are all different?

Official Answer

Source: physical PDF page 12 (2017-midterm.pdf).

4^{50}.

Supplementary Explanation

Answer. 4^{50}.

Derivation/proof. A distribution is a function from the set of fifty candies to the four labeled boxes. Each candy has four possible images, without any capacity or nonempty-box condition. The product rule therefore gives fifty factors of four.

The labels A, B, C, D distinguish the four boxes. Since empty boxes are allowed, each ball can choose any label independently, with no surjectivity restriction.

C008

Basic Counting

2016 Midterm | Question 1(a) | Source page 3 | Difficulty 1 | 2 marks
Q-2016-MID-P01-a | Answer provenance: official

> Consider a web system for online shopping that generates an order id for each business. The order id is a sequence of 7 characters where each character could be either one of the 26 Upper Case English Alphabets (A to Z) or one of the 10 digits (0 to 9).
>
> (a) How many possible order id's are there?

Official Answer

Source: PDF physical page 3 (2016-midterm.pdf).

(a) 36^7

Supplementary Explanation

Answer. 36^7.

Derivation/proof. There are $26 + 10 = 36$ choices for each of seven positions. Applying the product rule gives $\prod_{i=1}^{7} 36 = 36^7$.

The number of options stays at 36 at every position; their sevenfold product is 36^7.

C009

Basic Counting

2015 Midterm | Question 6(a) | Source page 11 | Difficulty 1 | 2 marks
Q-2015-MID-P06-a | Answer provenance: official

> In a computer system, a password is a string of 8 characters, where each character is either a digit or a lowercase letter from the English alphabet. How many possible passwords are there under each of the following conditions?
>
> (a) The first character must be a letter.
>
> Briefly explain how your answers are derived.

Official Answer

Source: PDF physical page 11 (2015-midterm.pdf).

(a) 26×36^7. There are 26 possibilities for the first character and 36 possibilities for each of the other characters.

Supplementary Explanation

Answer. $26 \cdot 36^7$.

Derivation/proof. Choose the first letter in 26 ways. Each remaining position independently has $26 + 10 = 36$ choices, so the product rule gives $26 \cdot 36^7$.

There are $10 + 26 = 36$ available characters in every unrestricted position.

C010

Basic Counting

2025 Spring Final | Question 7 | Source page 10 | Difficulty 2 | 1 mark
Q-2025S-FINAL-P07-minimal | Answer provenance: official

> A drawer in a dark room contains 100 red socks, 80 green socks, 60 blue socks and 40 black socks. A youngster selects socks one at a time from the drawer but is unable to see the color of the socks drawn. (A pair of socks is two socks of the same color. No sock may be counted in more than one pair.) Suppose your answer is n.
>
> while $n-1$ socks are not enough.

Official Answer

Source: physical PDF page 10 (COMP2711_S2025_Final.pdf).

On the other hand, 22 will not do: if the numbers of red, green, blue, and black socks are $5, 5, 5, 7$, then four are unpaired, leaving 9 pairs. Thus 23 is the minimum.

Supplementary Explanation

Answer. Twenty-two socks do not guarantee ten pairs.

Derivation/proof. With $n = 23$, test $n - 1 = 22$. Choose five red, five green, five blue and seven black socks: $5 + 5 + 5 + 7 = 22$, within every inventory bound. The maximum disjoint pairs by color are $\lfloor 5/2 \rfloor = 2$, two, two, and $\lfloor 7/2 \rfloor = 3$, giving nine pairs in total. The remaining four socks all have different colors, so no further pair is possible. This valid counterexample disproves sufficiency of twenty-two.

C011

Basic Counting

2025 Spring Final | Question 7 | Source page 10 | Difficulty 2 | 1 mark
Q-2025S-FINAL-P07-sufficient | Answer provenance: official

> A drawer in a dark room contains 100 red socks, 80 green socks, 60 blue socks and 40 black socks. A youngster selects socks one at a time from the drawer but is unable to see the color of the socks drawn. (A pair of socks is two socks of the same color. No sock may be counted in more than one pair.) Suppose your answer is n.
>
> You need to argue that n socks guarantee at least 10 pairs,

Official Answer

Source: physical PDF page 10 (COMP2711_S2025_Final.pdf).

For any selection, at most one sock of each color will be left unpaired, and this happens if and only if an odd number of socks of that color is selected. Thus 24 socks suffice: at most 4 will be unpaired, leaving at least 20 in pairs. However, 23 will do! Since 23 is not the sum of four odd numbers, at most 3 socks out of 23 will be unpaired.

Supplementary Explanation

Answer. Twenty-three socks always contain at least ten pairs.

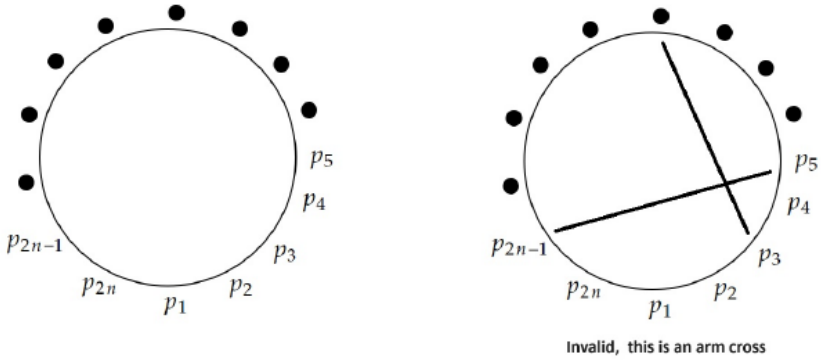
Derivation/proof. Use $n = 23$. Let the color counts be r,g,b,k. After pairing within colors, the number left over is $u = (r \bmod 2) + (g \bmod 2) + (b \bmod 2) + (k \bmod 2) \leq 4$. Since $23 - u$ is even, u is odd, hence at most three. The number of disjoint same-color pairs is $\sum \lfloor \text{color count}/2 \rfloor = (23 - u)/2 \geq 10$. This holds for every permitted distribution, establishing sufficiency.

C012

Basic Counting

2023 Fall Final | Question 9(a) | Source page 14 | Difficulty 2 | 2 marks
Q-2023F-FINAL-P09-a | Answer provenance: official

Assume that there are $2n$ people sitting around a round table. We want to split them into n pairs to engage in handshakes so that no arms cross, given that nobody is allowed to shake hands with more than one person at once. Let a_{2n} denote the number of ways that we can pair them legally without any arm crossing. You can see an invalid handshake in the figure below. Please answer the following questions by referring to the figure:



Is it possible to pair p_1, p_5 together for a handshake? Please explain why in at most 3 lines.

Official Answer

Source: physical PDF page 14 (COMP2711_F2023_Final.pdf).

> (a) No, because then on their right side they have $\{p_2, p_3, p_4\}$ which one of them has be paired with someone from the other side and this cause cross arm.

Supplementary Explanation

Answer. No.

Derivation/proof. The chord p_1p_5 leaves the three people p_2, p_3, p_4 on one side. An odd set cannot be perfectly paired internally, so one must pair with someone on the other side. That handshake crosses p_1p_5, contradicting legality.

This three-sentence proof uses the full-pairing requirement in the shared prompt; allowing an unmatched person would be a different problem.

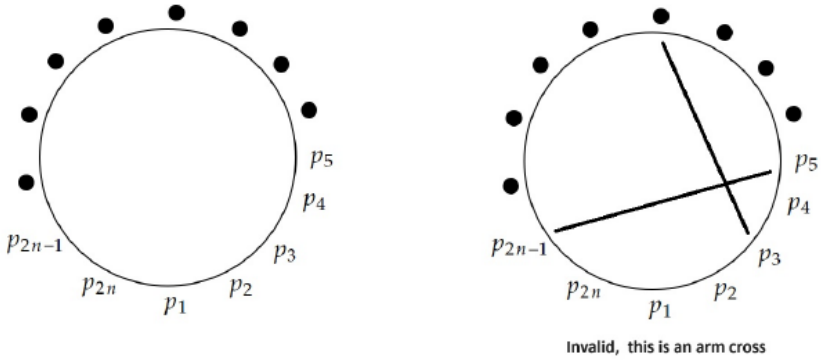
The side containing p_2, p_3, p_4 cannot be internally paired because three is odd. A pairing to the other side violates the noncrossing condition.

C013

Basic Counting

2023 Fall Final | Question 9(b) | Source page 14 | Difficulty 2 | 2 marks
Q-2023F-FINAL-P09-b | Answer provenance: official

Assume that there are $2n$ people sitting around a round table. We want to split them into n pairs to engage in handshakes so that no arms cross, given that nobody is allowed to shake hands with more than one person at once. Let a_{2n} denote the number of ways that we can pair them legally without any arm crossing. You can see an invalid handshake in the figure below. Please answer the following questions by referring to the figure:



Assume that you want to first pair p_1 with someone. List all potential candidates who can be paired with p_1 initially?

Official Answer

Source: physical PDF page 14 (COMP2711_F2023_Final.pdf).

> (b) It can only be paired with p_j where j is even. Otherwise the number of people in one side is odd and will guarantee an arm cross.

Supplementary Explanation

Answer. $p_2, p_4, p_6, \ldots, p_{2n}$.

Derivation/proof. If p_1 pairs with p_j, the intervening people $p_2, \ldots, p_{j-1}$ number $j-2$, and the other side has $2n-j$ people. A legal complete pairing cannot connect the two sides across the fixed chord, so both counts must be even; this is equivalent to j being even. Conversely, for each even j, pair consecutive people on each side. Those adjacent chords stay in their own sides and do not cross, so every listed candidate is feasible.

The two side sizes are $j-2$ and $2n-j$. If j is even, pair adjacent people on each side to construct a valid completion. This includes an empty side for $j=2$ or $j=2n$.

C014

Basic Counting

2023 Fall Midterm | Question 9(d)(i) | Source page 19 | Difficulty 2 | 1 mark
Q-2023F-MID-P09-d-i | Answer provenance: official

> Each time, a robot can move X cm to the right (negative X means the robot move to the left). X is a discrete random variable taking values from $\mathbb{N}$ with the following PMF (probability mass function),
>
> $$p(X=i)=\begin{cases}\frac{2}{3} & \text{if } i=1,\\ \frac{1}{3} & \text{if } i=-1,\\ 0 & \text{otherwise.}\end{cases}$$
>
> For (d): This robot is standing at position 0. It stops once it reaches to position $+1$ cm. Before stopping, the robot moves T times where T is a random variable.
>
> (i) When k is a positive integer, calculate $P(T=2k)$.

Official Answer

Source: physical PDF page 20 (COMP2711_F2023_Midterm.pdf).

> (d) Denote every right move to be R and every left move to be L. Since the robot will stop once it is in position +1 cm, the last move can only be from 0 to +1 cm. Thus, the last move is R, and before the last move, we need to have the same number of R and L. Denote the subsequence excluding the last move to be sequence S. L is legal if when we read from left to right, we have always encountered at least as many L as R.
>
> (i) Since S has the same number of R and L, the length of S must be even. Adding the last move R, it is impossible for T to be an even number. Thus $P(T=2k)=0$.

Supplementary Explanation

Answer. $P(T=2k)=0$ for every positive integer k.

Derivation/proof. After t moves, the position is the sum of t numbers each equal to +1 or −1. Both possible increments are congruent to 1 modulo 2, so the position is congruent to t modulo 2. Reaching +1 therefore requires odd t. In particular, no trajectory can first reach +1 at time 2k, and the event is empty. Its probability is zero under any dependence structure of the successive moves.

The event $\{T=2k\}$ is empty, so its probability must be zero. The proof does not use independence between different moves or eventual arrival; trajectories that never arrive cannot create an even hitting time.

C015

Basic Counting

2022 Final Part 1 | Question 3(c) | Source page 2 | Difficulty 2 | 3 marks
Q-2022-FINAL1-P03-c | Answer provenance: official

> Consider a web system for online shopping that will generate an ID for each order. The order ID is a sequence of 7 characters where each character could be either one of the 26 upper case English letters (A to Z) or one of the 10 digits (0 to 9).
>
> How many possible order IDs are there if the leftmost character must be a digit and no two consecutive characters are the same? (E.g. 3ABA3C6 is valid. C3ARX7B and 3ARRX7B are invalid)

Official Answer

Source: physical PDF page 3 (2022-final-part-1.pdf).

> (c) $10 \cdot 35^6$. The leftmost character has 10 choices, then the second leftmost character has 35 choices because it cannot be the same as the previous character, and so on.

Supplementary Explanation

Answer. $10 \cdot 35^6$.

Derivation/proof. Choose the leftmost digit in 10 ways. At every later position only one of the36 symbols is forbidden: the symbol immediately before it. Thus each has 35 choices regardless of earlier repeated symbols. Multiplication over the six remaining positions gives 10·35^6. The valid example's repeated3 shows distant repetition is allowed.

The valid example $3ABA3C6$ repeats 3. The restriction is adjacent inequality, not global nonrepetition.

C016

Basic Counting

2022 Final Part 1 | Question 3(d) | Source page 2 | Difficulty 2 | 3 marks
Q-2022-FINAL1-P03-d | Answer provenance: official

> Consider a web system for online shopping that will generate an ID for each order. The order ID is a sequence of 7 characters where each character could be either one of the 26 upper case English letters (A to Z) or one of the 10 digits (0 to 9).
>
> How many possible order IDs are there if characters must alternative between a letter and a digit? (E.g. A3R4B7Q and 4V7X3V7 are valid, B7XB4Q3 is invalid)

Official Answer

Source: physical PDF page 3 (2022-final-part-1.pdf).

> (d) $10^4 \cdot 26^3 + 10^3 \cdot 26^4$. It can either be four digits and three letters, or three digits and four letters. The positions of digits and alphabets are fixed in either case.

Supplementary Explanation

Answer. $26^4 \cdot 10^3 + 10^4 \cdot 26^3$.

Derivation/proof. If the ID starts with a letter, its type pattern is L-D-L-D-L-D-L, with four letter slots and three digit slots, giving 26^4·10^3. If it starts with a digit, the complementary pattern gives 10^4·26^3. These two cases cannot overlap and include every alternating ID. Repetition within one type is allowed by the displayed examples.

The digit-first pattern is D-L-D-L-D-L-D. The sample $4V7X3V7$ confirms that same-type positions may repeat.

C017

Basic Counting

2020 Final Part 2 | Question 2 | Source page 2 | Difficulty 2 | 5 marks
Q-2020-FINAL2-P02 | Answer provenance: official

> Three seminars were held by CSE department last week. 70% of students attended the seminar on Monday, 80% of students attended the seminar on Wednesday, and 90% of students attended the seminar on Friday. Prove that at least 40% of students attended all three seminars.

Official Answer

Source: physical PDF page 2 (2020-final-part-2.pdf).

U - Set of all students; A - Set of students having attended the seminar on Monday, $|A| = 70\%|U|$; B - Set of students having attended the seminar on Wednesday, $|B| = 80\%|U|$; C - Set of students having attended the seminar on Friday, $|C| = 90\%|U|$.

$$\begin{aligned} |A \cap B \cap C| &= |U| - |\overline{A} \cup \overline{B} \cup \overline{C}| \\ &\geq |U| - (|\overline{A}| + |\overline{B}| + |\overline{C}|) \\ &= |U| - (30\%|U| + 20\%|U| + 10\%|U|) \\ &= 40\%|U|. \end{aligned}$$

The inequality uses the fact that $|X \cup Y \cup Z| \leq |X| + |Y| + |Z|$.

Supplementary Explanation

Answer. At least 40% attended all three.

Derivation/proof. Let U be the students and A,B,C the three attendance sets. Their complements have sizes .3|U|,.2|U|,.1|U|. Anyone not attending all three lies in $\overline{A} \cup \overline{B} \cup \overline{C}$, whose size is at most the sum .6|U|. Therefore $|A \cap B \cap C| = |U| - |\overline{A} \cup \overline{B} \cup \overline{C}| \geq .4|U|$. No independence between attendance events is needed.

The union of the three absence sets has size at most $(0.3 + 0.2 + 0.1)|U| = 0.6|U|$. Taking its complement yields $|A \cap B \cap C| \geq |U| - 0.6|U| = 0.4|U|$, so at least forty percent attend all three.

C018

Basic Counting

2020 Final Part 2 | Question 5(c) | Source page 3 | Difficulty 2 | 2 marks
Q-2020-FINAL2-P05-c | Answer provenance: official

> There is a random undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> If G has exactly k connected components, what is the minimum possible value of X?

Official Answer

Source: physical PDF page 4 (2020-final-part-2.pdf).

Suppose that the k connected components are $C_1, C_2, \ldots, C_k$. Since C_i is a connected component with v_i vertices, the minimum number of edges to connect them is $v_i - 1$, which is a tree. The minimum of X is equal to all components reach minimal the same time.

$$\min(X) = \min_{C_i} \sum_{i=1}^{k} \min(C_i) = \min_{C_i} \sum_{i=1}^{k} (v_i - 1) = n - k.$$

Supplementary Explanation

Answer. $n - k$.

Derivation/proof. Let component i have v_i vertices. Connectivity requires at least v_i−1 edges: starting at one vertex, each additional vertex in a spanning traversal needs an edge to the reached set. Summing gives at least $\sum(v_i - 1) = n - k$. Attain this with a tree on each component, including isolated vertices as one-vertex trees. Thus the lower bound is exact.

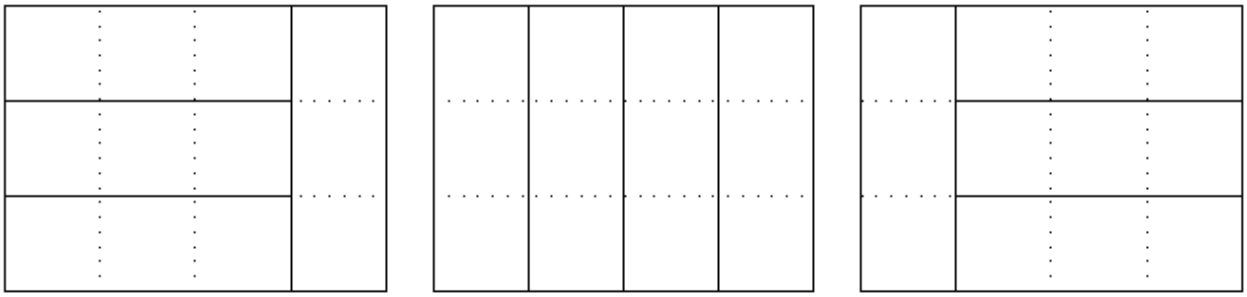
The component lower bound follows by growing a connected reached set: each newly reached vertex needs a connecting edge. Taking a tree in every component attains all bounds simultaneously.

The component sizes satisfy $\sum_i v_i = n$. A tree in component i uses $v_i - 1$ edges, so the total minimum is $\sum_i (v_i - 1) = n - k$.

C019

Basic Counting

2018 Final | Question 2(a) | Source page 4 | Difficulty 2
Q-2018-FINAL-P02-a-E3 | Answer provenance: official

> Let E_n be the number of ways in which you can tile a 3-by-n board with 1-by-3 pieces. For example, when $n = 4$ the board is 3-by-4 and $E_4 = 3$. The 3 ways are shown as follows:
>
> 
>
> (a) Calculate E_1, E_2, and E_3. [Atomic request: E_3.]

Official Answer

Source: physical PDF page 4 (2018-final.pdf).

1, 1, 2.

Supplementary Explanation

Answer. $E_3 = 2$.

Derivation/proof. If a vertical piece occurs, it fills a complete column. No horizontal piece can then cross all three columns, so the other columns must also be vertical. If no vertical piece occurs, each of the three rows must be one horizontal piece. These two disjoint cases are the all-vertical and all-horizontal tilings.

The three printed values are respectively E_1, E_2, E_3; this atomic question asks for the third.

On a three-column board the tiles are either all vertical or all horizontal. A horizontal tile forces the other two horizontal rows; otherwise all three columns are vertical, giving exactly two tilings.

C020

Basic Counting

2018 Final | Question 10(c) | Source page 14 | Difficulty 2
Q-2018-FINAL-P10-c | Answer provenance: official

> Given a $n \times p$ matrix A and a $p \times m$ matrix B, the following algorithm computes the $n \times m$ product matrix $C = A \cdot B$.
>
> Matrix-Mult(A,B)
>
> ```
> for i = 1 to n do
>     for j = 1 to m do
>         C[i,j] = 0
>         for k = 1 to p do
>             C[i,j] = C[i,j] + A[i,k] · B[k,j]
>         end for
>     end for
> end for
> return C
> ```
>
> c) Express the total number of multiplications that the algorithm does, expressed as a function of n, m, p.

Official Answer

Source: physical PDF page 15 (2018-final.pdf).

The first loop is executed n times, therefore the second is executed nm times, and the third nmp times. At every iteration of the innermost loop a multiplication is executed, therefore the answer is nmp.

Supplementary Explanation

Answer. nmp.

Derivation/proof. For each of n rows and m columns, the algorithm runs the inner loop p times. Every inner iteration contains exactly one scalar multiplication $A[i,k]B[k,j]$. Hence the exact multiplication count is $n \cdot m \cdot p$. Initializing to zero and adding do not contribute to the requested multiplication count.

For each output entry, sum the terms $A[i,k] \cdot B[k,j]$ over the common inner dimension. The output row is indexed by i and its column by j.

C021

Basic Counting

2018 Midterm | Question 3(a) | Source page 5 | Difficulty 2 | 2 marks
Q-2018-MID-P03-a | Answer provenance: official

> Consider the set $S = \{1, 2, \ldots, 2n\}$, and subsets $P = \{p_1, p_2, \ldots, p_m\} \subseteq S$ such that they satisfy the following criterion:
>
> $\forall p_i \in P \forall p_j \in P (i \neq j \rightarrow p_i \nmid p_j)$
>
> In the above, $p_i \nmid p_j$ means p_j is not divisible by p_i.
>
> (a) Find a subset $P \subseteq S$ satisfying the above criterion such that $|P| = n$.

Official Answer

Source: physical PDF page 5 (2018-midterm.pdf).

$P = \{n+1, n+2, \ldots, 2n\}$. Answer for a specific n is not accepted. (2 marks) Other answers are possible.

Supplementary Explanation

Answer. $P = \{n+1, n+2, \ldots, 2n\}$.

Derivation/proof. This set contains exactly $2n - (n+1) + 1 = n$ elements. If $a < b$ belong to it and $a \mid b$, their distinct positive values imply $b \geq 2a \geq 2(n+1) > 2n$, contradicting $b \leq 2n$. Also a larger positive integer cannot divide a smaller positive integer, so both ordered directions of the criterion hold.

The set size is $2n - (n+1) + 1 = n$. A larger positive integer cannot divide a smaller positive integer either, so the ordered-pair criterion holds in both directions.

C022

Basic Counting

2018 Midterm | Question 4(a) | Source page 6 | Difficulty 2
Q-2018-MID-P04-a-proof | Answer provenance: official

> Consider finite sets S and T, and a function $f : S \to T$.
>
> "If f is injective, then $|T| \geq |S|$."
>
> Then, prove the contrapositive directly.

Official Answer

Source: physical PDF page 6 (2018-midterm.pdf).

The contrapositive is "If $|T| < |S|$, then f is not injective". We can prove it by the pigeonhole principle. f maps pigeons in S to holes in T. Since $|T| < |S|$, there must be two pigeons that share a same hole, say $a, b \in S$. Then $a \neq b$ but $f(a) = f(b)$, so f is not injective.

Supplementary Explanation

Answer. If $|T| < |S|$, the function f is not injective.

Derivation/proof. Assume $|S| > |T|$. Regard the elements of S as pigeons and the elements of T as holes, sending s to hole $f(s)$. By pigeonhole principle, two distinct elements $a, b \in S$ share an image: $f(a) = f(b)$. This is precisely failure of injectivity.

To show failure of injectivity, the two inputs must satisfy $a \neq b$ while their outputs agree; equality of outputs for the same input would prove nothing.

C023

Basic Counting

2018 Midterm | Question 8(b) | Source page 10 | Difficulty 2
Q-2018-MID-P08-b | Answer provenance: official

> You are given a suitcase that has a combination lock. It consists of five dials, each of which can be rotated to take a value from 0–9. The suitcase only opens if you guess the correct 5-digit solution.
>
> Briefly justify your answers.
>
> (b) How many different solutions are there, if you know that no two adjacent dials can have the same digit?

Official Answer

Source: physical PDF page 10 (2018-midterm.pdf).

$10 \cdot 9^4$. All digits are possible for the first dial. For each subsequent dial, the digit chosen in the previous dial is not a valid option.

Supplementary Explanation

Answer. $10 \cdot 9^4$.

Derivation/proof. Choose the first digit in ten ways. For each of the next four dials, precisely one digit is forbidden: the digit on the immediately previous dial. Thus each has nine choices, even if it repeats a nonadjacent digit. Multiplication gives $10 \cdot 9^4$.

One may reuse the first digit on the third dial: only the preceding digit is forbidden. The last two positions consequently each have nine choices, not nine followed by eight.

C024

Basic Counting

2017 Final | Question 3(a) | Source page 5 | Difficulty 2
Q-2017-FINAL-P03-a | Answer provenance: official

> For each of the following parts, find the "best" big-O notation to describe the complexity of the algorithm. Choose your answers from the following:
>
> $O(1), O(\log n), O(n), O(n \log n), O(n^2), O(n^3), \ldots, O(2^n), O(n!)$
>
> Assume that the amount of time to print any string is a constant, i.e. $O(1)$.
>
> (a) An algorithm that prints all bit strings of length n that end with 1.

Official Answer

Source: physical PDF page 5 (2017-final.pdf).

2^n.

Supplementary Explanation

Answer. $O(2^n)$ (in fact $\Theta(2^n)$).

Derivation/proof. The last bit is fixed and the other $n-1$ bits have two choices each, so 2^{n-1} strings must be printed. Under the stipulated constant cost per string, this output work is $\Theta(2^{n-1}) = \Theta(2^n)$; enumeration with constant amortized work per output attains it.

C025

Basic Counting

2017 Midterm | Question 7 | Source page 9 | Difficulty 2 | 7 marks
Q-2017-MID-P07 | Answer provenance: official

> A computer randomly prints three-digit codes, with no repeated digits in any code (for example, 387, 072, 760). What is the minimum number of codes that must be printed in order to guarantee that at least six of the codes are identical?

Official Answer

Source: physical PDF page 9 (2017-midterm.pdf).

There are $10 \cdot 9 \cdot 8 = 720$ different codes. By the generalized pigeonhole principle, to guarantee that at least six identical codes will be printed, the minimum number n of printed codes is such that $\lceil n/720 \rceil = 6$, which is $n = 5 \cdot 720 + 1 = 3601$.

Supplementary Explanation

Answer. 3601.

Derivation/proof. There are $10 \cdot 9 \cdot 8 = 720$ code types because a leading zero is allowed. With at most five copies of each type, at most $5 \cdot 720 = 3600$ codes can be printed. Thus the next code forces a sixth copy. Conversely, printing exactly five copies of every type shows 3600 does not suffice.

The random generation does not change this worst-case guarantee.

C026

Basic Counting

2016 Midterm | Question 1(c) | Source page 3 | Difficulty 2 | 3 marks
Q-2016-MID-P01-c | Answer provenance: official

> Consider a web system for online shopping that generates an order id for each business. The order id is a sequence of 7 characters where each character could be either one of the 26 Upper Case English Alphabets (A to Z) or one of the 10 digits (0 to 9).
>
> (c) How many possible order id's are there if the leftmost character must be a digit and no two consecutive characters are the same? (E.g., 3ABA3C6 is valid. C3ARX7B and 3ARRX7B are invalid)

Official Answer

Source: PDF physical page 3 (2016-midterm.pdf).

(c) $10 \cdot 35^6$. The leftmost character has 10 choices, then the second leftmost character has 35 choices because it cannot be the same as it's adjacent character, and so on.

Supplementary Explanation

Answer. $10 \cdot 35^6$.

Derivation/proof. Choose the leading digit in ten ways. At each of the remaining six positions, exactly one of 36 characters equals its predecessor and is excluded. The multiplication rule thus gives $10 \cdot 35^6$.

Earlier nonadjacent characters may recur, as the A and 3 do in the valid example.

After the first position there are always 35 choices, because only one immediate predecessor is excluded.

C027

Basic Counting

2016 Midterm | Question 1(d) | Source page 3 | Difficulty 2 | 3 marks
Q-2016-MID-P01-d | Answer provenance: official

> Consider a web system for online shopping that generates an order id for each business. The order id is a sequence of 7 characters where each character could be either one of the 26 Upper Case English Alphabets (A to Z) or one of the 10 digits (0 to 9).
>
> (d) How many possible order id's are there if the order id is alternative between alphabet and digit? (E.g., A3R4B7Q and 4V7X3V7 are valid, B7XB4Q3 is invalid)

Official Answer

Source: PDF physical page 3 (2016-midterm.pdf).

(d) $10^4 \cdot 26^3 + 10^3 \cdot 26^4$. It can either be four digits and three alphabets or three digits and four alphabets. The position of digits and alphabets are fixed in either case.

Supplementary Explanation

Answer. $26^4 10^3 + 10^4 26^3$.

Derivation/proof. Pattern L-D-L-D-L-D-L gives $26^4 10^3$ possibilities; pattern D-L-D-L-D-L-D gives $10^4 26^3$. Repetition within a category is allowed. The patterns cannot overlap, so add the two counts.

C028

Basic Counting

2015 Midterm | Question 6(b) | Source page 11 | Difficulty 2 | 3 marks
Q-2015-MID-P06-b | Answer provenance: official

> In a computer system, a password is a string of 8 characters, where each character is either a digit or a lowercase letter from the English alphabet. How many possible passwords are there under each of the following conditions?
>
> (b) The first character must be a letter, and there must be at least one digit.
>
> Briefly explain how your answers are derived.

Official Answer

Source: PDF physical page 11 (2015-midterm.pdf).

(b) $26 \times 36^7 - 26^8$. There are 26^8 passwords without digits.

Supplementary Explanation

Answer. $26 \cdot 36^7 - 26^8$.

Derivation/proof. The unrestricted first-letter count is $26 \cdot 36^7$. A password in this set violates the digit requirement exactly when all eight characters are letters, giving 26^8 bad strings. Subtraction gives the answer.

C029

Basic Counting

2015 Midterm | Question 6(c) | Source page 11 | Difficulty 2 | 4 marks
Q-2015-MID-P06-c | Answer provenance: official

> In a computer system, a password is a string of 8 characters, where each character is either a digit or a lowercase letter from the English alphabet. How many possible passwords are there under each of the following conditions?
>
> (c) The first character must be a letter, and there must be at least two digits.
>
> Briefly explain how your answers are derived.

Official Answer

Source: PDF physical page 11 (2015-midterm.pdf).

(c) $26 \times 36^7 - 26^8 - 7 \times 10 \times 26^7$. The are 26^8 passwords without digits. The number of passwords with exactly 1 digit is: $7 \times 10 \times 26^7$, because there are 7 possible positions for the digit, the digit has 10 possibilities, and the other 7 characters must be letters.

Supplementary Explanation

Answer. $26 \cdot 36^7 - 26^8 - 7 \cdot 10 \cdot 26^7$.

Derivation/proof. Start from $26 \cdot 36^7$. Subtract the 26^8 all-letter strings. For exactly one digit, choose its position among positions 2–8 (7 ways), its value (10 ways), and all seven letters (26^7 ways). These two excluded cases are disjoint and exhaust passwords with fewer than two digits.

C030

Basic Counting

2015 Midterm | Question 7(c) | Source page 12 | Difficulty 2 | 4 marks
Q-2015-MID-P07-c | Answer provenance: official

> You are going to distribute kn cash coupons to kn members in your company such that each member will get one cash coupon, where $k, n \geq 1$. The values of the kn cash coupons are distinct. Suppose there are n ranks of seniority in the company, and there are exactly k members at each rank. If a more senior member receives a cash coupon worth less than the cash coupon given to a more junior member, you will receive a complaint.
>
> (c) Suppose you have an unlimited number of cash coupons, instead of kn cash coupons. Each member may receive 0 or more cash coupons, and you distribute the cash coupons randomly. For each rank r, let t_r be the total number of coupons members at the rank receive altogether. What is the minimum number of cash coupons you will need to distribute so that there is at least one rank r for which $t_r \geq 2k$?
>
> Briefly explain how your answers are derived.

Official Answer

Source: PDF physical page 12 (2015-midterm.pdf).

(c) $(2k-1)n+1$. This is the same as finding the minimum number x of pigeons such that at least one of the n pigeonholes contain at least $2k$ objects. By the pigeonhole principle, find the minimum x such that $\lceil x/n \rceil = 2k$. It is easy to see that this equality holds for $x \in \{(2k-1)n+1, \ldots, 2kn\}$. Thus, the minimum value of $x = (2k-1)n+1$.

Supplementary Explanation

Answer. $n(2k-1)+1$.

Derivation/proof. A distribution with every $t_r \leq 2k-1$ contains at most $n(2k-1)$ coupons. Therefore $n(2k-1)+1$ forces some $t_r \geq 2k$. Sharpness follows by giving exactly $2k-1$ coupons to each rank, which avoids the event using one fewer coupon.

The sharpness construction is admissible because a single member of each rank may receive all $2k-1$ coupons; members are allowed any nonnegative number.

C031

Basic Counting

2025 Fall Final | Question 2(b) | Source page 5 | Difficulty 3 | 5 marks
Q-2025F-FINAL-P02-b | Answer provenance: official

> (b) [5 points] Consider $A = \{1, 2, 3, 4, 5\}$. Imagine picking 17 different subsets from A at random. Prove that at least two of them are disjoint.

Official Answer

Source: physical PDF page 5 (COMP2711_F2025_Final.pdf).

(b) The power set of A has $2^5 = 32$ subsets. For each subset $S \subseteq A$, its complement S^c is disjoint with S (i.e., $S \cap S^c = \varnothing$).

Also the complement of any subset is unique (i.e., there is a bijection between any subset and its complement).

Thus, these 32 subsets can be paired into 16 groups of (subsets and their complement). Based on pigeonhole principle, if we choose 17 subsets, two of them will be from the same pair which means they are disjoint.

Supplementary Explanation

Answer. Among the 17 distinct subsets, at least two are disjoint.

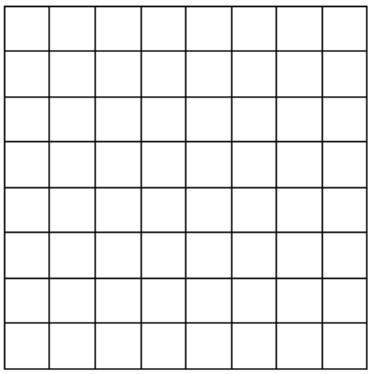
Derivation/proof. The power set of A has $2^5 = 32$ subsets. Pair each $S \subseteq A$ with its unique complement $A \setminus S$. They are disjoint and cannot be equal: otherwise element 1 would belong and not belong to the same set. Taking the complement twice returns S, so these subsets partition into exactly 16 disjoint pairs. Selecting 17 distinct subsets forces both members of some pair to be selected, by the pigeonhole principle. They are complements and hence disjoint. The random selection distribution has no effect on this deterministic conclusion.

C032

Basic Counting

2025 Fall Midterm | Question 7 | Source page 17 | Difficulty 3 | 7 marks
Q-2025F-MID-P07 | Answer provenance: official

> Consider an 8×8 grid of 1×1 cells. If 17 cells are colored red at random, prove that there exist at least two red cells that share a common corner.



Official Answer

Source: physical PDF page 17 (COMP2711_F2025_Midterm.pdf).

> We split the 8×8 grid into 16 2×2 grids as follows: page=17 (COMP2711_F2025_Midterm.pdf) Note that any two cells from a 2×2 Grid contains a shared common corner. If we color 17 1×1 cells, it means we color at least two cells from one 2×2 grid which have a shared common corner.

Supplementary Explanation

Answer. Two red cells must share a corner.

Derivation/proof. Pair adjacent rows into four groups and adjacent columns into four groups, producing $4 \cdot 4 = 16$ disjoint 2-by-2 blocks. Seventeen red cells distributed among sixteen blocks force at least one block to contain two red cells. All four cells in a block contain its central grid point, so any two distinct red cells within it share that corner. This proves the claim for every selection of seventeen cells; “at random” does not require a probability calculation.

C033

Basic Counting

2025 Spring Final | Question 3(b) | Source page 4 | Difficulty 3 | 6 marks
Q-2025S-FINAL-P03-b | Answer provenance: official

> For a positive integer p, define the positive integer n to be p-safe if n differs in absolute value by more than 2 from all multiples of p. For example, the set of 10-safe numbers are $3, 4, 5, 6, 7, 13, 14, 15, 16, 17, \cdots$.
>
> (b) [6 pts] Find the number of positive integers less than 1001 ($1001 = 7 \cdot 11 \cdot 13$) that are simultaneously 7-safe, 11-safe, and 13-safe. Please justify your answer.

Official Answer

Source: physical PDF page 4 (COMP2711_S2025_Final.pdf).

(b) Consider the mapping from x to the 3-tuple $(a = x \bmod 7, b = x \bmod 11, c = x \bmod 13)$. For any x that is 7-safe, 11-safe, and 13-safe, we must have $a \in \{3, 4\}$, $b \in \{3, 4, 5, 6, 7, 8\}$, $c \in \{3, 4, 5, 6, 7, 8, 9, 10\}$. By the Chinese Remainder Theorem, for each such triple (a, b, c), there is also a unique positive integer $x < 1001$ such that $a = x \bmod 7, b = x \bmod 11, c = x \bmod 13$. So there is a one-to-one correspondence between such triples and the set of positive integers less 1001 that are 7-safe, 11-safe, and 13-safe. Thus, the answer is

$$2 \cdot 6 \cdot 8 = 96.$$

Supplementary Explanation

Answer. 96.

Derivation/proof. The allowed residues are $\{3, 4\}$ modulo seven, $\{3, 4, 5, 6, 7, 8\}$ modulo eleven, and $\{3, 4, 5, 6, 7, 8, 9, 10\}$ modulo thirteen. These sets have two, six and eight members. The moduli are pairwise coprime, so CRT assigns each triple a unique class modulo 1001 and distinct triples cannot overlap. Zero satisfies none of the allowed residue conditions; hence each class has exactly one representative with $1 \leq x < 1001$. The product rule yields $2 \cdot 6 \cdot 8 = 96$.

C034

Basic Counting

2025 Spring Final | Question 4(b) | Source page 5 | Difficulty 3 | 5 marks
Q-2025S-FINAL-P04-b | Answer provenance: official

> The least common multiple (LCM) of two integers x and y is the smallest positive integer that is a multiple of both x and y. As an example, the least common multiple of 8 and 10 is 40.
>
> (b) [5 pts] Determine the number of ordered pairs of positive integers (a, b) such that the least common multiple of a and b is $2^3 \cdot 5^7 \cdot 11^{13}$. Briefly explain your answer.

Official Answer

Source: physical PDF page 5 (COMP2711_S2025_Final.pdf).

(b) Both a and b are divisors of $2^3 5^7 11^{13}$, and so $a = 2^x 5^y 11^z$ and $b = 2^s 5^t 11^u$ for some nonnegative integers x, y, z, s, t, u. Because $2^3 5^7 11^{13}$ is the least common multiple, $\max(x, s) = 3$, $\max(y, t) = 7$, and $\max(z, u) = 13$. Hence (x, s) can be equal to $(0, 3), (1, 3), (2, 3), (3, 3), (3, 2), (3, 1)$, or $(3, 0)$, so there are 7 choices for (x, s). Similarly, there are 15 and 27 choices for (y, t) and (z, u), respectively. By the product rule, there are $7 \times 15 \times 27 = 2835$ ordered pairs of positive integers (a, b) having $2^3 5^7 11^{13}$ as their least common multiple.

Supplementary Explanation

Answer. 2835.

Derivation/proof. Since both numbers divide the target, write $a = 2^u 5^v 11^w$ and $b = 2^s 5^t 11^z$ with nonnegative exponents. Require $\max(u, s) = 3$, $\max(v, t) = 7$ and $\max(w, z) = 13$. For target exponent e, there are $(e + 1)^2$ candidate pairs; subtract the e^2 pairs with both exponents smaller than e. This leaves $(e + 1)^2 - e^2 = 2e + 1$. The three primes therefore contribute seven, fifteen and twenty-seven ordered choices, giving $7 \cdot 15 \cdot 27 = 2835$. No other prime may appear, since that would change the LCM.

C035

Basic Counting

2025 Spring Final | Question 7 | Source page 10 | Difficulty 3 | 4 marks
Q-2025S-FINAL-P07-N | Answer provenance: official

> A drawer in a dark room contains 100 red socks, 80 green socks, 60 blue socks and 40 black socks. A youngster selects socks one at a time from the drawer but is unable to see the color of the socks drawn. (A pair of socks is two socks of the same color. No sock may be counted in more than one pair.) Suppose your answer is n.
>
> What is the smallest number of socks that must be selected to guarantee that the selection contains at least 10 pairs?

Official Answer

Source: physical PDF page 10 (COMP2711_S2025_Final.pdf).

First Solution: For any selection, at most one sock of each color will be left unpaired, and this happens if and only if an odd number of socks of that color is selected. Thus 24 socks suffice: at most 4 will be unpaired, leaving at least 20 in pairs. However, 23 will do! Since 23 is not the sum of four odd numbers, at most 3 socks out of 23 will be unpaired. On the other hand, 22 will not do: if the numbers of red, green, blue, and black socks are $5, 5, 5, 7$, then four are unpaired, leaving 9 pairs. Thus 23 is the minimum.

Second Solution: Proceed inductively. If we require only one pair, then it suffices to select 5 socks. Moreover, selecting 4 socks doesn't guarantee a pair since we might select one sock of each color. If we require two pairs, then it suffices to select 7 socks: any set of 7 socks must contain a pair; if we remove this pair, then the remaining 5 socks will contain a second pair as shown above. On the other hand, 6 socks might contain 3 greens, 1 black, 1 red and 1 blue - hence only one pair. Thus 7 socks is the smallest number to guarantee two pairs. Similar reasoning shows that we must draw 9 socks to guarantee 3 pairs, and in general, $2p + 3$ socks to guarantee p pairs. This formula is easily proved by mathematical induction. Thus 23 socks are needed to guarantee 10 pairs.

Marking Scheme

- 4 pts for correct n
- 1 pt for argument that n is sufficient
- 1 pt for argument that n-1 is not sufficient
- If n is wrong, 1-2 pts.

Supplementary Explanation

Answer. $n = 23$.

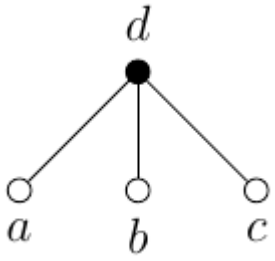
Derivation/proof. Pair the socks separately by color and let u be the number left over. Each color leaves at most one, so $u \leq 4$, with u having the same parity as the total. For twenty-three socks, u is odd, hence $u \leq 3$ and at least $(23 - 3)/2 = 10$ pairs exist. However, twenty-two socks may have color counts $(5, 5, 5, 7)$, giving only $2 + 2 + 2 + 3 = 9$ pairs without exceeding any inventory. Thus twenty-three guarantees the target and twenty-two does not; the minimum is twenty-three.

C036

Basic Counting

2025 Spring Final | Question 12(c) | Source page 16 | Difficulty 3 | 5 marks
Q-2025S-FINAL-P12-c | Answer provenance: official

(c) [5 pts] Consider a simple undirected graph G with n labeled vertices, where vertex i has degree d_i. Derive an expression for the total number of ordered paths of length 2 in G, in terms of the degrees $d_1, d_2, \ldots, d_n$. For this question, you may use the summation sign Σ to express your answer. Note that the origin and destination of the path must be different. As an example, the ordered paths of length 2 in the graph below are: $\{(adb), (bda), (adc), (cda), (bdc), (cdb)\}$.



Official Answer

Source: physical PDF page 17 (COMP2711_S2025_Final.pdf).

(c) Note that each path consists of 3 nodes. By assuming each node i to be the middle node, the number of possible ordered paths is equal to the number of ways to choose an ordered pair of distinct neighbors of i. Since i has d_i neighbors, the number of ordered pairs of distinct neighbors is $d_i(d_i - 1)$. So the answer to the question is:

$$\sum_{i=1}^{n} d_i(d_i - 1)$$

Also accept:

$$2\sum_{i=1}^{n} \binom{d_i}{2}$$

Supplementary Explanation

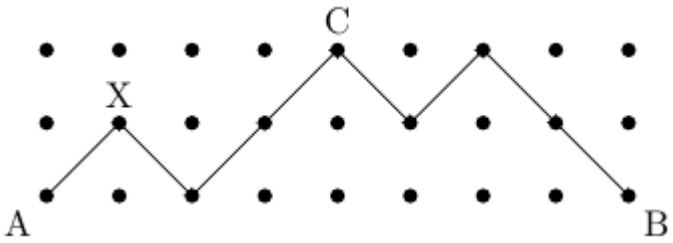
Answer. $\sum_{i=1}^{n} d_i(d_i - 1)$.

Derivation/proof. Fix middle vertex i. There are d_i choices for the starting neighbor and $d_i - 1$ choices for a different ending neighbor, giving $d_i(d_i - 1)$ ordered paths through i. Every length-two path has exactly one middle vertex, so summing over i does not duplicate any ordered path. In the example, the center has degree three and every other vertex degree one. The count is $3 \cdot 2 = 6$, matching the six listed ordered triples.

C037

Basic Counting

2024 Fall Midterm | Question 4(b)(i) | Source page 8 | Difficulty 3 | 3 marks
Q-2024F-MID-P04-b-i | Answer provenance: official

> Consider the following picture. You are only allowed to move ↗ (one to the right and up) and ↘ (one to the right and down) only. In how many ways you can get from A to B such that:
>
> 
>
>
> As you can see in this picture, This is a valid path from A to B.
>
> (i) There are no restrictions. Briefly explain why. Simply counting all possible ways will not give you full credits. Hint: Note that in order to get from A to B you have to pass from some certain points like X.

Official Answer

Source: physical PDF page 9 (COMP2711_F2024_Midterm.pdf).

> (b) (i) 2^3. Note that the path has to pass from the middle point on even columns. So for columns with odd index (from 3) we have two options to either go up or down. Thus the answer is 2^3.

Supplementary Explanation

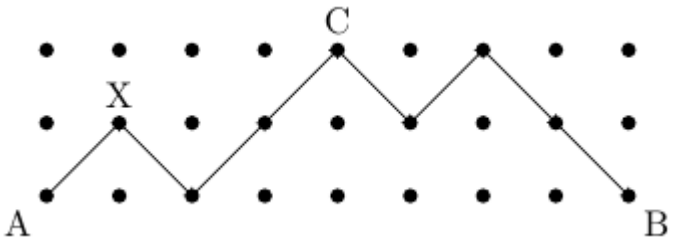
Answer. $2^3 = 8$.

Derivation/proof. Number the nine columns from left to right. Both A and B lie on the bottom row. Every move advances one column and changes height by one, so columns two, four, six and eight must lie on the middle row. The initial move from A to X and the last move down to B are forced. Columns three, five and seven may each be upper or lower, followed by a forced return to the middle. These three choices are independent, so $2 \cdot 2 \cdot 2 = 8$. Every choice triple gives one legal path.

C038

Basic Counting

2024 Fall Midterm | Question 4(b)(ii) | Source page 8 | Difficulty 3 | 3 marks
Q-2024F-MID-P04-b-ii | Answer provenance: official

> Consider the following picture. You are only allowed to move ↗ (one to the right and up) and ↘ (one to the right and down) only. In how many ways you can get from A to B such that:
>
> 
>
>
> As you can see in this picture, This is a valid path from A to B.
>
> (ii) You want to pass from C. Briefly explain why. Simply counting all possible ways will not give you full credits.

Official Answer

Source: physical PDF page 9 (COMP2711_F2024_Midterm.pdf).

> (ii) We can simply see that for the fifth column we have to move up. So the total possible ways is 2^2.

Supplementary Explanation

Answer. $2^2 = 4$.

Derivation/proof. The source places C on the top row of the fifth column. As before, columns two, four, six and eight are middle-row positions, and columns three, five and seven select the branches. Passing through C forces the fifth-column choice upward. Columns three and seven remain independent upper-or-lower choices. Each pair gives one path through C, so $2 \cdot 1 \cdot 2 = 4$.

C039

Basic Counting

2024 Fall Midterm | Question 8 | Source page 13 | Difficulty 3 | 8 marks
Q-2024F-MID-P08 | Answer provenance: derived

> Assume 16 teams. Each team plays exactly 5 games. A win gains 3 points, draw gains 1 point and loss gains 0 point. Prove that at least two teams will have the same number of points.

Official Answer

Source: physical PDF page 13 (COMP2711_F2024_Midterm.pdf).

> Answer: If a team with 15 points exists, the next team must have at least a loss, which means the next possible value for total points is 12 (i.e., 13 and 14 cannot occur). If no team achieves 15, the next maximum is 13 points for team(s) with for 4 wins and a tie (i.e., 14 and 15 cannot occur). In any case, the options for total points is less than 15, while we have 16 teams.

Supplementary Explanation

Answer. At least two teams have equal points.

Derivation/proof. A team scores $s = 3w + d$, where $w, d, \ell \geq 0$ and $w + d + \ell = 5$, so $0 \leq s \leq 15$. If $w \leq 4$, then $s \leq 3w + (5 - w) = 2w + 5 \leq 13$. If $w = 5$, then $d = 0$ and $s = 15$. Therefore fourteen points is impossible, and every score lies in $\{0, 1, \ldots, 13, 15\}$, at most fifteen possibilities. Sixteen teams occupy at most fifteen score classes, so the pigeonhole principle gives two with the same score. The argument does not depend on overlapping schedules.

C040

Basic Counting

2023 Fall Final | Question 2(b) | Source page 4 | Difficulty 3 | 3 marks
Q-2023F-FINAL-P02-b | Answer provenance: official

> How many 4-digit positive integers a exists such that both a and $2 \times a$ have only even digits? For example, 2004 is a valid number, but $a = 4282$ is not because $2a = 2 \cdot 4282 = 8564$ which contains the odd digit "5". (Notice that the leftmost digit in a 4-digit number cannot be 0)

Official Answer

Source: physical PDF page 4 (COMP2711_F2023_Final.pdf), physical PDF page 5 (COMP2711_F2023_Final.pdf).

> (b) Notice that when we have a carryover thing it changes the oddity of the next digit. So for example if we have a valid number a like $a = 2822$. When we multiply it by 2 we get the invalid number 5644 and that is because we got a carryover when multiplying 8 by 2. So in order to have a valid number a such that $2a$ is also valid, they should not produce any carryover value. Which means they can be either $0, 2, 4$. So the number $abcd$ can have $2 \times 3 \times 3 \times 3 = 54$ possible options.

Supplementary Explanation

Answer. $2 \cdot 3^3 = 54$.

Derivation/proof. Each digit of a is in $\{0, 2, 4, 6, 8\}$. If any digit is 6 or 8, doubling generates a carry of 1. At the next position, twice an even digit plus this carry is odd; if the carry exits the leading position, it becomes the odd leading digit 1. Hence no carry is allowed, so all four digits must lie in $\{0, 2, 4\}$. Conversely, these digits generate no carries and double to even digits. The leading digit has two choices, 2 or 4; each remaining digit has three, giving 54.

A carry of one makes the next doubled digit odd; if it exits the leading position it creates the odd leading digit 1. Hence every digit belongs to $\{0, 2, 4\}$, with only $2, 4$ allowed first, giving $2 \cdot 3 \cdot 3 \cdot 3 = 54$. Conversely, these choices never generate a carry and both numbers have only even digits.

C041

Basic Counting

2023 Fall Final | Question 6(a) | Source page 10 | Difficulty 3 | 6 marks
Q-2023F-FINAL-P06-a | Answer provenance: official

> Let a, b, c, d be integers. Show that the product
>
> $$(a-b)(a-c)(a-d)(b-c)(b-d)(c-d)$$
>
> is divisible by 12.
>
> (Hint: You might want to prove that there exist at least two numbers among a, b, c, d such that they have the same remainder mod 3 and this might help you with the main proof)

Official Answer

Source: physical PDF page 10 (COMP2711_F2023_Final.pdf).

> (a) Out of 4 different numbers, two of them have the same remainder mod 3 based on pigeonhole principle. So one of those six elements are divisible by 3. Now out of these four numbers, either we have two odd and two even or there exists at least three numbers with same oddity. In each case we can get two different pair of numbers that their difference is an even number. So it would be dividable by 4. So the total would be dividable by 12.

Supplementary Explanation

Answer. $12 \mid (a-b)(a-c)(a-d)(b-c)(b-d)(c-d)$.

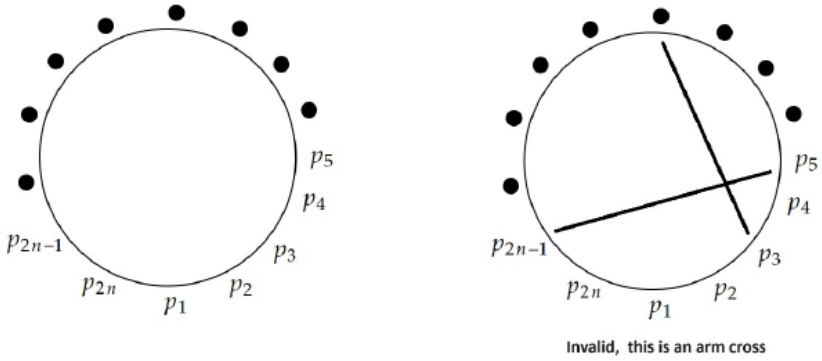
Derivation/proof. Among four integers, two share a residue modulo 3, so their difference is a factor divisible by 3. For parity, either there are two even and two odd numbers, giving one even difference in each pair, or at least three integers have the same parity, giving at least two distinct even differences among those three. Thus two factors supply a factor of 4. Since $\gcd(3,4)=1$, the product is divisible by 12. Repeated integer values cause a zero factor, and 12 divides zero, so distinctness is not required.

There are only three residue classes modulo three, so two inputs share a class. The parity cases provide two even factors, whose product is divisible by four. If two inputs coincide, the product is zero and is still divisible by twelve.

C042

Basic Counting

2023 Fall Final | Question 9(c) | Source page 14 | Difficulty 3 | 5 marks
Q-2023F-FINAL-P09-c | Answer provenance: official

> Assume that there are $2n$ people sitting around a round table. We want to split them into n pairs to engage in handshakes so that no arms cross, given that nobody is allowed to shake hands with more than one person at once. Let a_{2n} denote the number of ways that we can pair them legally without any arm crossing. You can see an invalid handshake in the figure below. Please answer the following questions by referring to the figure:
>
> 
>
>
> Assume p_1 is paired with p_j initially. Derive the number of ways to pair up the remaining people in terms of a_i.
>
> (Hint: Notice that this handshake will create two separate groups of people on the “left” side and the “right” side of it)

Official Answer

Source: physical PDF page 14 (COMP2711_F2023_Final.pdf).

> (c) In this case we have one group of size $j-2$ on one side and a group of size $2n-j$ on the other side. So We have $a_{j-2} \cdot a_{2n-j}$ number of ways to pair the remaining people in a valid way.

Supplementary Explanation

Answer. $a_{j-2}a_{2n-j}$ for even $j \in \{2, 4, \ldots, 2n\}$; zero for odd j.

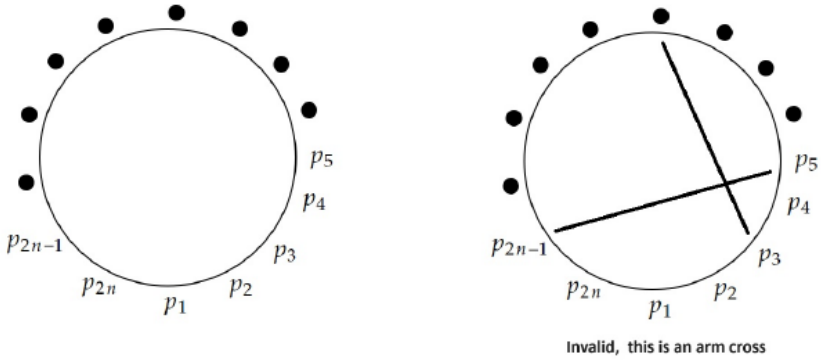
Derivation/proof. A chord from p_1 to p_j separates $p_2, \ldots, p_{j-1}$ from $p_{j+1}, \ldots, p_{2n}$. The groups have sizes $j-2$ and $2n-j$. If j is odd, both are odd and no complete internal pairing exists. For even j, there are a_{j-2} noncrossing pairings in the first group and a_{2n-j} in the second. Any pair of such choices combines with the fixed chord uniquely and introduces no crossings between regions. Conversely, every legal pairing containing the fixed chord restricts to those two choices. This bijection proves the product.

The count is $a_{j-2}a_{2n-j}$ for even j, and is zero for odd j. There is a bijection between completions and ordered pairs of valid side matchings; an empty side contributes its unique empty matching, counted by $a_0 = 1$.

C043

Basic Counting

2023 Fall Final | Question 9(d) | Source page 14 | Difficulty 3 | 3 marks
Q-2023F-FINAL-P09-d | Answer provenance: official

> Assume that there are $2n$ people sitting around a round table. We want to split them into n pairs to engage in handshakes so that no arms cross, given that nobody is allowed to shake hands with more than one person at once. Let a_{2n} denote the number of ways that we can pair them legally without any arm crossing. You can see an invalid handshake in the figure below. Please answer the following questions by referring to the figure:
>
> 
>
>
> Show the recursive relation of a_{2n}. You do not have to represent it in closed form. You can assume $a_0 = 1$.

Official Answer

Source: physical PDF page 14 (COMP2711_F2023_Final.pdf).

> (d) We just need to calculate the summation in which the first person is paired with $p_2, p_4, p_6, \cdots, p_{2n}$.

Supplementary Explanation

Answer. $a_0 = 1$, and $a_{2n} = \sum_{k=1}^{n} a_{2k-2}a_{2n-2k}$ for $n \geq 1$.

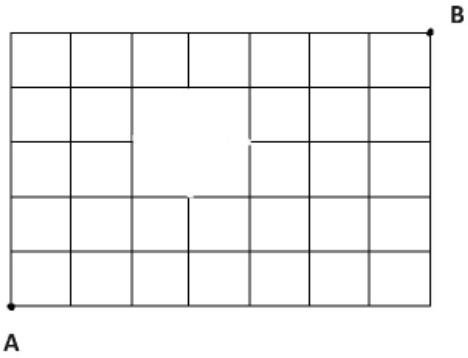
Derivation/proof. The feasible partners of p_1 are exactly p_{2k} with $1 \leq k \leq n$. For this fixed partner, the two blocks have $2k-2$ and $2n-2k$ people, so part (c) supplies the product $a_{2k-2}a_{2n-2k}$. Summing these disjoint and exhaustive classes gives the recurrence. The empty pairing counts as one, so $a_0 = 1$. Checks: $a_2 = 1$, $a_4 = 1+1 = 2$, and $a_6 = 2+1+2 = 5$. No closed form is needed.

The recurrence is $a_{2n} = \sum_{k=1}^{n} a_{2k-2}a_{2n-2k}$ with $a_0 = 1$. It gives $a_2 = 1$, $a_4 = 2$, and $a_6 = 2+1+2 = 5$.

C044

Basic Counting

2023 Fall Midterm | Question 4(a) | Source page 8 | Difficulty 3 | 4 marks
Q-2023F-MID-P04-a | Answer provenance: official

> Consider the following "incomplete" 5×7 grid. We are standing at position A and we want to go to position B. We are only allowed to go right or up. In how many ways can we go from A to B?
>
> 
>

Official Answer

Source: physical PDF page 9 (COMP2711_F2023_Midterm.pdf).

> (a) Basically the question is the same as asking how many ways to go from A to B without passing through C, which can be calculated as $\binom{12}{5} - \binom{6}{3} \cdot \binom{6}{2}$.
>
> page=9 (COMP2711_F2023_Midterm.pdf)

Supplementary Explanation

Answer. $\binom{12}{5} - \binom{6}{3}\binom{6}{2} = 792 - 300 = 492.$

Derivation/proof. In the full grid, every path has seven right steps and five up steps, giving $\binom{12}{5}$. The four absent incident segments all meet the missing vertex $C = (3, 3)$, so a monotone full-grid path is forbidden exactly when it visits C. The portion from A to C has three right and three up steps, counted by $\binom{6}{3}$. From C to B there are four right and two up steps, counted by $\binom{6}{2}$. The split at C is unique, so subtract their product to obtain 492.

The full-grid count is 792, while the two pieces contribute $20 \cdot 15 = 300$, giving $792 - 300 = 492$.

C045

Basic Counting

2023 Fall Midterm | Question 4(b) | Source page 8 | Difficulty 3 | 4 marks
Q-2023F-MID-P04-b | Answer provenance: official

> Consider the following 3×3 grid. You are sitting in your car on the bottom left cell and the aim is to reach the top right cell (shown by a star) as shown in Figure 1a. You want to color every cell in this 3×3 grid except the top right cell with colors either Blue(B) or Red(R) (i.e., you will color 8 cells). If your car is on a Red cell you move one cell to the right and if your car is standing on a Blue cell you move one cell up. How many different colorings exist such that you can reach to the top right cell? Note that you need to write down the combinatorial format and explain your solution.
>
> As an example, Figure 1b shows a valid coloring as you will reach to the top right cell as shown. However, Figure 1c shows an invalid coloring as you will not be able to reach to the top right cell.

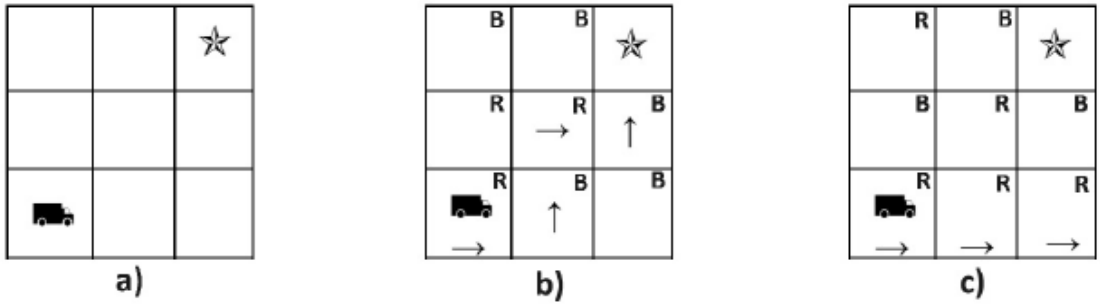


Figure 1: R/B denote the red/blue colors of the cells

Official Answer

Source: physical PDF page 9 (COMP2711_F2023_Midterm.pdf).

> (b) Notice that if you consider any path, the coloring of that path is unique. We have $\binom{4}{2}$ different paths from bottom left to top right with a unique coloring. There will be 4 cells left that can accept any color and it does not effect. So the final answer is $\binom{4}{2} \cdot 2^4$.

Supplementary Explanation

Answer. $\binom{4}{2}\, 2^4 = 96$.

Derivation/proof. There are $\binom{4}{2}$ monotone paths of four moves from the bottom-left cell to the top-right cell. On a chosen path, each of its four nonterminal cells has its color forced by the next move: R for right or B for up. Exactly four of the eight colored cells are not departure cells of this path, and each has two free colors. Multiply to get $6 \cdot 16 = 96$. A coloring determines a unique trajectory, so two different successful paths cannot count the same coloring.

There are $\binom{4}{2} = 6$ paths, with $2^4 = 16$ color extensions each. One coloring cannot belong to two distinct successful paths.

C046

Basic Counting

2021 Final Part 1 | Question 6 | Source page 3 | Difficulty 3 | 10 marks
Q-2021-FINAL1-P06 | Answer provenance: official

> A string is called E-String if the string consists of only digits in $\{0, 2, 4, 7, 8, 9\}$ and the sum of the digits is even. E.g. "079" is an E-String because $0 + 7 + 9 = 16$ is even. "708" is not an E-String because $7 + 0 + 8 = 15$ is odd. "107" is not an E-String because the digit 1 is not allowed. The empty string is also considered an E-string.
>
> How many E-Strings of length n are there? Give a closed-form solution. [Hint: use a recurrence.]

Official Answer

Source: physical PDF page 4 (2021-final-part-1.pdf).

> Let $E(n)$ denote the number of E-Strings of length n.
>
> **Base case:** $E(0) = 1$
>
> **Recursive case:** For $n > 0$, $E(n) = 2 \cdot (6^{n-1} - E(n-1)) + 4 \cdot E(n-1) = 2 \cdot 6^{n-1} + 2 \cdot E(n-1)$.
>
> $$
> \begin{aligned}
> > E(n) &= 2 \cdot 6^{n-1} + 2 \cdot E(n-1) \\
> &> = 2 \cdot 6^{n-1} + 2 \cdot (2 \cdot 6^{n-2} + 2 \cdot E(n-2)) \\
> &> = 2 \cdot 6^{n-1} + 2^2 \cdot 6^{n-2} + 2^2 \cdot E(n-2) \\
> &> \quad [redacted] \\
> &> = \vdots \sum_{i=1}^{n} 2^i \cdot 6^{n-i} + 2^n \cdot E(0) \\
> &> = \sum_{i=1}^{n} 2^n \cdot 3^{n-i} + 2^n \\
> &> = 2^n \sum_{i=0}^{n-1} 3^i + 2^n \\
> &> = 2^n \frac{3^n - 1}{3 - 1} + 2^n \\
> &> = 2^{n-1}(3^n - 1) + 2^n. >
> \end{aligned}
> $$

Supplementary Explanation

Answer. $E_n = (6^n + 2^n)/2$.

Derivation/proof. There are four even digits and two odd digits. With E0=1, append an even digit to an even-sum prefix (4E_(n−1) choices) or an odd digit to an odd-sum prefix (2(6^(n−1)−E_(n−1)) choices). Thus E_n=2E_(n−1)+2·6^(n−1). The proposed closed form equals 1 at n=0. Substitution gives 2[(6^(n−1)+2^(n−1))/2]+2·6^(n−1)=3·6^(n−1)+2^(n−1)=(6^n+2^n)/2. This proves it by recurrence induction; E1=4 also checks the alphabet count.

The even digits are $0, 2, 4, 8$ and the odd digits are $7, 9$. Explicitly,

$$E_n = 4E_{n-1} + 2(6^{n-1} - E_{n-1}) = 2E_{n-1} + 2 \cdot 6^{n-1}.$$

Unrolling gives

$$E_n = 2^n E_0 + \sum_{i=1}^{n} 2^i 6^{n-i} = 2^n + 2^n \sum_{j=0}^{n-1} 3^j = 2^n + 2^n(3^n - 1)/2 = (6^n + 2^n)/2.$$

The final answer contains no summation, as required by the paper.

As required, the final answer contains no summation.

C047

Basic Counting

2020 Final Part 1 | Question 4(b) | Source page 3 | Difficulty 3 | 4 marks
Q-2020-FINAL1-P04-b | Answer provenance: official

> How many integers $0 \leq n < 1000$ are there such that
>
> $n \equiv 1 \pmod 2$, $n \equiv 2 \pmod 3$, and $n \equiv 3 \pmod 5$?

Official Answer

Source: physical PDF page 3 (2020-final-part-1.pdf).

By Chinese remainder theorem, we know that $n \equiv 1 \pmod 2$, $n \equiv 2 \pmod 3$, and $n \equiv 3 \pmod 5$ is equivalent to $n = 23 + 30k$ for an integer k. Since $0 \leq n < 1000$, we have $0 \leq k \leq 32$. Therefore the answer is 33.

Supplementary Explanation

Answer. 33.

Derivation/proof. The last condition gives residues 3,8,13,18,23,28 modulo 30. Requiring oddness leaves 3,13,23; requiring remainder 2 modulo 3 leaves only 23. As 2,3,5 are pairwise coprime this is the unique CRT class: $n = 23 + 30k$. The range imposes $0 \leq k \leq 32$, giving 33 integers (last 983).

The allowed integer parameter runs from zero through thirty-two inclusive. Therefore its count is $32 - 0 + 1 = 33$, retaining both endpoints.

C048

Basic Counting

2020 Final Part 1 | Question 4(c) | Source page 3 | Difficulty 3 | 7 marks
Q-2020-FINAL1-P04-c | Answer provenance: official

> How many integers $0 \leq n < 1000$ are there such that
>
> n is divisible by 9 but it has no digit 9?

Official Answer

Source: physical PDF page 3 (2020-final-part-1.pdf).

Let $n = 100x_1 + 10x_2 + x_3$, i.e. n is represented by 3 digits $\overline{x_1x_2x_3}$. Then $x_i \in \{0, 1, \cdots, 8\}$ for $i = 0, 1, 2$, and $9 \mid 100x_1 + 10x_2 + x_3$, or equivalently $9 \mid x_1 + x_2 + x_3$. Since $0 \leq x_1 + x_2 + x_3 \leq 24$, we have $x_1 + x_2 + x_3 = 0, 9$ or 18. (1) $x_1 + x_2 + x_3 = 0$. There is only one solution: $x_1 = x_2 = x_3 = 0$. (2) $x_1 + x_2 + x_3 = 9$. By the combinations with repetition theorem, there are $\binom{9+3-1}{3-1} = 55$ non-negative solutions to this equation. However we need to exclude three solutions $(9, 0, 0)$, $(0, 9, 0)$ and $(0, 0, 9)$. Therefore the number of solutions is 52. (3) $x_1 + x_2 + x_3 = 18$. First note that there is no solution when $x_3 = 0$ or 1. Suppose $x_3 = i$ where $i \in \{2, 3, \cdots, 8\}$, the solutions to the equation $x_1 + x_2 = 18 - i$ are $(j, 18 - i - j)$ where $j \in \{10 - i, 11 - i, \cdots, 8\}$, i.e. there are $i - 1$ solutions to this equation. Therefore the number of solutions to $x_1 + x_2 + x_3 = 18$ is $\sum_{i=2}^{8}(i - 1) = 28$. Therefore we conclude that the answer is $1 + 52 + 28 = 81$. Alternative Solution: We also find the number of solutions to $9 \mid x_1 + x_2 + x_3$ where $x_i \in \{0, 1, \cdots, 8\}$ for $i = 1, 2, 3$. For any x_1 and x_2, $9 \mid x_1 + x_2 + x_3$ is equivalent to $x_3 \equiv 9 - x_1 - x_2 \pmod 9$, which has exactly one solution in $\{0, 1, \cdots, 8\}$. Therefore the number of solutions is $9 \times 9 = 81$.

Supplementary Explanation

Answer. 81.

Derivation/proof. Represent each integer uniquely as a three-digit string abc with leading zeros allowed. This does not introduce any digit 9. Each digit lies in {0,...,8}. Since $100 \equiv 10 \equiv 1 \pmod 9$, divisibility is $a + b + c \equiv 0$. For each of the 9^2 choices of a,b there is exactly one c in {0,...,8} satisfying this congruence. Therefore 81 strings, including 000, qualify.

After choosing the first two residues, the third must satisfy $x_3 \equiv -x_1 - x_2 \pmod 9$. Each residue has one representative in the allowed range, so there are $9^2 = 81$ choices in total.

C049

Basic Counting

2019 Final | Question 7 | Source page 9 | Difficulty 3 | 8 marks
Q-2019-FINAL-P07 | Answer provenance: official

> Let P be a set of 5 distinct points in the plane with integer coordinates. Draw a line segment between every two points of P (note that a total of 10 line segments will be drawn). Show that at least one of these line segments has its midpoint with integer coordinates.

Official Answer

Source: physical PDF page 9 (2019-final.pdf).

The midpoint of the segment whose endpoints are (a, b) and (c, d) is $((a + c)/2, (b + d)/2)$. We are concerned only with integer values of the original coordinates. Clearly the coordinates of these fractions will be integers as well if and only if a and c have the same parity (both odd or both even) and b and d have the same parity. Thus what matters in this problem is the parities of the coordinates. There are four possible pairs of parities: (odd, odd), (odd, even), (even, odd), and (even, even). Since we are given five points, the pigeonhole principle guarantees that at least two of them will have the same pair of parities. The midpoint of the segment joining these two points will therefore have integer coordinates.

Supplementary Explanation

Answer. At least one of the ten segments has an integer-coordinate midpoint.

Derivation/proof. Classify a point (a, b) by $(a \bmod 2, b \bmod 2)$. There are four classes: $(0, 0), (0, 1), (1, 0), (1, 1)$. Five points force two distinct points $(a, b), (c, d)$ in one class by pigeonhole principle. Then a+c and b+d are even, so the midpoint $((a + c)/2, (b + d)/2)$ has both coordinates integral. The required segment is present because every pair is joined.

Every pair of points is joined in the question, so the guaranteed same-parity pair supplies one of the ten drawn segments.

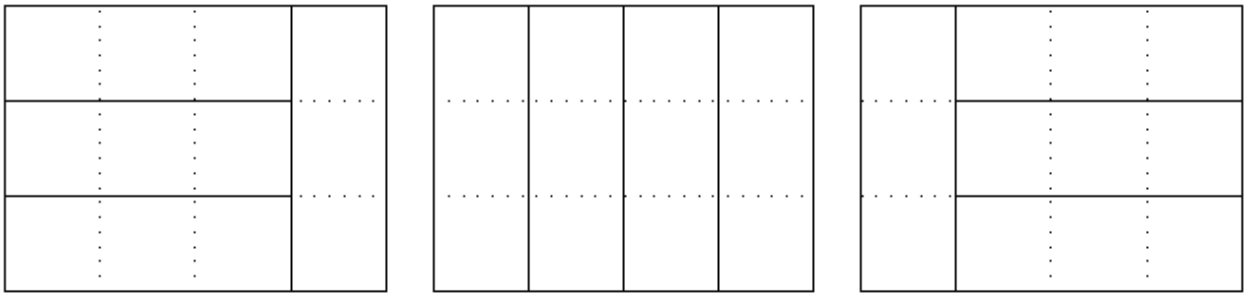
If $a \equiv b \pmod 2$ and $b \equiv c \pmod 2$, then adding the two even differences gives $a \equiv c \pmod 2$. This verifies transitivity of the parity relation.

C050

Basic Counting

2018 Final | Question 2(b) | Source page 4 | Difficulty 3
Q-2018-FINAL-P02-b | Answer provenance: official

Let E_n be the number of ways in which you can tile a 3-by-n board with 1-by-3 pieces. For example, when $n = 4$ the board is 3-by-4 and $E_4 = 3$. The 3 ways are shown as follows:



(b) Show that $E_n = E_{n-1} + E_{n-3}$. (Hint: You may use a counting argument.)

Official Answer

Source: physical PDF page 4 (2018-final.pdf).

Consider the top-rightmost tile of a 3-by-n board. If it is vertical, the remaining board is 3-by-$(n-1)$, so there are E_{n-1} ways to tile it. If it is horizontal, it must be followed by two more horizontal tiles below it, so there are E_{n-3} ways to tile it. Therefore $E_n = E_{n-1} + E_{n-3}$.

Supplementary Explanation

Answer. $E_n = E_{n-1} + E_{n-3}$ for $n \geq 4$ (also $n = 3$ with $E_0 = 1$).

Derivation/proof. Examine the tile covering the top-right cell. If vertical, it occupies the last column, leaving a 3-by-$(n-1)$ board. If horizontal, it covers the top cells of the last three columns. Each remaining cell in the rightmost column must then be covered horizontally: a vertical piece would overlap that top tile. Those two horizontal pieces also end at the boundary and fill the same three columns. This leaves a 3-by-$(n-3)$ board. Removing or adding the forced boundary tiles gives bijections to the two smaller tiling sets; the cases are disjoint and exhaustive.

With only positive-index initial values, use the recurrence for $n \geq 4$. Defining the empty-board count $E_0 = 1$ also makes it valid at $n = 3$.

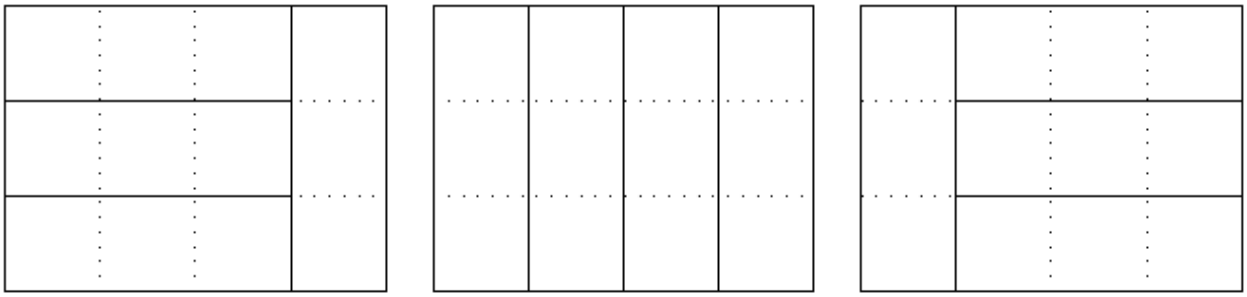
Separate tilings according to their right boundary. A vertical last tile leaves one fewer column, whereas horizontal tiles occupy the rightmost three columns. Removing and restoring these forced tiles is a one-to-one correspondence with the two smaller-board families.

C051

Basic Counting

2018 Final | Question 2(c) | Source page 4 | Difficulty 3
Q-2018-FINAL-P02-c | Answer provenance: official

Let E_n be the number of ways in which you can tile a 3-by-n board with 1-by-3 pieces. For example, when $n = 4$ the board is 3-by-4 and $E_4 = 3$. The 3 ways are shown as follows:



(c) Let F_n be the Fibonacci numbers. Prove that $E_n \leq F_n$ for all $n \geq 1$ using induction. You may use the result of (b).

Official Answer

Source: physical PDF page 4 (2018-final.pdf).

We prove this statement $P(n)$ using induction on three base cases. Observe that $E_1 = F_1$ ($P(1)$), $E_2 = F_2$ ($P(2)$), and $E_3 = F_3$ ($P(3)$). For the inductive step, suppose that $P(k-1)$, $P(k-2)$, and $P(k-3)$ are true. Then $E_n = E_{n-1} + E_{n-3} \leq F_{n-1} + F_{n-3} \leq F_{n-1} + F_{n-2} = F_n$. (For the last inequality, we use the fact that the Fibonacci sequence is non-decreasing.) So $P(k)$ is true, thus proving the statement $P(n)$ by induction.

Supplementary Explanation

Answer. $E_n \leq F_n$ for every $n \geq 1$, with $F_1 = F_2 = 1$.

Derivation/proof. The base values are $E_1 = 1 = F_1$, $E_2 = 1 = F_2$, $E_3 = 2 = F_3$. For $n \geq 4$, assume the inequality for all smaller positive indices. Then $E_n = E_{n-1} + E_{n-3} \leq F_{n-1} + F_{n-3} \leq F_{n-1} + F_{n-2} = F_n$. The middle inequality follows because the positive Fibonacci sequence is nondecreasing. Strong induction completes the proof.

The required initial matches are $E_1 = F_1 = 1$, $E_2 = F_2 = 1$, and $E_3 = F_3 = 2$. Since the recurrences also agree, strong induction proves equality for every later index.

C052

Basic Counting

2018 Final | Question 10(d) | Source page 14 | Difficulty 3
Q-2018-FINAL-P10-d | Answer provenance: official

> Given a $n \times p$ matrix A and a $p \times m$ matrix B, the following algorithm computes the $n \times m$ product matrix $C = A \cdot B$.
>
> Matrix-Mult(A,B)
>
> ```
> for i = 1 to n do
>     for j = 1 to m do
>         C[i,j] = 0
>         for k = 1 to p do
>             C[i,j] = C[i,j] + A[i,k] · B[k,j]
>         end for
>     end for
> end for
> return C
> ```
>
> d) Assume you are given three matrices M_1, M_2, M_3 where M_1 is 50×60, M_2 is 60×25, and M_3 is 25×8. You want to compute the product $M_1 \cdot M_2 \cdot M_3$. In which order will you do the pairwise matrix multiplications, if you want to minimize the total number of multiplications?

Official Answer

Source: physical PDF page 15 (2018-final.pdf).

There are two ways to do the multiplication. Either $M_1 \cdot M_2$ and then multiply the result with M_3 from the right, or $M_2 \cdot M_3$ and then multiply the result with M_1 from the left. For the first case, to multiply $M_1 \cdot M_2$ we need $50 \cdot 60 \cdot 25$ multiplications and the result is of size 50×25. Multiplying this with M_3 takes $50 \cdot 25 \cdot 8$ multiplications. Total multiplications is 85000. For the second case, to multiply $M_2 \cdot M_3$ we need $60 \cdot 25 \cdot 8$ multiplications and the result is of size 60×8. Multiplying this with M_1 takes $50 \cdot 60 \cdot 8$ multiplications. Total multiplications is 36000. Clearly, the second approach is much faster.

Supplementary Explanation

Answer. Compute M_2M_3 first, then $M_1(M_2M_3)$; cost 36000.

Derivation/proof. For $(M_1M_2)M_3$, the first product costs $50 \cdot 60 \cdot 25 = 75000$ and is 50×25; the second costs $50 \cdot 25 \cdot 8 = 10000$, totaling 85000. For $M_1(M_2M_3)$, the first costs $60 \cdot 25 \cdot 8 = 12000$ and is 60×8; the second costs $50 \cdot 60 \cdot 8 = 24000$, totaling 36000. These are the two parenthesizations preserving matrix order, so the second is minimal.

For three matrices in fixed order these are the only two binary parenthesizations.

Comparing the two legal parenthesizations gives $36000 < 85000$. Thus the product order with thirty-six thousand scalar multiplications is cheaper, even though both yield the same matrix.

C053

Basic Counting

2018 Midterm | Question 3(c) | Source page 5 | Difficulty 3 | 2 marks
Q-2018-MID-P03-c | Answer provenance: official

> Consider the set $S = \{1, 2, \ldots, 2n\}$, and subsets $P = \{p_1, p_2, \ldots, p_m\} \subseteq S$ such that they satisfy the following criterion:
>
> $\forall p_i \in P \forall p_j \in P (i \neq j \rightarrow p_i \nmid p_j)$
>
> In the above, $p_i \nmid p_j$ means p_j is not divisible by p_i.
>
> (c) Prove that there is no $P' \subseteq S$ satisfying the above criterion such that $|P'| > n$. (Hint: By contradiction, suppose $|P'| = n + 1$ satisfies the above criterion. Following (b), put each p_i in the pigeonhole c_i.)

Official Answer

Source: physical PDF page 5 (2018-midterm.pdf).

Let each hole be an odd number in S. The largest odd number in S is $2n - 1$, so the holes are $\{1, 3, 5, \ldots, 2n - 1\}$, so there are n holes, and the odd factor of each number in P' fits into one of these holes. Therefore, there are $n + 1$ numbers for n holes, and by the pigeonhole principle, there exists two different $p_i, p_j \in P'$ in the same hole, i.e. $c_i = c_j$. But we know that this is impossible from (b), giving a contradiction. (2 marks)

Supplementary Explanation

Answer. Every such subset has size at most n.

Derivation/proof. There are exactly n possible odd parts in S, namely $1, 3, \ldots, 2n - 1$. If P' had more than n elements, pigeonhole principle would place two distinct elements in the same odd-part class. Writing them as $c2^r, c2^s$ makes the smaller divide the larger, contradicting the criterion. Thus $|P'| \leq n$.

If there are more than n elements, select any $n + 1$ of them before applying pigeonhole principle.

C054

Basic Counting

2018 Midterm | Question 4(c) | Source page 6 | Difficulty 3
Q-2018-MID-P04-c | Answer provenance: official

> (c) Consider a lossless data compression algorithm, like zip or rar: compressed files can always be decompressed to obtain the original files. Prove that it is impossible for a lossless data compression algorithm to reduce the size of every file. (Hint: Use (a), and prove by contradiction on the set of all files of size n bits or less.)

Official Answer

Source: physical PDF page 6 (2018-midterm.pdf).

By contradiction, consider such a function f defined on $F_n \to T$, where F_n is the set of all files of size n bits or less. As f reduces the sizes of all files, T satisfies $T \subseteq F_{n-1}$, the size of all files of size $n-1$ bits or less. Clearly $F_{n-1} \subset F_n$, so $|T| < |F_n|$. However, since f is invertible, f must be injective. Therefore, using (a), we have $|T| \geq |F_{n-1}|$. This is a contradiction.

Supplementary Explanation

Answer. No lossless compressor can shorten every file.

Derivation/proof. Let F_n contain all bit strings of length at most n, including the empty string. Then $|F_n| = \sum_{i=0}^{n} 2^i = 2^{n+1} - 1$, whereas $|F_{n-1}| = 2^n - 1$. If every input became shorter, compression would map F_n into F_{n-1}. Lossless decoding implies injectivity: equal compressed outputs must decode to equal inputs. This contradicts pigeonhole principle because $|F_n| > |F_{n-1}|$.

More explicitly, $|T| \leq 2^n - 1 < 2^{n+1} - 1 = |F_n|$, while lossless decoding requires $|T| \geq |F_n|$.

A decoder must reconstruct every original string from its compressed representation. Thus two distinct inputs cannot share one output; injectivity and the pigeonhole principle supply the cardinality obstruction.

C055

Basic Counting

2023 Fall Midterm | Question 4(c) | Source page 8 | Difficulty 4 | 4 marks
Q-2023F-MID-P04-c | Answer provenance: official

> Imagine having a 4×4 grid. We want to write either $+1$ or -1 in its cells such that the sum of every row is equal to 0. Similarly we want the sum of every column to be 0 also. How many distinct "valid" tables exist?

Official Answer

Source: physical PDF page 9 (COMP2711_F2023_Midterm.pdf), physical PDF page 10 (COMP2711_F2023_Midterm.pdf).

> (c) The first row needs two $+1$ and two -1. So there are $\binom{4}{2}$ different ways to complete the first row. For the second row there are 3 general cases:
>
> Case 1: The columns of the second row have the exact same values as row one. For example, if row one is $\{+1, +1, -1, -1\}$ then the second row is also $\{+1, +1, -1, -1\}$. This case happens only in one valid permutation of the second row. In this case row 3 and row 4 have only a unique filling.
>
> Case 2: All columns of the second row have different values with row one. For example, if row one is $\{+1, +1, -1, -1\}$ then the second row is $\{-1, -1, +1, +1\}$. This case happens only in one valid permutation of the second row. In this case row 3 can accept any valid input which has $\binom{4}{2}$ different ways and the last row has a unique filling.
>
> Case 3: Half of the columns of the second row have different values with row one and half of them are the same. For example, the first row is $\{+1, +1, -1, -1\}$ and the second row is $\{-1, +1, +1, -1\}$. This case happens in four different permutations of the second row. In this case there are 2 ways to complete row 3 and row 4 (why?).
>
> So the total number of solutions will be $\binom{4}{2} \cdot (1 \cdot 1 + 1 \cdot 6 + 4 \cdot 2)$.

Supplementary Explanation

Answer. $\binom{4}{2}(1 \cdot 1 + 1 \cdot 6 + 4 \cdot 2) = 90$.

Derivation/proof. Choose the two +1 positions in row 1 in six ways. For a fixed row 1, row 2 can match it (one choice), complement it (one choice), or overlap in exactly one +1 position (four choices). If rows 1–2 match, rows 3–4 must both complement them: one completion. If they complement, each column needs one further +1, so row 3 has six choices and row 4 is forced. In the mixed case, one column already has two +1s, one has none, and two have one; row 3 must put -1 in the first, +1 in the second, and select one of the two remaining columns for its other +1. This yields two completions, with row 4 forced. Thus the total is $6(1 + 6 + 8) = 90$.

The second-row counts are one, one, and four, respectively. The third row controls the remaining freedom, and the fourth is forced. Thus $6(1 \cdot 1 + 1 \cdot 6 + 4 \cdot 2) = 90$.

C056

Basic Counting

2023 Fall Midterm | Question 4(d) | Source page 9 | Difficulty 4 | 4 marks
Q-2023F-MID-P04-d | Answer provenance: official

> Imagine having an equilateral triangle with each side being 3 cm. Prove that for all possible ways that you can place 10 points inside this equilateral triangle, there exist at least two points with their distance being less than 1 cm.

Official Answer

Source: physical PDF page 10 (COMP2711_F2023_Midterm.pdf).

> (d) We split the equilateral triangle of sides 3 into 9 equilateral triangles of side 1cm as follows:
>
> page=10 (COMP2711_F2023_Midterm.pdf)
>
> Now we want to put 10 points inside this triangle. Which means at least two points will be either inside one of the $1 \times 1 \times 1$ triangle cells (based on pigeonhole principle and that there are 10 points and 9 triangles) which means their distance is less than 1. Also it is possible for both points to be on the internal lines (not on the border lines though) but even in that case the results remains the same because the distance between any two points on the internal lines (and not on the border lines) are still less than 1 and also if one point is one the internal line and another point is completely inside the $1 \times 1 \times 1$ related triangle, their distance is still less than 1cm.

Supplementary Explanation

Answer. Some pair of the ten interior points has distance strictly less than 1 cm.

Derivation/proof. Subdivide the side-3 triangle into nine equilateral triangles of side 1 by drawing parallels through the trisection points. Assign each point on a shared edge to one adjacent cell, so every point belongs to exactly one of nine cells. Ten points force two into the same cell. A unit equilateral triangle has diameter 1, and equality is possible only between two distinct vertices. In this subdivision only one lattice vertex lies strictly inside the original triangle; every other small-triangle vertex is on the original boundary. Consequently two allowed interior points cannot be such a distinct vertex pair. Their distance is therefore strictly less than 1.

There are nine side-one triangles. Only one grid vertex lies strictly inside the original triangle, so the distance must be strictly less than one for the selected pair. If the outer boundary were allowed, the ten grid vertices would be a counterexample.

C057

Permutations and Combinations

2025 Fall Midterm | Question 4(a)(i) | Source page 12 | Difficulty 1 | 2 marks
Q-2025F-MID-P04-a-i | Answer provenance: official

> Answer the following questions. You can leave your answer in a combinatorial form without calculating the final answer. For example something like $5! \times 2^3 \times P(5,3) \times 2^7 \times \binom{20}{2}$ is totally fine. No explanations is needed. You are NOT allowed to use $\sum$.
>
> (a) [8 points, 2 pt each] Assume having 6 distinct teachers and 5 distinct students.
>
> (i) In how many ways they can stand in a line?

Official Answer

Source: physical PDF page 13 (COMP2711_F2025_Midterm.pdf).

> (a) (i) $11!$

Supplementary Explanation

Answer. $11!$.

Derivation/proof. There are $6 + 5 = 11$ distinct people, with a fixed direction along the line. Choose the first person in eleven ways, the second in ten ways, and continue to the last position. The product rule gives $11 \cdot 10 \cdots 1 = 11!$. Every permutation describes exactly one line and vice versa. No occupational grouping constraint is imposed.

C058

Permutations and Combinations

2022 Final Part 1 | Question 3(b) | Source page 2 | Difficulty 1 | 2 marks
Q-2022-FINAL1-P03-b | Answer provenance: official

> Consider a web system for online shopping that will generate an ID for each order. The order ID is a sequence of 7 characters where each character could be either one of the 26 upper case English letters (A to Z) or one of the 10 digits (0 to 9).
>
> How many possible order IDs are there if no two characters are the same? (E.g. A034NXD is valid, A034NAD is invalid)

Official Answer

Source: physical PDF page 2 (2022-final-part-1.pdf).

> (b) $P(36,7)$ or $7!\binom{36}{7}$

Supplementary Explanation

Answer. $P(36,7) = 36!/29!$.

Derivation/proof. The first position has 36 choices, then each used character becomes unavailable, leaving 35,34,33,32,31,30 choices for later positions. Multiplication gives 36!/29!, equivalent to C(36,7)·7!. The order matters: two IDs containing the same set of seven characters in different orders are distinct.

The explicit product is $36 \cdot 35 \cdot 34 \cdot 33 \cdot 32 \cdot 31 \cdot 30 = 36!/29!$, also $\binom{36}{7}7!$.

C059

Permutations and Combinations

2022 Final Part 2 | Question 2(a) | Source page 2 | Difficulty 1
Q-2022-FINAL2-P02-a-23 | Answer provenance: official

> Consider the following 5-door version of the Monty Hall problem. There are 5 doors, behind one of which (randomly with equal probability) there is a car (which you want), and behind the rest of which there are goats (which you don't want). You choose a door. Monty then opens 2 goat doors. Assume that Monty knows which door has the car, and will choose with equal probabilities from all his choices of which goat doors to open. Without loss of generality, assume your first guess is door 1. Let W be the random variable denoting the door with the car and M the random variable denoting the two doors that Monty opens. Note that M is a random variable whose value is a subset of $\{1, 2, 3, 4, 5\}$.
>
> What is $P(M = \{2, 3\} \mid W = 1)$?

Official Answer

Source: physical PDF page 2 (2022-final-part-2.pdf).

> (a) $P(M = \{2, 3\} \mid W = 1) = P(M = \{3, 5\} \mid W = 1) = 1/\binom{4}{2} = 1/6$.

Supplementary Explanation

Answer. $1/6$.

Derivation/proof. Given W=1, all four other doors 2,3,4,5 contain goats and remain eligible for Monty's opening. He selects an unordered two-door subset uniformly; there are C(4,2)=6 such pairs. {2,3} is one pair, so its conditional probability is 1/6. The two openings are not ordered outcomes.

There are $\binom{4}{2} = 6$ pairs and $P(M = \{2, 3\} \mid W = 1) = 1/6$. The eligible goat-door set is the same under either interpretation of whether the initially chosen door may be opened, because that door contains the car.

C060

Permutations and Combinations

2022 Final Part 2 | Question 2(a) | Source page 2 | Difficulty 1
Q-2022-FINAL2-P02-a-35 | Answer provenance: official

> Consider the following 5-door version of the Monty Hall problem. There are 5 doors, behind one of which (randomly with equal probability) there is a car (which you want), and behind the rest of which there are goats (which you don't want). You choose a door. Monty then opens 2 goat doors. Assume that Monty knows which door has the car, and will choose with equal probabilities from all his choices of which goat doors to open. Without loss of generality, assume your first guess is door 1. Let W be the random variable denoting the door with the car and M the random variable denoting the two doors that Monty opens. Note that M is a random variable whose value is a subset of $\{1, 2, 3, 4, 5\}$.
>
> What is $P(M = \{3, 5\} \mid W = 1)$?

Official Answer

Source: physical PDF page 2 (2022-final-part-2.pdf).

> (a) $P(M = \{2, 3\} \mid W = 1) = P(M = \{3, 5\} \mid W = 1) = 1/\binom{4}{2} = 1/6.$

Supplementary Explanation

Answer. $1/6$.

Derivation/proof. With the car fixed behind door 1, Monty can choose any pair from goat doors 2,3,4,5. Uniformity assigns equal weight1/C(4,2) to each of the six pairs, including {3,5}. This is a fresh pair selection event under the same condition, not a sequential probability after already opening {2,3}.

The pairs are $\{2, 3\}, \{2, 4\}, \{2, 5\}, \{3, 4\}, \{3, 5\}, \{4, 5\}$. The target occupies one of six pairs. Both interpretations of the initially chosen door give the same set when the car is at one.

C061

Permutations and Combinations

2020 Final Part 2 | Question 6(b) | Source page 4 | Difficulty 1 | 1 mark
Q-2020-FINAL2-P06-b | Answer provenance: official

> Consider the permutations of the 10 digits (0–9). We denote a permutation as A, whose elements are $A[0], A[1], \ldots, A[9]$. Calculate the number of permutations that satisfy each of the following conditions:
>
> (a) There is exactly one i such that $A[i] = i$. (b) There are exactly nine i's such that $A[i] = i$. (c) $A[0] = 1, A[1] = 0$. (d) $A[4] = 4, A[5] = 5$.
>
> You can express your answers using binomial coefficients, factorials, and D_n (derangement number). However, you should not have summation $\sum$ in your final answers.
>
> There are exactly nine i's such that $A[i] = i$.

Official Answer

Source: physical PDF page 5 (2020-final-part-2.pdf).

If 9 digits appear in their corresponding index positions, the remaining one digit must also appear in its index position, so the answer is 0.

Supplementary Explanation

Answer. 0.

Derivation/proof. If nine positions contain their own digits, exactly one digit and one position remain unused. Because all other digits are already at their matching positions, the remaining digit is the label of that remaining position. It too must be fixed, producing ten fixed points rather than exactly nine. Hence no such permutation exists.

If j is the sole unused position, all other digits already match their positions. At the empty position j the remaining digit must be j, producing ten rather than nine fixed points. Hence the answer is 0.

C062

Permutations and Combinations

2020 Final Part 2 | Question 6(c) | Source page 4 | Difficulty 1 | 1 mark
Q-2020-FINAL2-P06-c | Answer provenance: official

> Consider the permutations of the 10 digits (0–9). We denote a permutation as A, whose elements are $A[0], A[1], \ldots, A[9]$. Calculate the number of permutations that satisfy each of the following conditions:
>
> (a) There is exactly one i such that $A[i] = i$. (b) There are exactly nine i's such that $A[i] = i$. (c) $A[0] = 1, A[1] = 0$. (d) $A[4] = 4, A[5] = 5$.
>
> You can express your answers using binomial coefficients, factorials, and D_n (derangement number). However, you should not have summation $\sum$ in your final answers.
>
> $A[0] = 1, A[1] = 0$.

Official Answer

Source: physical PDF page 5 (2020-final-part-2.pdf).

Fix the first two digits, and the remaining 8 digits are randomly permuted, so the answer is 8!.

Supplementary Explanation

Answer. 8!.

Derivation/proof. The conditions place digit 1 at position 0 and digit 0 at position 1. The other eight positions receive digits 2 through 9 bijectively with no remaining restriction, giving 8! assignments. They are allowed to be fixed or not; a derangement restriction is not present in this part.

After fixing the swap of zero and one, the remaining digits are $2, 3, \ldots, 9$. These may occupy their original positions, so do not use D_8; all 8! permutations are allowed.

C063

Permutations and Combinations

2020 Final Part 2 | Question 6(d) | Source page 4 | Difficulty 1 | 1 mark
Q-2020-FINAL2-P06-d | Answer provenance: official

> Consider the permutations of the 10 digits (0–9). We denote a permutation as A, whose elements are $A[0], A[1], \ldots, A[9]$. Calculate the number of permutations that satisfy each of the following conditions:
>
> (a) There is exactly one i such that $A[i] = i$. (b) There are exactly nine i's such that $A[i] = i$. (c) $A[0] = 1, A[1] = 0$. (d) $A[4] = 4, A[5] = 5$.
>
> You can express your answers using binomial coefficients, factorials, and D_n (derangement number). However, you should not have summation $\sum$ in your final answers.
>
> $A[4] = 4, A[5] = 5$.

Official Answer

Source: physical PDF page 5 (2020-final-part-2.pdf).

Fix the middle two digits, and the remaining 8 digits are randomly permuted, so the answer is 8!.

Supplementary Explanation

Answer. 8!.

Derivation/proof. Positions 4 and 5 are already occupied by digits 4 and 5. Delete those two positions and digits; any bijection between the eight remaining positions and eight remaining digits extends uniquely to a valid permutation. There are 8! such bijections. Other fixed points are permitted because the condition does not say exactly two.

Every bijection of the eight remaining digits to the eight remaining positions has a unique extension after placing four and five in their prescribed positions. This establishes the count 8! without overcounting.

C064

Permutations and Combinations

2019 Final | Question 10(a) | Source page 12 | Difficulty 1 | 2 marks
Q-2019-FINAL-P10-a | Answer provenance: official

> Consider the equation $x_1 + x_2 + x_3 = 20$ where x_1, x_2, and x_3 are nonnegative integers.
>
> For this question, give exact numbers as the solution.
>
> (a) How many solutions are there?

Official Answer

Source: physical PDF page 12 (2019-final.pdf).

$C(3 + 20 - 1, 3 - 1) = C(22, 2) = 231.$

Supplementary Explanation

Answer. 231.

Derivation/proof. By stars and bars, solutions of $x_1 + x_2 + x_3 = 20$ correspond to twenty stars and two separators. Choose the separator positions among 22: $\binom{22}{2} = 22 \cdot 21/2 = 231$. Adjacent separators and endpoint separators account for zero coordinates.

The bars may be adjacent, which represents a zero occupancy between them. This is why ordinary stars and bars counts the nonnegative solutions including empty boxes.

C065

Permutations and Combinations

2018 Midterm | Question 7(a) | Source page 9 | Difficulty 1
Q-2018-MID-P07-a | Answer provenance: official

> A committee of 5 members needs to be formed in order to represent the employees in an office. There are 20 employees, 11 men and 9 women.
>
> Briefly justify your answers.
>
> (a) How many ways are there to form the committee if there are no restrictions in the gender of the members?

Official Answer

Source: physical PDF page 9 (2018-midterm.pdf).

This is simply $C(20, 5) = 15504$.

Supplementary Explanation

Answer. $\binom{20}{5} = 15504$.

Derivation/proof. Choose five of the twenty employees without ordering them. There are $20 \cdot 19 \cdot 18 \cdot 17 \cdot 16$ ordered selections, and each committee appears in 5! orders. Dividing by 5! gives $\binom{20}{5} = 15504$.

C066

Permutations and Combinations

2017 Final | Question 3(b) | Source page 5 | Difficulty 1
Q-2017-FINAL-P03-b | Answer provenance: official

> For each of the following parts, find the "best" big-O notation to describe the complexity of the algorithm. Choose your answers from the following:
>
> $O(1), O(\log n), O(n), O(n \log n), O(n^2), O(n^3), \ldots, O(2^n), O(n!)$
>
> Assume that the amount of time to print any string is a constant, i.e. $O(1)$.
>
> (b) An algorithm that prints all subsets of size two of the set $\{1, 2, 3, \ldots, n\}$.

Official Answer

Source: physical PDF page 5 (2017-final.pdf).

n^2.

Supplementary Explanation

Answer. $O(n^2)$ (in fact $\Theta(n^2)$).

Derivation/proof. Each output is an unordered pair $\{i, j\}$ with $1 \le i < j \le n$. There are $\binom{n}{2} = n(n-1)/2$ such pairs. A nested enumeration over $i < j$ prints each once, with constant cost per output, so the total is $\Theta(n^2)$.

C067

Permutations and Combinations

2017 Midterm | Question 6(a) | Source page 8 | Difficulty 1
Q-2017-MID-P06-a | Answer provenance: official

> Five couples go to the movies together and sit in a row of ten seats. In how many ways can the 10 people be arranged if
>
> (a) They may sit in any order.

Official Answer

Source: physical PDF page 8 (2017-midterm.pdf).

10!.

Supplementary Explanation

Answer. 10!.

Derivation/proof. There are ten choices for the first seat, nine for the second, and successively one for the last. The product $10 \cdot 9 \cdots 1 = 10!$ counts each seating once because its ordered occupants identify it uniquely.

C068

Permutations and Combinations

2017 Midterm | Question 9(a) | Source page 11 | Difficulty 1
Q-2017-MID-P09-a | Answer provenance: official

> Six women and nine men are on the faculty of a department. The department needs to form a committee of five members.
>
> (a) How many committees are possible if the committee must have two women and three men?

Official Answer

Source: physical PDF page 11 (2017-midterm.pdf).

$\binom{6}{2}\binom{9}{3}$.

Supplementary Explanation

Answer. $\binom{6}{2}\binom{9}{3}$.

Derivation/proof. Choose the two women from the six available in $\binom{6}{2}$ ways and the three men from the nine in $\binom{9}{3}$ ways. Each pair of choices gives a unique five-person committee, so multiply; no ordering of members is recorded.

Numerically, $\binom{6}{2}\binom{9}{3} = 15 \cdot 84 = 1260$.

The committee is selected without ordering: changing the order in which the same five people are selected produces the same committee, which is why the count uses a binomial coefficient.

C069

Permutations and Combinations

2016 Midterm | Question 1(b) | Source page 3 | Difficulty 1 | 2 marks
Q-2016-MID-P01-b | Answer provenance: official

> Consider a web system for online shopping that generates an order id for each business. The order id is a sequence of 7 characters where each character could be either one of the 26 Upper Case English Alphabets (A to Z) or one of the 10 digits (0 to 9).
>
> (b) How many possible order id's are there if no two characters are the same? (E.g., A034NXD is valid and A034NAD is invalid)

Official Answer

Source: PDF physical page 3 (2016-midterm.pdf).

(b) $36^{\underline{7}}$ or $7!\binom{36}{7}$

Supplementary Explanation

Answer. $36 \cdot 35 \cdot 34 \cdot 33 \cdot 32 \cdot 31 \cdot 30 = 7!\binom{36}{7}$.

Derivation/proof. After i distinct characters have been used, the next position has $36 - i$ options. Multiplying for $i = 0, \ldots, 6$ gives $36!/29!$, also equal to choosing seven characters and permuting them.

The sequence of available counts is $36, 35, 34, 33, 32, 31, 30$. The repeated A in the source example violates the requirement.

Choosing the seven distinct characters first and ordering them gives $\binom{36}{7}7!$.

C070

Permutations and Combinations

2015 Midterm | Question 7(a) | Source page 12 | Difficulty 1 | 3 marks
Q-2015-MID-P07-a | Answer provenance: official

> You are going to distribute kn cash coupons to kn members in your company such that each member will get one cash coupon, where $k, n \geq 1$. The values of the kn cash coupons are distinct. Suppose there are n ranks of seniority in the company, and there are exactly k members at each rank. If a more senior member receives a cash coupon worth less than the cash coupon given to a more junior member, you will receive a complaint.
>
> (a) How many ways are there to distribute the kn cash coupons if you don't care about receiving complaints?
>
> Briefly explain how your answers are derived.

Official Answer

Source: PDF physical page 12 (2015-midterm.pdf).

(a) $(kn)!$, the number of permutation of the kn cash coupons.

Supplementary Explanation

Answer. $(kn)!$.

Derivation/proof. Order the members once. The first can receive any of kn coupons, the next any of $kn-1$, down to one choice for the last. Multiplying gives $(kn)!$.

The product of successive choices is $(kn)(kn-1)\cdots 1 = (kn)!$.

Every coupon is distinct and each member receives exactly one, so an allocation is a bijection.

C071

Permutations and Combinations

2014 Midterm | Question 1(a) | Source page 3 | Difficulty 1
Q-2014-MID-P01-a | Answer provenance: official

> A by-election is being held to fill 5 seats in a council that have recently become vacant. A total of 9 candidates have registered to run in the election. The candidates are divided into 3 groups based on their political affiliations with: 4 in Group A, 3 in Group B, and 2 in Group C.
>
> (a) In how many ways can a voter choose 5 candidates?
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 3 (2014-midterm.pdf).

(a) $\binom{9}{5} = 126$.

Supplementary Explanation

Answer. $\binom{9}{5} = 126$.

Derivation/proof. The ballot is an unordered selection of five of the nine distinct candidates, so it has $\binom{9}{5}$ possibilities.

$\binom{9}{5} = 9!/(5!4!) = 126$. There is no additional factor $5!$.

C072

Permutations and Combinations

2014 Midterm | Question 2(a) | Source page 4 | Difficulty 1
Q-2014-MID-P02-a | Answer provenance: official

> There are n families joining a party. Each family consists of 3 people, a mother, a father and a child. In the party, there are $3n$ seats placed in a line.
>
> (a) Suppose everyone is allowed to sit anywhere. How many different seating arrangements are there?
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 4 (2014-midterm.pdf).

(a) $(3n)!$

Supplementary Explanation

Answer. $(3n)!$.

Derivation/proof. All $3n$ people are distinct and every permutation of them in the $3n$ seats is allowed.

The sequential choices are $3n, 3n-1, \ldots, 1$, whose product is $(3n)!$.

C073

Permutations and Combinations

2014 Midterm | Question 4(a) | Source page 6 | Difficulty 1
Q-2014-MID-P04-a | Answer provenance: official

> An anagram is a distinct ordering of the letters. For example, the word “eat” has six anagrams “eat”, “eta”, “ate”, “aet”, “tea” and “tae”, while the word “too” has only three anagrams “too”, “oto” and “oot”.
>
> (a) How many anagrams does the word “java” have?
>
> Explain your answers.

Official Answer

Source: PDF physical page 6 (2014-midterm.pdf).

(a) This is the same as the number of ways to label 4 positions according to the following rules: Label 2 positions with 'a', 1 positions with 'j', and 1 position with 'v'. So, the answer is:

$$\frac{4!}{2!1!1!} = 12.$$

Supplementary Explanation

Answer. $4!/2! = 12$.

Derivation/proof. There are four positions with two indistinguishable a's. Divide the 4! permutations of labelled letters by the 2! swaps that do not change the word.

C074

Permutations and Combinations

2013 Midterm | Question 1(a) | Source page 3 | Difficulty 1
Q-2013-MID-P01-a | Answer provenance: official

> Twelve Easter eggs are numbered $1, 2, \ldots, 12$. Eggs 1 to 4 are colored red, eggs 5 to 8 are colored green, and eggs 9 to 12 are colored purple. A child is allowed to choose 5 of the eggs.
>
> (a) In how many different ways can the child choose 5 eggs?
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 3 (2013-midterm.pdf).

(a) There are totally 12 eggs. The child chooses 5 of them. The number of different ways he can do that is:

$$\binom{12}{5}.$$

Supplementary Explanation

Answer. $\binom{12}{5} = 792$.

Derivation/proof. A selection is an unordered 5-subset of the twelve labelled eggs, so the binomial rule gives $\binom{12}{5} = 792$.

$\binom{12}{5} = 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8/(5 \cdot 4 \cdot 3 \cdot 2 \cdot 1) = 792$. Dividing by 5! removes the different orderings of the same selected set.

C075

Permutations and Combinations

2013 Midterm | Question 4(a) | Source page 6 | Difficulty 1
Q-2013-MID-P04-a | Answer provenance: official

> There are 8 boys and 8 girls to pair up in a dance party.
>
> (a) In how many different ways can the boys be paired up with the girls?
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 6 (2013-midterm.pdf).

(a) Let us number of the boys from 1 to 8. Boy number 1 can be paired up with any of the 8 girls; boy number 2 can be paired up any of the remaining 7 girls; and so on. So the answer is: 8!.

Supplementary Explanation

Answer. 8!.

Derivation/proof. Order the girls assigned to the eight labelled boys: 8 choices for the first boy, then 7, down to 1, giving 8!.

Each choice sequence determines exactly one complete pairing, giving $8 \cdot 7 \cdots 1 = 8!$.

C076

Permutations and Combinations

2025 Fall Final | Question 3(a) | Source page 7 | Difficulty 2 | 3 marks
Q-2025F-FINAL-P03-a | Answer provenance: official

> We have 14 volleyball players, all of different heights. All 14! permutations of rankings (tallest = rank 1 to shortest = rank 14) are equally likely. Answer the following questions and briefly explain. You do not need to calculate the final numerical answer.
>
> (a) [3 points] Given A is taller than B, what is the probability that A is tallest?

Official Answer

Source: physical PDF page 7 (COMP2711_F2025_Final.pdf).

(a) Given $A > B$, permutations satisfying this: $14!/2$. If A is tallest, arrange others: 13! permutations.

$$\Pr(A \text{ tallest} \mid A > B) = \frac{13!}{14!/2} = \frac{2}{14} = \frac{1}{7}.$$

Supplementary Explanation

Answer. $1/7$.

Derivation/proof. All 14! rankings are equally likely. Swapping A and B gives a bijection between rankings with $A > B$ and those with $B > A$, so the conditional sample space contains $14!/2$ rankings. If A is first, $A > B$ holds automatically and the other 13 players have 13! arrangements. Thus the conditional probability is $13!/(14!/2) = 2/14 = 1/7$. The denominator counts rankings satisfying the condition, not all rankings.

C077

Permutations and Combinations

2025 Fall Midterm | Question 4(a)(ii) | Source page 12 | Difficulty 2 | 2 marks
Q-2025F-MID-P04-a-ii | Answer provenance: official

> Answer the following questions. You can leave your answer in a combinatorial form without calculating the final answer. For example something like $5! \times 2^3 \times P(5,3) \times 2^7 \times \binom{20}{2}$ is totally fine. No explanations is needed. You are NOT allowed to use $\sum$.
>
> (a) [8 points, 2 pt each] Assume having 6 distinct teachers and 5 distinct students.
>
> (ii) In how many ways, all teachers are adjacent to each other?

Official Answer

Source: physical PDF page 13 (COMP2711_F2025_Midterm.pdf).

> (ii) $6! \times 6!$

Supplementary Explanation

Answer. $6! \times 6!$.

Derivation/proof. Compress the six teachers into one consecutive block. Together with the five distinct students, there are six distinct objects, arranged in 6! ways. Independently arrange the teachers within their block in 6! ways, giving $6! \cdot 6!$. Every valid line uniquely compresses to this outer order and inner order.

C078

Permutations and Combinations

2025 Fall Midterm | Question 4(a)(iii) | Source page 12 | Difficulty 2 | 2 marks
Q-2025F-MID-P04-a-iii | Answer provenance: official

> Answer the following questions. You can leave your answer in a combinatorial form without calculating the final answer. For example something like $5! \times 2^3 \times P(5,3) \times 2^7 \times \binom{20}{2}$ is totally fine. No explanations is needed. You are NOT allowed to use $\sum$.
>
> (a) [8 points, 2 pt each] Assume having 6 distinct teachers and 5 distinct students.
>
> (iii) In how many ways, all teachers are adjacent and also all students are adjacent to each other?

Official Answer

Source: physical PDF page 13 (COMP2711_F2025_Midterm.pdf).

> (iii) $2! \times 5! \times 6!$

Supplementary Explanation

Answer. $2! \times 5! \times 6!$.

Derivation/proof. Compress the teachers and students into two blocks. The teacher block may come first or second, giving 2! outer orders. There are 6! teacher orders and 5! student orders inside, so the total is $2! \cdot 5! \cdot 6!$. Expansion recovers exactly one line, and every valid line has this decomposition.

C079

Permutations and Combinations

2025 Fall Midterm | Question 4(a)(iv) | Source page 12 | Difficulty 2 | 2 marks
Q-2025F-MID-P04-a-iv | Answer provenance: official

> Answer the following questions. You can leave your answer in a combinatorial form without calculating the final answer. For example something like $5! \times 2^3 \times P(5,3) \times 2^7 \times \binom{20}{2}$ is totally fine. No explanations is needed. You are NOT allowed to use $\sum$.
>
> (a) [8 points, 2 pt each] Assume having 6 distinct teachers and 5 distinct students.
>
> (iv) In how many ways, no two teacher stand next to each other.

Official Answer

Source: physical PDF page 13 (COMP2711_F2025_Midterm.pdf).

> (iv) $6! \times 5!$

Supplementary Explanation

Answer. $6! \times 5!$.

Derivation/proof. First arrange the students in 5! ways. There are six gaps, including both ends. At most one teacher may occupy each gap, and six teachers must fill all six gaps. Assigning the distinct teachers takes 6! ways, giving $5! \cdot 6!$. The occupation pattern is necessarily TSTSTSTSTST.

C080

Permutations and Combinations

2025 Fall Midterm | Question 4(c)(i) | Source page 13 | Difficulty 2 | 2 marks
Q-2025F-MID-P04-c-i | Answer provenance: official

> (c) [4 point, 2 pts each] Imagine there are two sets $A = \{a_1, a_2, \cdots, a_7\}$ and $B = \{b_1, b_2, \cdots, b_{10}\}$. Answer the following questions:
>
> (i) In how many ways we can have 7 unordered pairs (e.g., (a_1, b_1) is the same as (b_1, a_1)) such that each pair contains an element from A and an element from B while the elements can not be reused?

Official Answer

Source: physical PDF page 13 (COMP2711_F2025_Midterm.pdf).

> (c) (i) $P(10, 7)$

Supplementary Explanation

Answer. $10!/3!$.

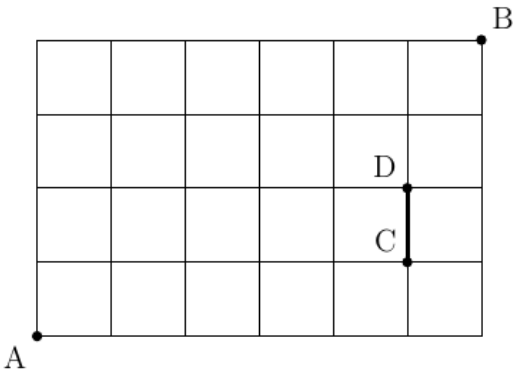
Derivation/proof. Seven pairs use all seven A elements. In fixed label order, choose the partner for a_1 in ten ways, for a_2 in nine ways, and continue through four choices for a_7. The product is $10 \cdot 9 \cdots 4 = 10!/3!$. This label order only encodes the set of pairs and does not introduce an ordering of the pairs themselves. Each pair has a distinguishable A member and B member, so there is no need to divide by 2^7 or 7!.

C081

Permutations and Combinations

2024 Fall Midterm | Question 4(a)(i) | Source page 8 | Difficulty 2 | 2 marks
Q-2024F-MID-P04-a-i | Answer provenance: official

In the following picture, starting from A we wish to reach B, by taking edges Up or Right. How many different paths (two paths are different if they differ in at least one edge) exist from A to B such that:



(i) There are no restrictions. No explanation is needed. Hint: Any path from A to B is a combination of Up and Right moves for a total of 10.

Official Answer

Source: physical PDF page 8 (COMP2711_F2024_Midterm.pdf).

> Answer: (a) (i) $\binom{10}{4}$ Alternative explanation: We have to select with repetition 4 Up edges from the 7 columns: $\binom{7+4-1}{4} = \binom{10}{4}$.

Supplementary Explanation

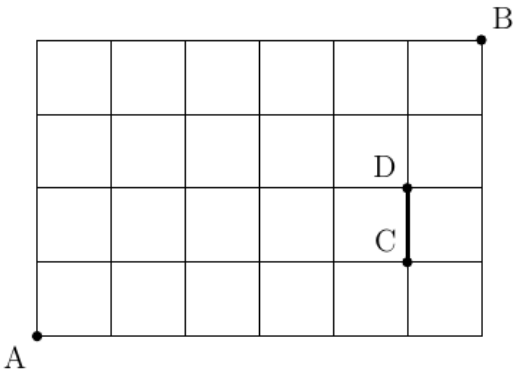
Answer. $\binom{10}{4} = 210$.

Derivation/proof. The source grid is six columns wide and four rows high. Thus every path from A to B has six Right moves and four Up moves. Choose the four Up positions in a sequence of ten moves; the others are Right, giving $\binom{10}{4} = 210$. Move sequences correspond to paths edge by edge, and distinct sequences differ in at least one edge.

C082

Permutations and Combinations

2024 Fall Midterm | Question 4(a)(ii) | Source page 8 | Difficulty 2 | 2 marks
Q-2024F-MID-P04-a-ii | Answer provenance: official

> In the following picture, starting from A we wish to reach B, by taking edges Up or Right. How many different paths (two paths are different if they differ in at least one edge) exist from A to B such that:
>
> 
>
>
> (ii) You have to pass from CD edge. No explanation is needed.

Official Answer

Source: physical PDF page 9 (COMP2711_F2024_Midterm.pdf).

> (ii) $\binom{6}{1} \cdot \binom{3}{1}$

Supplementary Explanation

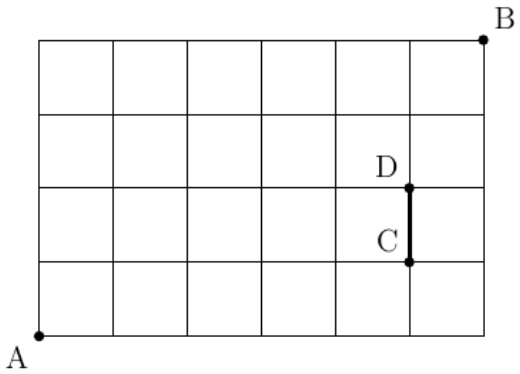
Answer. $\binom{6}{1}\binom{3}{1} = 18$.

Derivation/proof. Taking $A = (0, 0)$, the diagram gives $C = (5, 1)$, $D = (5, 2)$ and $B = (6, 4)$. A path through CD splits uniquely into A to C, the fixed upward edge CD, and D to B. The first segment has five right moves and one up, giving $\binom{6}{1} = 6$ choices. The last has one right and two up, giving $\binom{3}{1} = 3$ choices. The edge has one choice, so multiplication gives $6 \cdot 1 \cdot 3 = 18$.

C083

Permutations and Combinations

2024 Fall Midterm | Question 4(a)(iii) | Source page 8 | Difficulty 2 | 2 marks
Q-2024F-MID-P04-a-iii | Answer provenance: official

> In the following picture, starting from A we wish to reach B, by taking edges Up or Right. How many different paths (two paths are different if they differ in at least one edge) exist from A to B such that:
>
> 
>
>
> (iii) You can not pass from CD edge. No explanation is needed.

Official Answer

Source: physical PDF page 9 (COMP2711_F2024_Midterm.pdf).

> (iii) $\binom{10}{4} - \binom{6}{1} \cdot \binom{3}{1}$

Supplementary Explanation

Answer. $\binom{10}{4} - \binom{6}{1}\binom{3}{1} = 192$.

Derivation/proof. All paths use six right and four up moves, so there are $\binom{10}{4} = 210$. Paths using CD take five right and one up move to C, then CD, then one right and two up moves to B, giving $\binom{6}{1}\binom{3}{1} = 18$. Using and avoiding the edge are disjoint exhaustive cases. Therefore the desired count is $210 - 18 = 192$. The forbidden object is the edge, not every path visiting C or D.

C084

Permutations and Combinations

2024 Fall Midterm | Question 6(a) | Source page 11 | Difficulty 2 | 4 marks
Q-2024F-MID-P06-a | Answer provenance: official

> Assume that in any decimal number the leftmost digit cannot be 0. How many 6-digit numbers exist such that their digits from left to right:
>
> (a) Are strictly decreasing (e.g., 864320). Briefly explain why. Hint: Any six different digits give you a unique number with strictly decreasing digits.

Official Answer

Source: physical PDF page 11 (COMP2711_F2024_Midterm.pdf).

> Answer: (a) By choosing any 6 digits from $\{0, 1, 2, \cdots, 9\}$ we can form a unique 6 digit number that its digits is strictly decreasing from left to right. so the answer is $\binom{10}{6}$.

Supplementary Explanation

Answer. $\binom{10}{6} = 210$.

Derivation/proof. Strict decrease requires six distinct digits. Choose them from zero through nine in $\binom{10}{6}$ ways, then sort them in decreasing order. The largest digit is first; six distinct digits cannot all be zero, so the first digit is nonzero. Conversely, every strictly decreasing six-digit number determines one six-element digit set. Hence the count is $\binom{10}{6} = 210$.

C085

Permutations and Combinations

2024 Fall Midterm | Question 6(b) | Source page 11 | Difficulty 2 | 3 marks
Q-2024F-MID-P06-b | Answer provenance: official

> Assume that in any decimal number the leftmost digit cannot be 0. How many 6-digit numbers exist such that their digits from left to right:
>
> (b) Are strictly increasing (e.g., 136789). Briefly explain why.

Official Answer

Source: physical PDF page 11 (COMP2711_F2024_Midterm.pdf).

> (b) Similar to part (a), but we cant pick 0 because then it has to be the leftmost digit which is impossible. So the answer is $\binom{9}{6}$.

Supplementary Explanation

Answer. $\binom{9}{6} = 84$.

Derivation/proof. Strict increase requires six distinct digits in ascending order. If zero were selected, it would be the smallest digit and therefore the leading digit, which is forbidden. Thus choose six digits from one through nine, in $\binom{9}{6} = 84$ ways. Each chosen set has exactly one increasing arrangement and a nonzero first digit, so no factorial multiplier is needed.

C086

Permutations and Combinations

2024 Fall Midterm | Question 6(c) | Source page 11 | Difficulty 2 | 4 marks
Q-2024F-MID-P06-c | Answer provenance: official

> Assume that in any decimal number the leftmost digit cannot be 0. How many 6-digit numbers exist such that their digits from left to right:
>
> (c) Are deceasing, but potentially contain repeated digits (e.g., 997770). Briefly explain why. Hint: Only the number of occurrences of each digit is important.

Official Answer

Source: physical PDF page 11 (COMP2711_F2024_Midterm.pdf).

> (c) We can convert this question to another question so that we count the frequency of each digit. In other words: $x_0 + x_1 + x_2 + \cdots + x_9 = 6$. this equation has $\binom{15}{9}$ non-negative integer solutions. However we need to remove 000000 case which gives us $\binom{15}{9} - 1$.

Supplementary Explanation

Answer. $\binom{15}{9} - 1 = 5004$.

Derivation/proof. With repetition, "deceasing" means nonincreasing as illustrated. Let x_i be the frequency of digit i, so $x_0 + \cdots + x_9 = 6$ and all $x_i \geq 0$. Each frequency vector gives one descending arrangement. Stars and bars gives $\binom{6+10-1}{10-1} = \binom{15}{9} = 5005$ nonnegative solutions. Only $x_0 = 6$ with all other entries zero is invalid, producing 000000. Any nonzero digit becomes the leading digit. Thus $5005 - 1 = 5004$.

C087

Permutations and Combinations

2024 Fall Midterm | Question 6(d) | Source page 11 | Difficulty 2 | 3 marks
Q-2024F-MID-P06-d | Answer provenance: official

> Assume that in any decimal number the leftmost digit cannot be 0. How many 6-digit numbers exist such that their digits from left to right:
>
> (d) Are increasing, but potentially contain repeated digits (e.g., 111144). Briefly explain why.

Official Answer

Source: physical PDF page 11 (COMP2711_F2024_Midterm.pdf).

> (d) Similar to part (c). However we should not pick any 0. This means $x_1 + x_2 + \cdots + x_9 = 6$ which has $\binom{14}{8}$ different non-negative integer solutions.

Supplementary Explanation

Answer. $\binom{14}{8} = 3003$.

Derivation/proof. Allowing repetitions makes "increasing" mean nondecreasing. Any zero would appear first, so $x_0 = 0$. The remaining frequencies satisfy $x_1 + \cdots + x_9 = 6$ with all entries nonnegative. Every vector gives one valid ascending six-digit number. Stars and bars yields $\binom{6+9-1}{9-1} = \binom{14}{8} = 3003$. No all-zero case needs subtracting because zero was excluded altogether.

C088

Permutations and Combinations

2022 Final Part 1 | Question 2 | Source page 2 | Difficulty 2 | 8 marks
Q-2022-FINAL1-P02 | Answer provenance: official

> At a gathering of 30 people, there are 20 people who all know each other, while the other 10 people don't know anyone. People who know each other hug, and people who do not know each other shake hands. How many handshakes occur within the group?

Official Answer

Source: physical PDF page 2 (2022-final-part-1.pdf).

> One way to solve this problem is as follows. Each one of the ten people has to shake hands with all the 20 other people they don't know. So $10 \cdot 20 = 200$. From there, we calculate how many handshakes occurred between the people who don't know each other. This is simply counting how many ways to choose two people to shake hands, or $\binom{10}{2} = 45$. Thus the answer is $200 + 45 = 245$.
>
> Another way to consider this problem is as follows. We can focus on how many handshakes the 10 people get. The 1st person gets 29 handshakes. 2nd get 28 ... And the 10th receives 20 handshakes. We can write this as the sum of an arithmetic sequence.
>
> $$\frac{10(20+29)}{2} = 5 \cdot 49 = 245.$$
>
> Therefore the answer is 245.

Supplementary Explanation

Answer. 245.

Derivation/proof. Each of the ten strangers shakes hands with each of the twenty mutually acquainted people, giving 10·20=200 cross-group handshakes. Within the ten strangers every unordered pair also shakes, contributing C(10,2)=45. The twenty-person group contributes none because they hug. These cases are disjoint and exhaustive, so total 245. Check by complement: C(30,2)−C(20,2)=435−190=245.

The two disjoint classes give $10 \cdot 20 = 200$ and $\binom{10}{2} = 45$. The independent check is $435 - 190 = 245$. Each unordered pair is counted once.

C089

Permutations and Combinations

2022 Final Part 2 | Question 6(a) | Source page 3 | Difficulty 2
Q-2022-FINAL2-P06-a-expectation | Answer provenance: official

> There is a random simple undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the expectation of X?

Official Answer

Source: physical PDF page 4 (2022-final-part-2.pdf).

> (a) X_e is the indicator random variable that indicates the event of a pair of vertices $e = (u, v)$ having an edge. X_e is a Bernoulli trial, thus $E(X_e) = p$, $V(X_e) = p(1-p)$.
>
> There are $\binom{n}{2}$ pairs of vertices in total. Thus $E(X) = \binom{n}{2}p$ and $V(X) = \binom{n}{2}p(1-p)$.

Supplementary Explanation

Answer. $E[X] = \binom{n}{2}p$.

Derivation/proof. Index the unordered pairs by e and define I_e=1 for an existing edge and 0 otherwise. The edge count is X=∑I_e. Each I_e has mean p. Summing by linearity over N=n(n−1)/2 pairs gives E[X]=Np. This matches the binomial mean for X distributed as Binomial(N,p); linearity alone already suffices for this calculation.

Write $X = \sum_e I_e$. Since $E[I_e] = 1 \cdot p + 0 \cdot (1-p) = p$,

$$E[X] = \sum_e p = Np = \binom{n}{2}p.$$

Linearity itself does not require independence.

C090

Permutations and Combinations

2022 Final Part 2 | Question 6(a) | Source page 3 | Difficulty 2
Q-2022-FINAL2-P06-a-variance | Answer provenance: official

> There is a random simple undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the variance of X?

Official Answer

Source: physical PDF page 4 (2022-final-part-2.pdf).

> (a) X_e is the indicator random variable that indicates the event of a pair of vertices $e = (u, v)$ having an edge. X_e is a Bernoulli trial, thus $E(X_e) = p$, $V(X_e) = p(1-p)$.
>
> There are $\binom{n}{2}$ pairs of vertices in total. Thus $E(X) = \binom{n}{2}p$ and $V(X) = \binom{n}{2}p(1-p)$.

Supplementary Explanation

Answer. $\mathrm{Var}(X) = \binom{n}{2}p(1-p)$.

Derivation/proof. For one edge indicator I, E[I²]=E[I]=p, so Var(I)=p−p². In the stated random graph distinct edge indicators are independent, giving covariance zero for every distinct pair. The variance of their sum is therefore N(p−p²), with N=C(n,2). At p=0 or1 the answer is zero, correctly reflecting a deterministic graph.

Because $I_e^2 = I_e$, its variance is $p - p^2 = p(1-p)$. Independence makes the covariance zero for each distinct pair; hence

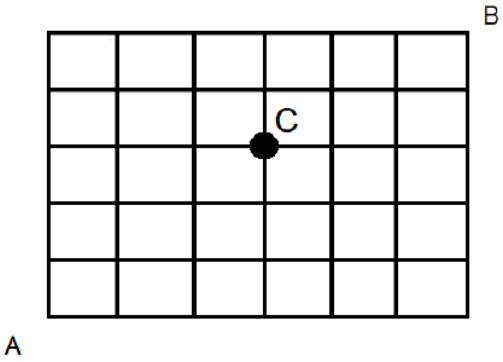
$$\mathrm{Var}(X) = \sum_e \mathrm{Var}(I_e) = \binom{n}{2}p(1-p).$$

Equal marginal probabilities alone would not justify dropping covariance terms.

C091

Permutations and Combinations

2020 Final Part 1 | Question 3(a) | Source page 2 | Difficulty 2 | 2 marks
Q-2020-FINAL1-P03-a | Answer provenance: official

> You are staying on point A and you are allowed to walk either up or to the right. Answer these questions:
>
> 
>
>
> In how many ways you can go to point B?

Official Answer

Source: physical PDF page 2 (2020-final-part-1.pdf).

$\binom{11}{5}$.

Supplementary Explanation

Answer. $\binom{11}{5} = 462$.

Derivation/proof. The figure has six columns and five rows between A and B. Every allowed path therefore has exactly six right steps and five up steps, in an eleven-step sequence. Choosing the five positions for up steps uniquely specifies the path, so there are $\binom{11}{5}$ paths. No other move is permitted.

C092

Permutations and Combinations

2020 Final Part 2 | Question 5(b) | Source page 3 | Difficulty 2
Q-2020-FINAL2-P05-b-expectation | Answer provenance: official

> There is a random undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the expectation of X?

Official Answer

Source: physical PDF page 4 (2020-final-part-2.pdf).

X_e is the indicator random variable that indicates the event, pair of vertices $e = (u, v)$ has an edge. X_e is a Bernoulli trial, thus $E(X_e) = p, V(X_e) = p(1-p)$. There are $\binom{n}{2}$ pairs of vertices in total. So that $E(X) = \binom{n}{2}p$ and $V(X) = \binom{n}{2}p(1-p)$.

Supplementary Explanation

Answer. $E[X] = \binom{n}{2}p$.

Derivation/proof. For each of the $\binom{n}{2}$ unordered vertex pairs e let I_e be 1 when its edge exists and 0 otherwise. Then $X = \sum_e I_e$ and $E[I_e] = 1 \cdot p + 0 \cdot (1-p) = p$. By linearity, $E[X] = \sum_e p = \binom{n}{2}p$. Independence is not needed for this expectation step.

There is one indicator for each potential unordered edge. Therefore $E(X) = \sum_e E(I_e) = \binom{n}{2}p$ by linearity, without any need for independence in this particular step.

C093

Permutations and Combinations

2020 Final Part 2 | Question 5(b) | Source page 3 | Difficulty 2
Q-2020-FINAL2-P05-b-variance | Answer provenance: official

> There is a random undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the variance of X?

Official Answer

Source: physical PDF page 4 (2020-final-part-2.pdf).

X_e is the indicator random variable that indicates the event, pair of vertices $e = (u, v)$ has an edge. X_e is a Bernoulli trial, thus $E(X_e) = p, V(X_e) = p(1-p)$. There are $\binom{n}{2}$ pairs of vertices in total. So that $E(X) = \binom{n}{2}p$ and $V(X) = \binom{n}{2}p(1-p)$.

Supplementary Explanation

Answer. $\mathrm{Var}(X) = \binom{n}{2}p(1-p)$.

Derivation/proof. Write X as the sum of independent edge indicators I_e. Because $I_e^2 = I_e$, each has variance $E[I_e^2] - E[I_e]^2 = p - p^2 = p(1-p)$. Distinct indicators are independent by the graph model, so their covariances vanish. Adding $\binom{n}{2}$ identical variance terms gives the result.

C094

Permutations and Combinations

2019 Final | Question 8(c) | Source page 10 | Difficulty 2 | 2 marks
Q-2019-FINAL-P08-c | Answer provenance: official

> How many strings of six lowercase letters from the English alphabet contain
>
> (c) the letters a and b in consecutive positions with a preceding b, with all the letters distinct?

Official Answer

Source: physical PDF page 10 (2019-final.pdf).

First choose the position for the a; this can be done in 5 ways, since the b must follow it. There are four remaining positions, and these can be filled in $P(24, 4)$ ways, since there are 24 letters left (no repetitions being allowed this time). Therefore the answer is $5P(24, 4) = 1275120$.

Supplementary Explanation

Answer. $5P(24, 4) = 1275120$.

Derivation/proof. The a in the forced block ab may start at positions 1 through 5, giving five choices. Its two positions are fixed and ordered. Fill the remaining four labeled positions with four distinct letters from the other 24 in $P(24, 4) = 24 \cdot 23 \cdot 22 \cdot 21 = 255024$ ways. Multiply to obtain 1275120.

$P(24, 4) = 24 \cdot 23 \cdot 22 \cdot 21 = 255024$, and each valid string has one unique ab starting position.

The forced block ab can start at positions $1, 2, 3, 4, 5$ in a six-letter string. Its starting position is unique because repeated letters are prohibited, so the five choices may be multiplied by the arrangements of the other four letters.

C095

Permutations and Combinations

2019 Final | Question 8(d) | Source page 10 | Difficulty 2 | 3 marks
Q-2019-FINAL-P08-d | Answer provenance: official

> How many strings of six lowercase letters from the English alphabet contain
>
> (d) the letters a and b, where a is somewhere to the left of b in the string, with all the letters distinct?

Official Answer

Source: physical PDF page 10 (2019-final.pdf).

First choose the positions for the a and b; this can be done in $C(6,2)$ ways, since once we pick two positions, we put the a in the left-most and the b in the other. There are four remaining positions, and these can be filled in $P(24,4)$ ways, since there are 24 letters left (no repetitions being allowed this time). Therefore the answer is $C(6,2)P(24,4) = 3825360$.

Supplementary Explanation

Answer. $\binom{6}{2}P(24,4) = 3825360$.

Derivation/proof. Choose the two positions occupied by a and b in $\binom{6}{2} = 15$ ways. Their roles are determined: put a in the earlier position and b in the later. Fill the four remaining positions by an ordered selection of distinct letters from the other 24, giving $P(24,4) = 255024$. The product is 3825360.

$\binom{6}{2} = 15$ and $P(24,4) = 255024$, so $15 \cdot 255024 = 3825360$.

The required occurrences need not be adjacent. Choosing their positions handles separated occurrences as well as consecutive ones without imposing an extra block condition.

C096

Permutations and Combinations

2019 Final | Question 9(a) | Source page 11 | Difficulty 2 | 2 marks
Q-2019-FINAL-P09-a | Answer provenance: official

> A professor packs her collection of 40 (distinguishable) issues of a mathematics journal in four boxes with 10 issues per box. How many ways can she distribute the journals if
>
> (a) each box is numbered, so that they are distinguishable?

Official Answer

Source: physical PDF page 11 (2019-final.pdf).

$40!/10!^4$.

Supplementary Explanation

Answer. $\frac{40!}{(10!)^4}$.

Derivation/proof. Choose ten issues for box 1, then ten of the remaining thirty for box 2, then ten of the remaining twenty for box 3; the last ten go to box 4. This gives $\binom{40}{10}\binom{30}{10}\binom{20}{10}\binom{10}{10} = 40!/(10!)^4$. Order within a box does not matter, but box labels distinguish the four groups.

C097

Permutations and Combinations

2019 Final | Question 9(b) | Source page 11 | Difficulty 2 | 2 marks
Q-2019-FINAL-P09-b | Answer provenance: official

> A professor packs her collection of 40 (distinguishable) issues of a mathematics journal in four boxes with 10 issues per box. How many ways can she distribute the journals if
>
> (b) the boxes are identical, so that they cannot be distinguished?

Official Answer

Source: physical PDF page 11 (2019-final.pdf).

Each distribution into identical boxes gives rise to $4! = 24$ distributions into labeled boxes, since once we have made the distribution into unlabeled boxes we can arbitrarily label the boxes. Therefore the answer is the same as the answer in part (a) divided by 24, namely $(40!/10!^4)/4!$.

Supplementary Explanation

Answer. $\frac{40!}{(10!)^4 \cdot 4!}$.

Derivation/proof. For labeled boxes, the count is $40!/(10!)^4$. Each unlabeled distribution consists of four distinct, nonempty sets of ten distinguishable issues. Assigning box labels to those sets can be done in exactly 4! ways, and no nontrivial label permutation fixes the distribution because the disjoint nonempty sets differ. Divide by 4! to count each unlabeled partition once.

Every box has ten issues and is nonempty; allowing empty boxes would require rechecking this uniform-labeling argument.

The forty people are pairwise distinct. Assigning them to four equal groups gives $\frac{40!}{(10!)^4\,4!}$ after division by the internal orders and then by the permutation of the four unlabeled groups.

C098

Permutations and Combinations

2019 Final | Question 10(b) | Source page 12 | Difficulty 2 | 2 marks
Q-2019-FINAL-P10-b | Answer provenance: official

> Consider the equation $x_1 + x_2 + x_3 = 20$ where x_1, x_2, and x_3 are nonnegative integers.
>
> For this question, give exact numbers as the solution.
>
> (b) How many solutions are there so that $x_1 \geq 9$?

Official Answer

Source: physical PDF page 12 (2019-final.pdf).

Let $x_1' = x_1 - 9$. The problem becomes the number of solutions to the equation $x_1' + x_2 + x_3 = 11$, which is $C(11 + 3 - 1, 3 - 1) = C(13, 2) = 78$.

Supplementary Explanation

Answer. 78.

Derivation/proof. Set $y_1 = x_1 - 9 \geq 0$. The equation becomes $y_1 + x_2 + x_3 = 11$, a bijective change of variables with inverse $x_1 = y_1 + 9$. Stars and bars gives $\binom{13}{2} = 13 \cdot 12/2 = 78$.

After reserving the required minimum allocations, stars and bars gives $\binom{13}{2} = 78$ arrangements of the remainder.

C099

Permutations and Combinations

2019 Final | Question 10(c) | Source page 12 | Difficulty 2 | 2 marks
Q-2019-FINAL-P10-c | Answer provenance: official

> Consider the equation $x_1 + x_2 + x_3 = 20$ where x_1, x_2, and x_3 are nonnegative integers.
>
> For this question, give exact numbers as the solution.
>
> (c) How many solutions are there so that $x_1 \geq 9$ and $x_2 \geq 9$?

Official Answer

Source: physical PDF page 12 (2019-final.pdf).

Same concept as part (b). The answer is $C(2 + 3 - 1, 3 - 1) = C(4, 2) = 6$.

Supplementary Explanation

Answer. 6.

Derivation/proof. Write $y_1 = x_1 - 9, y_2 = x_2 - 9$. Then $y_1 + y_2 + x_3 = 20 - 18 = 2$ with all variables nonnegative. Its number of solutions is $\binom{4}{2} = 6$. Equivalently the possibilities distribute two units as one coordinate 2 (three choices) or two coordinates 1 (three choices).

As a separate enumeration, two units may both occupy one coordinate in three ways or occupy two distinct coordinates in three ways, giving six.

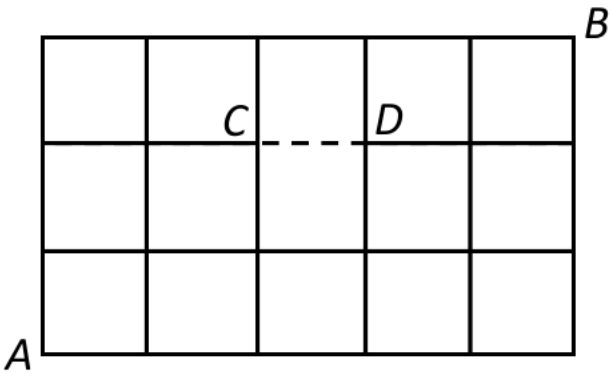
The allowed residual cases give six possibilities. Enumerating those cases separately is valid because the cases have different occupancy vectors and therefore do not overlap.

C100

Permutations and Combinations

2018 Final | Question 11(a) | Source page 16 | Difficulty 2
Q-2018-FINAL-P11-a | Answer provenance: official

In the following shape, consider paths that start from A and finish at B following the lines. Every path may only have steps going up or right. E.g., one path would be $\{\uparrow, \uparrow, \uparrow, \rightarrow, \rightarrow, \rightarrow, \rightarrow, \rightarrow\}$. How many distinct such paths are there if:



a) All lines in the rectangle are possible?

Official Answer

Source: physical PDF page 16 (2018-final.pdf).

The distance from A to B is 8 steps, 5 right, and 3 up. The answer is $C(8, 5) = 56$.

Supplementary Explanation

Answer. $\binom{8}{5} = 56$.

Derivation/proof. Moving from A to B requires exactly five right steps and three up steps. Any ordering of these eight steps remains inside the rectangle because neither coordinate can exceed its final total before all such steps have been used. Choose the five right-step positions among eight, giving $\binom{8}{5} = 56$.

Recording a path as five right steps and three up steps gives a one-to-one correspondence between paths and words with those fixed multiplicities. Choosing the three up-step positions counts every path once.

C101

Permutations and Combinations

2018 Midterm | Question 7(b) | Source page 9 | Difficulty 2
Q-2018-MID-P07-b | Answer provenance: official

> A committee of 5 members needs to be formed in order to represent the employees in an office. There are 20 employees, 11 men and 9 women.
>
> Briefly justify your answers.
>
> (b) How many ways are there to form the committee if at least one woman and one man must be in it?

Official Answer

Source: physical PDF page 9 (2018-midterm.pdf).

There are $C(11,5) = 462$ ways to form a committee entirely from men and $C(9,5) = 126$ ways to form a committee entirely from women. Note that no committee can consist entirely of women and entirely of men at the same time. Therefore the correct answer is $C(20,5) - C(11,5) - C(9,5) = 14916$.

Supplementary Explanation

Answer. $\binom{20}{5} - \binom{11}{5} - \binom{9}{5} = 14916$.

Derivation/proof. Among all 15504 committees, the forbidden ones are all male, counted by $\binom{11}{5} = 462$, or all female, counted by $\binom{9}{5} = 126$. No nonempty committee belongs to both forbidden cases. Therefore subtract each once: $15504 - 462 - 126 = 14916$.

C102

Permutations and Combinations

2018 Midterm | Question 7(c) | Source page 9 | Difficulty 2
Q-2018-MID-P07-c | Answer provenance: official

> A committee of 5 members needs to be formed in order to represent the employees in an office. There are 20 employees, 11 men and 9 women.
>
> Briefly justify your answers.
>
> (c) How many ways are there to form the committee if there must be more women than men?

Official Answer

Source: physical PDF page 9 (2018-midterm.pdf).

Let us consider all the distinct cases that lead to more women than men. There are $C(9,5)$ ways to form a committee from five women and no man, $C(9,4)\cdot C(11,1)$ ways to form a committee with four women and one man, and $C(9,3)\cdot C(11,2)$ ways to form a committee with three women and two men. In all other cases, there will not be more women than men in the committee. Therefore the correct answer is $C(9,5)+C(9,4)\cdot C(11,1)+C(9,3)\cdot C(11,2)=6132$.

Supplementary Explanation

Answer. $\binom{9}{5}+\binom{9}{4}\binom{11}{1}+\binom{9}{3}\binom{11}{2}=6132$.

Derivation/proof. Let w be the number of women. Since there are five members, $w>5-w$ means $w=3,4,5$. The corresponding counts are $\binom{9}{3}\binom{11}{2}=4620$, $\binom{9}{4}\binom{11}{1}=1386$, and $\binom{9}{5}=126$. Their sum is 6132.

The three case counts sum to $4620+1386+126=6132$. Cases with $w\le 2$ have fewer women and are excluded. The printed source says “will not be more women”; the earlier note's claim of a “more” typo is not supported by the source and is not adopted.

C103

Permutations and Combinations

2018 Midterm | Question 8(c) | Source page 10 | Difficulty 2
Q-2018-MID-P08-c | Answer provenance: official

> You are given a suitcase that has a combination lock. It consists of five dials, each of which can be rotated to take a value from 0–9. The suitcase only opens if you guess the correct 5-digit solution.
>
> Briefly justify your answers.
>
> (c) Assume that the lock has a weird flaw: The exact order in which you put the digits does not matter. In this case how many different solutions are there? (Note that all permuted versions of a given combination still counts as one solution. For example, if the correct solution is 12345, then 34125,51243,23145,etc., all count as the same solution.)

Official Answer

Source: physical PDF page 10 (2018-midterm.pdf).

Since the order of the digits does not matter, this can be seen as a classic “stars-and-bars” problem. We need 10 gaps, each one for one of the digits 0–9, thus the number of bars is $10 - 1 = 9$. The numbers of stars is 5, one for the chosen digit in each of the dials. The standard formula for this problem gives $C(14, 5) = 2002$.

Supplementary Explanation

Answer. $\binom{14}{5} = 2002$.

Derivation/proof. For each digit $d = 0, \ldots, 9$, let x_d be its multiplicity. An unordered setting is uniquely described by $x_0 + \cdots + x_9 = 5$ with each $x_d \geq 0$. Five stars and nine bars encode these ten counts. Thus there are $\binom{14}{9} = \binom{14}{5} = 2002$ settings. Part (b)'s adjacency restriction is not carried into this separate part.

C104

Permutations and Combinations

2017 Final | Question 6(b) | Source page 10 | Difficulty 2 | 2 marks
Q-2017-FINAL-P06-b | Answer provenance: official

> 64 fair coins are flipped independently randomly and then arranged in an 8 by 8 block randomly without looking. Each row contains eight coins and there are eight rows. If all eight coins in a row have the same symbol (heads or tails), we call it a homogeneous row. Let X denote the number of homogeneous rows in the block.
>
> (b) Calculate $p(X = 1)$.

Official Answer

Source: physical PDF page 10 (2017-final.pdf).

For any given row, the chance it is homogeneous is $(1/2)^8 \cdot 2 = (1/2)^7$ (all heads or all tails). The chance that it is not homogeneous is $(1 - (1/2)^7)$. Overall, seven rows must not be homogeneous and one row must be homogeneous, so $p(X = 1) = 8(1/2)^7(1 - (1/2)^7)^7$.

Supplementary Explanation

Answer. $8(1/128)(127/128)^7$.

Derivation/proof. A given row is homogeneous in two of its 2^8 equally likely configurations, so its probability is $q = 2/2^8 = 1/128$. Distinct rows use disjoint independent coins. Choose which of the eight rows succeeds, multiply by q for it and $(1 - q)^7$ for the failures. The eight choices are disjoint, giving $8q(1 - q)^7$.

Randomly arranging the coins without observing their faces preserves the independent fair distribution at all sixty-four positions.

A row is homogeneous with probability $q = 2(1/2)^8 = 1/128$, since either all eight coins are heads or all are tails. The independent eight rows then give $P(X = 1) = 8(1/128)(127/128)^7$ by selecting its unique homogeneous row.

C105

Permutations and Combinations

2017 Midterm | Question 6(b) | Source page 8 | Difficulty 2
Q-2017-MID-P06-b | Answer provenance: official

> Five couples go to the movies together and sit in a row of ten seats. In how many ways can the 10 people be arranged if
>
> (b) All the men sit together and all the women sit together.

Official Answer

Source: physical PDF page 8 (2017-midterm.pdf).

$2 \cdot 5!^2$.

Supplementary Explanation

Answer. $2(5!)^2$.

Derivation/proof. Treat all men as one block and all women as another. The two blocks can appear in 2! orders. Independently permute the five men inside their block in 5! ways and the five women in 5! ways. Multiplication gives $2!(5!)(5!)$.

The two group blocks have 2! possible orders, and each group has 5! internal orders. Their product is $2! \cdot 5! \cdot 5! = 2(5!)^2$; each arrangement determines exactly these choices.

C106

Permutations and Combinations

2017 Midterm | Question 6(c) | Source page 8 | Difficulty 2
Q-2017-MID-P06-c | Answer provenance: official

> Five couples go to the movies together and sit in a row of ten seats. In how many ways can the 10 people be arranged if
>
> (c) Each couple sits together (i.e., for each couple, the two people are in adjacent seats).

Official Answer

Source: physical PDF page 8 (2017-midterm.pdf).

$5! \cdot 2^5$.

Supplementary Explanation

Answer. $5!2^5$.

Derivation/proof. Collapse each couple into a labeled block. Arrange the five blocks in 5! orders. Each couple independently chooses which partner occupies its left seat, giving 2^5 internal choices. Every allowed seating has exactly one such decomposition.

There is a unique block decomposition for each admissible arrangement. Collapsing the required adjacent people to their blocks, and then expanding with their internal orders, are inverse operations, so the block count neither omits nor duplicates arrangements.

C107

Permutations and Combinations

2017 Midterm | Question 6(d) | Source page 8 | Difficulty 2
Q-2017-MID-P06-d | Answer provenance: official

> Five couples go to the movies together and sit in a row of ten seats. In how many ways can the 10 people be arranged if
>
> (d) One couple is arguing and they refuse to sit together. The other couples can sit in any way (i.e., together or not).

Official Answer

Source: physical PDF page 8 (2017-midterm.pdf).

$10! - 9 \cdot 8! \cdot 2$.

Supplementary Explanation

Answer. $10! - 2 \cdot 9! = 10! - 9 \cdot 8! \cdot 2$.

Derivation/proof. Start with all 10! seatings. For forbidden seatings, collapse the specified arguing couple into one block with two orientations. Together with the other eight people there are nine distinct objects, giving $2 \cdot 9!$ forbidden arrangements. Subtract this complement; no condition is imposed on the other couples.

C108

Permutations and Combinations

2017 Midterm | Question 9(b) | Source page 11 | Difficulty 2
Q-2017-MID-P09-b | Answer provenance: official

> Six women and nine men are on the faculty of a department. The department needs to form a committee of five members.
>
> (b) How many committees are possible if the committee must have at least two women?

Official Answer

Source: physical PDF page 11 (2017-midterm.pdf).

$\binom{6}{2}\binom{9}{3} + \binom{6}{3}\binom{9}{2} + \binom{6}{4}\binom{9}{1} + \binom{6}{5}\binom{9}{0}$.

Supplementary Explanation

Answer. $\binom{6}{2}\binom{9}{3} + \binom{6}{3}\binom{9}{2} + \binom{6}{4}\binom{9}{1} + \binom{6}{5}\binom{9}{0}$.

Derivation/proof. Classify a committee by its number k of women. The allowed values are $k = 2, 3, 4, 5$. For each, choose k of the six women and $5 - k$ of the nine men, giving $\binom{6}{k}\binom{9}{5-k}$. The four cases are disjoint and exhaustive, so sum them.

Numerically, the four cases give $1260 + 720 + 135 + 6 = 2121$.

Partition committees by the exact number k chosen from the six-person group. The disjoint cases sum to $\sum_{k=2}^{5} \binom{6}{k}\binom{9}{5-k}$, equivalent to subtracting the zero- and one-member cases from all committees.

C109

Permutations and Combinations

2017 Midterm | Question 10(b) | Source page 12 | Difficulty 2
Q-2017-MID-P10-b | Answer provenance: official

> Assume you have 50 candies and four boxes, labeled A, B, C, and D.
>
> (b) How many ways can you put the candies in the boxes, assuming that the candies are identical?

Official Answer

Source: physical PDF page 12 (2017-midterm.pdf).

$\binom{50+4-1}{3}$.

Supplementary Explanation

Answer. $\binom{53}{3}$.

Derivation/proof. Let x_A, x_B, x_C, x_D be nonnegative box counts, with sum 50. Represent a distribution by fifty stars and three separators; the numbers of stars before and between separators specify the labeled boxes. Choosing the three separator positions among 53 gives $\binom{53}{3}$.

Write the occupancies as nonnegative integers satisfying $x_A + x_B + x_C + x_D = 50$. A stars-and-bars arrangement and its occupancy vector determine each other, establishing a one-to-one correspondence even when some boxes are empty.

C110

Permutations and Combinations

2017 Midterm | Question 10(c) | Source page 12 | Difficulty 2
Q-2017-MID-P10-c | Answer provenance: derived

> Assume you have 50 candies and four boxes, labeled A, B, C, and D.
>
> (c) How many ways can you put the candies in the boxes, assuming that the candies are identical and each box must have at least five candies put into it?

Official Answer

Source: physical PDF page 12 (2017-midterm.pdf).

$\binom{35+4-1}{3}$.

Supplementary Explanation

Answer. $\binom{33}{3} = 5456$.

Derivation/proof. Write each box count as $x_i = 5 + y_i$, where $y_i \geq 0$. Then $y_1 + y_2 + y_3 + y_4 = 50 - 4 \cdot 5 = 30$. Stars and bars gives $\binom{30+4-1}{3} = \binom{33}{3} = 5456$. Independently, direct enumeration of nonnegative quadruples summing to 30 gives the same 5456; the bijection adds five to each coordinate.

For the independent enumeration, let $y_1 = 0, \ldots, 30$, $y_2 = 0, \ldots, 30 - y_1$, and $y_3 = 0, \ldots, 30 - y_1 - y_2$; y_4 is determined by the sum. This counts 5456 quadruples.

The four required allocations reserve twenty balls in total, leaving $50 - 20 = 30$ freely distributed balls. Apply stars and bars to those thirty, rather than to the printed thirty-five.

C111

Permutations and Combinations

2016 Final | Question 7(a) | Source page 13 | Difficulty 2
Q-2016-FINAL-P07-a-recurrence | Answer provenance: official

> Assume that a party of n people wants to select some of their members to form a committee. The committee must have at least one member. Let $T(n)$ be the total number of possible committees that can be formed from a party of n people.
>
> (a) Define $T(n)$ as a recurrence equation, and prove that your recurrence equation is correct by induction.
>
> Atomic focus [editorial]: define the recurrence; its induction proof is a separate atom.

Official Answer

Source: PDF physical page 13 (2016-final.pdf).

(a) $T(1) = 1$, and $T(n) = 2T(n-1) + 1$ for $n > 1$.

The base case $T(1)$ is obviously true.

Assume that $T(n-1)$ counts the number of committees that are formed from $n-1$ people correctly for $n > 1$.

When there are n people, the number of possible committees that are formed without the n-th person is counted by $T(n-1)$. On the other hand, each committee without the n-th person union with the n-th person forms another possible committee. So, we have $2T(n-1)$ possible committees. There is only one case is missed, which is the committee that contains only the n-th person. Therefore, when there are n people, $T(n) = 2T(n-1) + 1$ counts all possible committees with at least one member from n people. By the principle of induction, we have proved that the recurrence equation is correct.

Supplementary Explanation

Answer. $T(1) = 1$ and $T(n) = 2T(n-1) + 1$ for $n > 1$.

Derivation/proof. With one person there is exactly one allowed committee. For $n > 1$, old nonempty committees contribute $T(n-1)$ without person n and another $T(n-1)$ with person n added. The remaining allowed committee is $\{n\}$, giving the recurrence. These cases are disjoint and exhaustive.

C112

Permutations and Combinations

2016 Midterm | Question 2(a) | Source page 4 | Difficulty 2 | 4 marks
Q-2016-MID-P02-a | Answer provenance: official

> Consider a university with 300 faculties, 500 teaching assistants, 600 postgraduate students and 10,000 undergraduate students. A tennis club would like to invite a group of guests to join an activity. The group of guests must consist of 4 faculties, 10 teaching assistants, 10 postgraduate students and 40 undergraduate students. Assume that each person initially has only one role (i.e. no one person can be both a postgraduate student and a teaching assistant, etc).
>
> (a) How many possible groups of guests are there?

Official Answer

Source: PDF physical page 4 (2016-midterm.pdf).

(a) $\binom{300}{4}\binom{500}{10}\binom{600}{10}\binom{10000}{40}$

Supplementary Explanation

Answer. $\binom{300}{4}\binom{500}{10}\binom{600}{10}\binom{10000}{40}$.

Derivation/proof. Select 4 of 300 faculty, 10 of 500 assistants, 10 of 600 postgraduates, and 40 of 10,000 undergraduates. The original roles are disjoint, so each combination of four selections gives one allowed guest group.

Each four-part selection determines exactly one guest set, with no invitation ordering.

C113

Permutations and Combinations

2015 Midterm | Question 7(b) | Source page 12 | Difficulty 2 | 4 marks
Q-2015-MID-P07-b | Answer provenance: official

> You are going to distribute kn cash coupons to kn members in your company such that each member will get one cash coupon, where $k, n \geq 1$. The values of the kn cash coupons are distinct. Suppose there are n ranks of seniority in the company, and there are exactly k members at each rank. If a more senior member receives a cash coupon worth less than the cash coupon given to a more junior member, you will receive a complaint.
>
> (b) How many ways are there to distribute the kn cash coupons such that you will not receive any complaints?
>
> Briefly explain how your answers are derived.

Official Answer

Source: PDF physical page 12 (2015-midterm.pdf).

(b) $(k!)^n$. The value of each cash coupon given to a person at the lowest rank should be less than the value of any cash coupon given to a person in the second lowest rank, and so on. Therefore, once sorted, the k coupons distributed to each rank are fixed. Within each rank, there are $k!$ ways to distribute the k cash coupons to the k members. By the product principle, the answer is $(k!)^n$.

Supplementary Explanation

Answer. $(k!)^n$.

Derivation/proof. If a lower rank received a coupon above any coupon at a higher rank, a complaint would occur. Hence the sorted coupons must split into n consecutive blocks of k, assigned in rank order. Each block can be bijected to its k distinct members in $k!$ ways. Multiply across ranks.

The lowest rank must receive the smallest k coupons; repeating this argument fixes every later block.

C114

Permutations and Combinations

2015 Midterm | Question 8(a) | Source page 13 | Difficulty 2 | 5 marks
Q-2015-MID-P08-a | Answer provenance: official

> A shopping bag contains 12 pair of gloves. Each pair consists of two different gloves, one for the left hand and another for the right hand, and the two gloves are in the same color. Moreover, different pairs are in different colors. You blindly take 6 gloves (not 6 pairs of gloves) from the bag. How many possible ways are there if
>
> (a) the number of colors of these 6 gloves is 3?
>
> Briefly explain how your answers are derived.

Official Answer

Source: PDF physical page 13 (2015-midterm.pdf).

(a) The number of colors of these 6 gloves is 3 means you have picked 3 pairs of gloves out of 12. So there are $\binom{12}{3}$ possible cases.

Supplementary Explanation

Answer. $\binom{12}{3} = 220$.

Derivation/proof. If the three selected colors contribute $g_1, g_2, g_3 \leq 2$ gloves each and sum to six, all $g_i = 2$. Choose those three colors from twelve; their gloves are then fixed.

The constraints $g_1 + g_2 + g_3 = 6$ and $g_i \leq 2$ force all three values to be two.

C115

Permutations and Combinations

2015 Midterm | Question 8(c) | Source page 13 | Difficulty 2 | 5 marks
Q-2015-MID-P08-c | Answer provenance: official

> A shopping bag contains 12 pair of gloves. Each pair consists of two different gloves, one for the left hand and another for the right hand, and the two gloves are in the same color. Moreover, different pairs are in different colors. You blindly take 6 gloves (not 6 pairs of gloves) from the bag. How many possible ways are there if
>
> (c) these 6 gloves have different colors?
>
> Briefly explain how your answers are derived.

Official Answer

Source: PDF physical page 13 (2015-midterm.pdf).

(c) These 6 gloves have different colors means these gloves are from different pairs. Firstly you pick 6 pairs of gloves out of 12 pairs. There are $\binom{12}{6}$ possibilities. Then you choose only one glove from each pair. So there are $\binom{12}{6} \cdot 2^6$ cases in total.

Supplementary Explanation

Answer. $\binom{12}{6}2^6 = 59136$.

Derivation/proof. A valid selection identifies a unique six-color subset, with $\binom{12}{6}$ choices. For each selected color exactly one of its left/right gloves is taken, giving 2^6 selections per subset. Multiplication yields the result.

No ordering of the selected gloves is counted, so no factor of 6! belongs in the answer.

C116

Permutations and Combinations

2014 Midterm | Question 1(b) | Source page 3 | Difficulty 2
Q-2014-MID-P01-b | Answer provenance: official

> A by-election is being held to fill 5 seats in a council that have recently become vacant. A total of 9 candidates have registered to run in the election. The candidates are divided into 3 groups based on their political affiliations with: 4 in Group A, 3 in Group B, and 2 in Group C.
>
> (b) In how many ways can a voter choose 5 candidates such that at least 3 candidates from Group A are chosen?
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 3 (2014-midterm.pdf).

(b) We partition this into two cases: choosing either 3 or 4, from group A and, respectively, 2 or 1 candidates from the combination of groups B and C. This is

$$\binom{4}{3}\binom{5}{2} + \binom{4}{4}\binom{5}{1} = 4 \cdot 10 + 1 \cdot 5 = 45.$$

Supplementary Explanation

Answer. $\binom{4}{3}\binom{5}{2} + \binom{4}{4}\binom{5}{1} = 45$.

Derivation/proof. The possible Group-A counts are three or four. Select the A candidates and fill the remaining places from the five candidates not in A; the cases are disjoint.

The cases contribute 40 and 5, giving $40 + 5 = 45$.

The three-A case alone contributes $\binom{4}{3}\binom{5}{2} = 40$.

C117

Permutations and Combinations

2014 Midterm | Question 2(b) | Source page 4 | Difficulty 2
Q-2014-MID-P02-b | Answer provenance: official

> There are n families joining a party. Each family consists of 3 people, a mother, a father and a child. In the party, there are $3n$ seats placed in a line.
>
> (b) Suppose the three people in each family must sit together. How many different seating arrangement are there?
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 4 (2014-midterm.pdf).

(b) $n!(3!)^n$. Consider a family as a unit. There are $n!$ ways to arrange the n units. In each unit there are $3!$ different seating orders.

Supplementary Explanation

Answer. $n!(3!)^n$.

Derivation/proof. Treat each family as one block and arrange the n blocks in $n!$ ways. Each block has $3!$ internal orders, independently.

Expanding each block uniquely recovers the seating, so no arrangement is duplicated.

C118

Permutations and Combinations

2014 Midterm | Question 2(d) | Source page 4 | Difficulty 2
Q-2014-MID-P02-d | Answer provenance: official

> There are n families joining a party. Each family consists of 3 people, a mother, a father and a child. In the party, there are $3n$ seats placed in a line.
>
> (d) Suppose each child must sit next to his/her mother. How many different seating arrangements are there?
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 4 (2014-midterm.pdf).

(d) $(2n)!2^n$. Consider the child and his/her mother as a unit. There are $2n$ units to arrange, and the number of ways is $(2n)!$. For each mother and child pair, there are 2 different seating orders.

Supplementary Explanation

Answer. $(2n)!2^n$.

Derivation/proof. Make each mother–child pair a block and keep each father as a singleton: $2n$ units arrange in $(2n)!$ ways. Every pair has two internal orders.

The two orders are mother-left/child-right and child-left/mother-right. Fathers need not sit near their families.

C119

Permutations and Combinations

2014 Midterm | Question 4(b) | Source page 6 | Difficulty 2
Q-2014-MID-P04-b | Answer provenance: official

> An anagram is a distinct ordering of the letters. For example, the word “eat” has six anagrams “eat”, “eta”, “ate”, “aet”, “tea” and “tae”, while the word “too” has only three anagrams “too”, “oto” and “oot”.
>
> (b) How many anagrams does the word “lollapalooza” have?
>
> Explain your answers.

Official Answer

Source: PDF physical page 6 (2014-midterm.pdf).

(b) This is the same as the number of ways to label 12 positions according to the following rules: Label 3 positions with 'a', 4 positions with 'l', 3 positions with 'o', 1 position with 'p', and 1 position with 'z'. So the answer is:

$$\frac{12!}{3!4!1!3!1!} = ?$$

Supplementary Explanation

Answer. $12!/(4!3!3!) = 554400$.

Derivation/proof. The twelve letters have multiplicities $l:4$, $a:3$, $o:3$, $p:1$, $z:1$. Divide $12!$ by factorials of repeated-letter multiplicities.

Temporarily labelling repeated letters gives $12!$ permutations. Removing the labels identifies $4!3!3!$ permutations for each word, giving $12!/(4!3!3!) = 554400$.

C120

Permutations and Combinations

2013 Midterm | Question 1(b) | Source page 3 | Difficulty 2
Q-2013-MID-P01-b | Answer provenance: official

> Twelve Easter eggs are numbered $1, 2, \ldots, 12$. Eggs 1 to 4 are colored red, eggs 5 to 8 are colored green, and eggs 9 to 12 are colored purple. A child is allowed to choose 5 of the eggs.
>
> (b) What is the answer to (a) if the child must choose at least one egg from each color?
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 3 (2013-midterm.pdf).

(b) The wanted number equals the total number of ways minus the number of ways to choose the eggs from only two of the three colors.

The number of ways to choose 5 eggs from among the red and green eggs is $\binom{8}{5}$. The numbers for the red-purple and green-purple combination are the same. The three subsets of choice plans are disjoint. So, the answer to the question is:

$$\binom{12}{5} - 3\binom{8}{5}.$$

Supplementary Explanation

Answer. $\binom{12}{5} - 3\binom{8}{5} = 624$.

Derivation/proof. Start with all $\binom{12}{5}$ selections. A bad selection omits red, green, or purple; after one colour is omitted only eight eggs remain. Choosing five eggs cannot omit two colours, so the three bad classes are disjoint and the subtraction is valid.

$\binom{8}{5} = 56$, so the subtraction gives $792 - 3 \cdot 56 = 624$.

C121

Permutations and Combinations

2013 Midterm | Question 1(c) | Source page 3 | Difficulty 2
Q-2013-MID-P01-c | Answer provenance: official

> Twelve Easter eggs are numbered $1, 2, \ldots, 12$. Eggs 1 to 4 are colored red, eggs 5 to 8 are colored green, and eggs 9 to 12 are colored purple. A child is allowed to choose 5 of the eggs.
>
> (c) What is the answer to (a) if the number of red eggs chosen must be more than the total number of green and purple eggs?
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 3 (2013-midterm.pdf).

(c) Here are are possible choice plans that satisfy the requirement:

3 red, 2 green or purple

4 red, 1 green or purple

So, the answer is:

$$\binom{4}{3}\binom{8}{2} + \binom{4}{4}\binom{8}{1}.$$

Supplementary Explanation

Answer. $\binom{4}{3}\binom{8}{2} + \binom{4}{4}\binom{8}{1} = 120$.

Derivation/proof. If r red eggs are selected then $r > 5 - r$, hence $r = 3$ or 4. The two disjoint cases are $(r, 5-r) = (3, 2)$ and $(4, 1)$, producing the displayed sum.

The disjoint cases contribute $4 \cdot 28 = 112$ and 8, giving $112 + 8 = 120$.

C122

Permutations and Combinations

2013 Midterm | Question 4(d) | Source page 6 | Difficulty 2
Q-2013-MID-P04-d | Answer provenance: official

> There are 8 boys and 8 girls to pair up in a dance party.
>
> (d) What is the answer for (a) if there are 16 boys and 8 girls. Note that in this case not all boys can have a dance partner.
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 7 (2013-midterm.pdf).

(d) There are totally 16 boys. Only 8 of them will have dance partners. The number of ways to pick 8 boys from among 16 is $\binom{16}{8}$. Let $b_1, b_2, \ldots, b_8$ be the 8 boys picked. Boy b_1 can be paired up with any of the 8 girls; Boy b_2 can be paired up any of the remaining 7 girls; and so on. So the answer is:

$$\binom{16}{8} 8!.$$

There is another way to solve the problem. Number the girls from 1 to 8. Girl number 1 can be paired up with any of the 16 boys; girl number 2 can then be paired up with with any of the remaining 15 boys; and so on. So the answer is $16 \times 15 \times \ldots \times 9 = 16^{\underline{8}}$, which is the same as $\binom{16}{8} 8!$.

Supplementary Explanation

Answer. $\binom{16}{8} 8! = 16!/8!$.

Derivation/proof. Choose the eight boys who receive partners, then biject them with the eight girls. This is $\binom{16}{8} 8! = 16 \cdot 15 \cdots 9$.

C123

Permutations and Combinations

2025 Fall Final | Question 3(b) | Source page 7 | Difficulty 3 | 3 marks
Q-2025F-FINAL-P03-b | Answer provenance: official

> We have 14 volleyball players, all of different heights. All 14! permutations of rankings (tallest = rank 1 to shortest = rank 14) are equally likely. Answer the following questions and briefly explain. You do not need to calculate the final numerical answer.
>
> (b) [3 points] Given A is taller than B, what is the probability that A is the third tallest?

Official Answer

Source: physical PDF page 7 (COMP2711_F2025_Final.pdf).

(b) Given $A > B$. If A is rank 3, B must be in ranks $4, \ldots, 14$: 11 choices. Then arrange remaining 12 players: 12!.

$$\Pr(A \text{ rank } 3 \mid A > B) = \frac{11 \cdot 12!}{14!/2} = \frac{11 \cdot 2}{14 \cdot 13} = \frac{11}{91}.$$

Supplementary Explanation

Answer. $11/91$.

Derivation/proof. The condition $A > B$ leaves $14!/2$ equally likely rankings. Fix A in third place. Then B can occupy ranks 4 through 14, giving $14 - 3 = 11$ choices. After fixing both positions, the remaining 12 players have 12! arrangements, so there are $11 \cdot 12!$ favorable rankings. Division gives $11 \cdot 12!/(14!/2) = 22/(14 \cdot 13) = 11/91$. Neither position above A may be given to B.

C124

Permutations and Combinations

2025 Fall Final | Question 3(c) | Source page 7 | Difficulty 3 | 3 marks
Q-2025F-FINAL-P03-c | Answer provenance: official

> We have 14 volleyball players, all of different heights. All 14! permutations of rankings (tallest = rank 1 to shortest = rank 14) are equally likely. Answer the following questions and briefly explain. You do not need to calculate the final numerical answer.
>
> (c) [3 points] Given A is taller than B and shorter than C, what is the probability that A is seventh tallest?

Official Answer

Source: physical PDF page 7 (COMP2711_F2025_Final.pdf).

(c) Given $C > A > B$. Permutations satisfying this: $14!/6$. If A is rank 7, C in ranks $1, \ldots, 6$ (6 choices), B in ranks $8, \ldots, 14$ (7 choices), arrange remaining 11: 11!.

$$\Pr(A \text{ rank } 7 \mid C > A > B) = \frac{6 \cdot 7 \cdot 11!}{14!/6} = \frac{252}{14 \cdot 13 \cdot 12} = \frac{3}{26}.$$

Supplementary Explanation

Answer. $3/26$.

Derivation/proof. The three distinct-height players have $3! = 6$ equally likely relative orders, so $C > A > B$ leaves $14!/6$ rankings. Fix A seventh. Player C has six choices among ranks 1–6; player B has seven choices among ranks 8–14. The other 11 players can be arbitrarily arranged, giving $6 \cdot 7 \cdot 11!$ favorable rankings. Hence the conditional probability is $6 \cdot 7 \cdot 11!/(14!/6) = 252/(14 \cdot 13 \cdot 12) = 3/26$. The position sets for C and B are disjoint, so multiplication does not create collisions.

C125

Permutations and Combinations

2025 Fall Final | Question 4(a) | Source page 8 | Difficulty 3 | 3 marks
Q-2025F-FINAL-P04-a | Answer provenance: official

> Assume there are 4 teachers (T), 5 engineers (E), 6 doctors (D) and 7 students (S). These $4 + 5 + 6 + 7 = 22$ people want to stand in a single queue (a line with a well-defined front/head and back/end). Every individual is a distinct person (i.e., they are all different people) Determine the number of ways to arrange them in the queue for each of the following conditions. No argument is needed. There is no need to calculate the exact numerical answer. You are not allowed write the answer in any form of summation.
>
> (a) [3 points] Every teacher stands in front of all doctors (i.e., the last teacher is before the first doctor in the queue).

Official Answer

Source: physical PDF page 8 (COMP2711_F2025_Final.pdf).

(a) $\binom{22}{12} \times 12! \times 6! \times 4!$

Supplementary Explanation

Answer. $\binom{22}{12}\, 12!\, 6!\, 4!$.

Derivation/proof. The five engineers and seven students form 12 people with unrestricted relative positions. Choose their positions among 22 in $\binom{22}{12}$ ways and arrange them in 12! ways. The remaining ten positions belong to teachers and doctors. Their earliest four positions must all be teachers, and the final six must all be doctors. The teachers can be ordered in 4! ways and the doctors in 6! ways. Thus the total is $\binom{22}{12} 12! 4! 6!$, without a summation.

C126

Permutations and Combinations

2025 Fall Final | Question 4(b) | Source page 8 | Difficulty 3 | 3 marks
Q-2025F-FINAL-P04-b | Answer provenance: official

> Assume there are 4 teachers (T), 5 engineers (E), 6 doctors (D) and 7 students (S). These $4 + 5 + 6 + 7 = 22$ people want to stand in a single queue (a line with a well-defined front/head and back/end). Every individual is a distinct person (i.e., they are all different people) Determine the number of ways to arrange them in the queue for each of the following conditions. No argument is needed. There is no need to calculate the exact numerical answer. You are not allowed write the answer in any form of summation.
>
> (b) [3 points] Every teacher has at least one doctor behind (i.e., no teacher is after the last doctor). Note that behind does not necessarily mean the previous person.

Official Answer

Source: physical PDF page 8 (COMP2711_F2025_Final.pdf).

(b) $\binom{22}{12} \times 12! \times 6 \times 9!$

Supplementary Explanation

Answer. $\binom{22}{12}\, 12!\, 6\, 9!$.

Derivation/proof. Choose positions for the 12 engineers and students, and arrange them, contributing $\binom{22}{12}12!$. Consider the relative order of the remaining ten teachers and doctors. If the last one is a teacher, that teacher has no doctor behind. If the last one is a doctor, every teacher has that doctor behind. Choose the last person from the six doctors; the remaining nine people can be arranged arbitrarily in 9! ways. The result is $\binom{22}{12}12! \cdot 6 \cdot 9!$. Doctors may also appear between teachers.

C127

Permutations and Combinations

2025 Fall Midterm | Question 4(c)(ii) | Source page 13 | Difficulty 3 | 2 marks
Q-2025F-MID-P04-c-ii | Answer provenance: official

> (c) [4 point, 2 pts each] Imagine there are two sets $A = \{a_1, a_2, \cdots, a_7\}$ and $B = \{b_1, b_2, \cdots, b_{10}\}$. Answer the following questions:
>
> (ii) In how many ways we can have 5 unordered pairs (e.g., (a_1, b_1) is the same as (b_1, a_1)) such that each pair contains an element from A and an element from B while the elements can not be reused?

Official Answer

Source: physical PDF page 13 (COMP2711_F2025_Midterm.pdf).

> (ii) $\binom{7}{5} \times \binom{10}{5} \times 5!$

Supplementary Explanation

Answer. $\binom{7}{5}\binom{10}{5}5!$.

Derivation/proof. Choose the five participating A elements in $\binom{7}{5}$ ways and five B elements in $\binom{10}{5}$ ways. Put the chosen A elements in label order and assign the chosen B elements by one of 5! permutations. Every pairing uniquely determines both selected subsets and the bijection between them, so the product counts each pairing once with no reused element.

C128

Permutations and Combinations

2025 Spring Final | Question 6(b) | Source page 9 | Difficulty 3 | 6 marks
Q-2025S-FINAL-P06-b | Answer provenance: official

> (b) [6 pts] How many 6-digit natural numbers (from 100000 to 999999) have digits that sum to 11? For example, the number 100019 sums to $1 + 0 + 0 + 0 + 1 + 9 = 11$ and would be counted. Briefly explain why.

Official Answer

Source: physical PDF page 9 (COMP2711_S2025_Final.pdf).

Let the 6-digit number be $\overline{a_1a_2a_3a_4a_5a_6}$, where $a_1 + a_2 + \cdots + a_6 = 11$ and $a_1 \geq 1$. Substitute $a_1 = a_1' + 1$, so $a_1' + a_2 + \cdots + a_6 = 10$, with $a_1', a_2, \ldots, a_6 \geq 0$. The number of non-negative solutions is $\binom{15}{5}$. Subtract invalid cases: 5 cases where any of $a_2, \ldots, a_6 = 10$, and 6 cases where $a_1' = 9$ or 10. Total valid numbers: $\binom{15}{5} - 11 = 2992$.

Supplementary Explanation

Answer. $\binom{15}{5} - 11 = 2992$.

Derivation/proof. Put $b_1 = a_1 - 1$. The digit sum becomes $b_1 + a_2 + \cdots + a_6 = 10$. Without upper bounds, stars and bars gives $\binom{15}{5} = 3003$ solutions. The actual bounds are $b_1 \leq 8$ and $a_i \leq 9$ for the other digits. When $b_1 \geq 9$, subtract nine to leave total one over six positions, yielding six invalid solutions. When any other digit is at least ten, it must be ten and every other position zero; this gives five invalid solutions. No two violations can overlap because the total is only ten. The valid count is $3003 - 6 - 5 = 2992$.

C129

Permutations and Combinations

2025 Spring Final | Question 12(b) | Source page 16 | Difficulty 3 | 4 marks
Q-2025S-FINAL-P12-b | Answer provenance: official

> (b) [4 pts] How many distinct simple undirected graphs exist on 7 labeled vertices $\{a, b, c, d, e, f, g\}$ such that $\deg(a) \geq 4$?

Official Answer

Source: physical PDF page 16 (COMP2711_S2025_Final.pdf), physical PDF page 17 (COMP2711_S2025_Final.pdf).

(b) First Solution: There are 2^{21} possible graphs in total. Among these, exactly $\binom{6}{3} \times 2^{15}$ graphs satisfy $\deg(a) = 3$; the remaining $2^{21} - \binom{6}{3} \times 2^{15}$ graphs are split equally between those with $\deg(a) \geq 4$ and those with $\deg(a) \leq 2$. This symmetry is due to each graph with $\deg(a) \geq 4$ has a complement graph with $\deg(a) \leq 2$. So, the answer to the question is:

$$\frac{2^{21} - \binom{6}{3} \times 2^{15}}{2} = 720896$$

Second Solution: We can count the number of graphs with $\deg(a) = 4$, $\deg(a) = 5$ and $\deg(a) = 6$ directly:

$$\left(\binom{6}{4} + \binom{6}{5} + \binom{6}{6}\right) \cdot 2^{15} = 720896$$

Supplementary Explanation

Answer. $\left(\binom{6}{4} + \binom{6}{5} + \binom{6}{6}\right)2^{15} = 720896$.

Derivation/proof. If $\deg(a) = d$ for $d = 4, 5, 6$, choose its neighbors from the other six vertices in $\binom{6}{d}$ ways. This fixes all potential edges incident to a. Among the remaining six vertices there are $\binom{6}{2} = 15$ possible edges, each independently present or absent, yielding 2^{15} graphs for each neighbor set. The three degree cases are disjoint, giving $(15 + 6 + 1)2^{15} = 22 \cdot 32768 = 720896$. Vertices are labeled, so distinct neighbor sets or edge sets are distinct graphs without division by isomorphism symmetries.

C130

Permutations and Combinations

2024 Fall Midterm | Question 5(a) | Source page 10 | Difficulty 3 | 6 marks
Q-2024F-MID-P05-a | Answer provenance: official

> (a) How many ways are there to place numbers 1 to 6 on a dice so that different rotations are considered equivalent? Explain your solution.

Official Answer

Source: physical PDF page 10 (COMP2711_F2024_Midterm.pdf).

> Answer: (a) Solution 1: Lets say we put "1" on top side. for the opposite side we have 5 possible ways. Now for the 4 remaining values on its side we have $(4-1)!$ different ways to put the numbers as they are in a round situation. So the total answer is $5 \cdot 3!$. Solution 2: we have 6! permutations in general but each case can be duplicate if it can be made from different rotations. So we can fix one side to be on top which as 6 options. And we can now rotate it 4 times clockwise. So we have to divide 6! by $6 \cdot 4$.

Supplementary Explanation

Answer. $5 \cdot 3! = 30$.

Derivation/proof. Rotate any labelled die to place one on the top face. The opposite face can take any of the five remaining numbers. Arrange the four other distinct numbers around the sides, identifying four rotations about the vertical axis: $4!/4 = 3! = 6$ cyclic arrangements. Thus the total is $5 \cdot 6 = 30$. As a check, there are 6! labellings and $6 \cdot 4 = 24$ orientations per rotation class. Distinct labels prevent any nontrivial rotation from fixing a labelling, so $6!/24 = 30$.

C131

Permutations and Combinations

2024 Fall Midterm | Question 5(b) | Source page 10 | Difficulty 3 | 6 marks
Q-2024F-MID-P05-b | Answer provenance: official

> (b) In how many ways we can pick 3 different numbers from $\{1, 2, 3, \ldots, 2n\}$ such that they form arithmetic sequence? Explain your solution. (Arithmetic sequence is a sequence of numbers such that the difference from any succeeding term to its preceding term remains constant throughout the sequence, For example 1, 4, 7 is an arithmetic sequence but 1, 5, 10 is not) Hint: Consider your sorted sequence to be a, b, c. What can you say about the oddity of a and c?

Official Answer

Source: physical PDF page 10 (COMP2711_F2024_Midterm.pdf).

> (b) As $b = \frac{a+c}{2}$, it means $a + c$ must be even. So they are either both odd or both even. b can be uniquely calculated when we have both a and c. Picking two different odd numbers can be done in $\binom{n}{2}$ ways and picking 2 different even numbers also can be done in $\binom{n}{2}$ ways. So the total possible ways are $\binom{n}{2} + \binom{n}{2} = 2\binom{n}{2}$.

Supplementary Explanation

Answer. $2\binom{n}{2} = n(n-1)$.

Derivation/proof. Write $a < b < c$. The condition $b - a = c - b$ is equivalent to $b = (a + c)/2$, so the endpoints have the same parity. There are n odd and n even numbers. Choose two odd endpoints in $\binom{n}{2}$ ways or two even endpoints in $\binom{n}{2}$ ways. Each pair determines a unique integer midpoint b, strictly between them and still in $1, \ldots, 2n$. Conversely, each valid three-element subset determines a unique endpoint pair. Thus the count is $2\binom{n}{2}$, without a permutation factor.

C132

Permutations and Combinations

2024 Fall Midterm | Question 9(a) | Source page 14 | Difficulty 3 | 7 marks
Q-2024F-MID-P09-a | Answer provenance: official

> (a) Consider the set $\{1, 2, \ldots, 2n-1\}$. Show that the number of subsets with at most $n-1$ elements is equal to $\frac{2^{2n-1}}{2}$. Hint: Think about the size of any random subset and the size of its complement.

Official Answer

Source: physical PDF page 14 (COMP2711_F2024_Midterm.pdf).

> Answer: For the first part note that the total number of subsets of $\{1, 2, \cdots, 2n-1\}$ is 2^{2n-1}. For each subset with at most $n-1$ elements, there is a complementary subset with n or more elements. Since the total number of subsets is 2^{2n-1} and they come in complementary pairs, exactly half, $\frac{2^{2n-1}}{2}$, have at most $n-1$ elements.

Supplementary Explanation

Answer. $2^{2n-1}/2 = 2^{2n-2}$.

Derivation/proof. Let U have $2n-1$ elements. Pair each subset A with its complement $U \setminus A$; their sizes sum to $2n-1$. If $|A| \le n-1$, the complement has at least n elements, and conversely a subset of size at least n has a complement of size at most $n-1$. Taking the complement twice returns A, and no subset equals its complement in a nonempty universe. Thus all 2^{2n-1} subsets form disjoint two-element pairs. Each pair contains exactly one small subset, giving half the total.

C133

Permutations and Combinations

2024 Spring Final | Question 4 | Source page 4 | Difficulty 3 | 10 marks
Q-2024S-FINAL-P04 | Answer provenance: official

> You want to walk from the point $(0,0)$ to the point $(8,8)$ on a square grid, where each point on the grid has integer coordinates. In each step, you can move from a point (i,j) to either $(i+1,j)$ or $(i,j+1)$, which means you can move one unit in the positive x-direction or one unit in the positive y-direction, respectively. The walk will take exactly 16 unit steps in total. Additionally, your path must include the grid position $(2,3)$ and you must avoid the grid position $(6,5)$. How many different paths can you take from $(0,0)$ to $(8,8)$ while satisfying these conditions?

Official Answer

Source: physical PDF page 4 (COMP2711_S2024_Final.pdf), physical PDF page 5 (COMP2711_S2024_Final.pdf).

> Answer: We say that a path is monotonic if at each step it either moves one unit in the positive x-direction or one unit in the positive y-direction. Let P be a path satisfying the requirements of the problem. Since it must include the point $(2,3)$, we can decompose P into two monotonic paths P_1 and P_2 as follows:
>
> $$\underbrace{(0,0) \to \cdots \to (2,3)}_{P_1} \underbrace{\to \cdots \to (8,8)}_{P_2}.$$
>
> Our answer will be $\#P_1 \cdot \#P_2$, where $\#P_1$ ($\#P_2$) is the number of possibilities for P_1 (P_2). It is clear that the requirement of P avoiding $(6,5)$ only restricts the possibilities for P_2 but not P_1. Thus, $\#P_1$ is simply the number of all monotonic paths from $(0,0)$ to $(2,3)$, which is given by $\binom{2+3}{2} = \binom{5}{2}$. We can count $\#P_2$ as $\#P_2 = \#P_{2,\text{all}} - \#P_{2,\text{bad}}$ where $\#P_{2,\text{all}}$ is the number of all monotonic paths from $(2,3)$ to $(8,8)$ and $\#P_{2,\text{bad}}$ is the number of 'bad' monotonic paths from $(2,3)$ to $(8,8)$ that go through $(6,5)$. Using the same reasoning for counting $\#P_1$, we get that:
>
> $$\#P_{2,\text{all}} = \binom{(8-2)+(8-3)}{8-2} = \binom{11}{6}.$$
>
> Similar to P, the paths counted by $\#P_{2,\text{bad}}$ have the form:
>
> $$\underbrace{(2,3) \to \cdots \to (6,5)}_{P_3} \underbrace{\to \cdots \to (8,8)}_{P_4},$$
>
> where P_3 and P_4 are monotonic paths. The number of possibilities for P_3 is $\binom{(6-2)+(5-3)}{6-2} = \binom{6}{4}$, and the number of possibilities for P_4 is $\binom{(8-6)+(8-5)}{8-6} = \binom{5}{2}$. Overall, we get that $\#P_{2,\text{bad}} = \binom{11}{6} - \binom{6}{4} \cdot \binom{5}{2}$, which implies that our final answer is: $\binom{5}{2} \cdot \left[\binom{11}{6} - \binom{6}{4} \cdot \binom{5}{2}\right]$.

Supplementary Explanation

Answer. $\binom{5}{2}\left[\binom{11}{6} - \binom{6}{4}\binom{5}{2}\right] = 3120$.

Derivation/proof. From $(0,0)$ to $(2,3)$ there are two right steps and three up steps, giving $\binom{5}{2} = 10$ paths. From there to $(8,8)$ there are six right steps and five up steps, giving $\binom{11}{6} = 462$ paths. A bad continuation through $(6,5)$ consists of four right and two up steps, then two right and three up steps. Their counts are $\binom{6}{4} = 15$ and $\binom{5}{2} = 10$, so there are 150 bad continuations. There are $462 - 150 = 312$ permissible continuations and $10 \cdot 312 = 3120$ complete paths. The unique decomposition at each specified point prevents duplication or omission.

C134

Permutations and Combinations

2024 Spring Final | Question 7 | Source page 8 | Difficulty 3 | 10 marks
Q-2024S-FINAL-P07 | Answer provenance: official

> A binary string is a string consisting of 0s and 1s. A run is a maximal substring of consecutive 1s. For example, the binary string 11010111011 has 4 runs, which are underlined. Given positive integers n, m, and k, determine the number of binary strings that simultaneously satisfy all of the following three conditions: the string contains exactly m 0s, the string contains exactly n 1s, and the string has exactly k runs of consecutive 1s.

Official Answer

Source: physical PDF page 8 (COMP2711_S2024_Final.pdf).

> Solution: We first lay out all the m 0s which are surrounded by boxes that will later hold 1s:
>
> $$\underbrace{\square\, 0\, \square\, 0\, \cdots\, 0\, \square\, 0\, \square}_{m \text{ 0s and } m+1 \text{ squares}}.$$
>
> Note that we have boxes between 0s, as well as to the left of the leftmost 0 and to the right of the rightmost 0. We define a box to hold an arbitrary number of 1s. Observe the following: any string with k runs chooses exactly k out of the $m+1$ boxes and maps the k runs to the k boxes by putting at least one 1 in each such box while leaving the remaining $m+1-k$ boxes empty. Thus, we can choose the positions of runs with respect to the 0s in $\binom{m+1}{k}$ ways. After choosing the k boxes that hold the k runs, it remains to distribute the n 1s among the k runs. We resolve this issue using the stars and bars kind of reasoning. By definition, each box must have at least one 1, and we want the sum of the number of 1s in the k boxes to be n. The number of ways to distribute the n 1s over the k runs is, therefore, the same as the number of permutations of $n-k$ stars and $k-1$ bars, which is given by $\binom{(n-k)+(k-1)}{k-1} = \binom{n-1}{k-1}$. Therefore, we conclude that the final answer is $\binom{m+1}{k} \cdot \binom{n-1}{k-1}$.

Supplementary Explanation

Answer. $\binom{m+1}{k}\binom{n-1}{k-1}$ if $1 \le k \le \min(n, m+1)$, and 0 otherwise.

Derivation/proof. The m zeros create $m+1$ gaps including both ends. Exactly k runs means choosing k distinct gaps for nonempty strings of ones, in $\binom{m+1}{k}$ ways. In left-to-right order their lengths are $r_1, \ldots, r_k \ge 1$ with sum n. Choosing $k-1$ breakpoints from the $n-1$ spaces between a row of n ones gives $\binom{n-1}{k-1}$ positive length assignments. Positions and lengths uniquely reconstruct the string, so multiply. If $k > n$ or $k > m+1$, there are insufficient ones or gaps and the count is zero.

C135

Permutations and Combinations

2022 Final Part 2 | Question 6(b) | Source page 3 | Difficulty 3 | 4 marks
Q-2022-FINAL2-P06-b | Answer provenance: official

> There is a random simple undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> If G has exactly m connected components $C_1, C_2, \ldots, C_m$, where C_i is a connected component with v_i vertices, what is the maximum possible value of X?

Official Answer

Source: physical PDF page 4 (2022-final-part-2.pdf).

> (b) For connected component C_i, the maximum number of edges is $\binom{v_i}{2}$, forming a complete graph.
>
> The maximum of X is attained when all components take the maximal value at the same time.
>
> $$\max(X) = \sum_{i=1}^{m} \max(C_i) = \sum_{i=1}^{m} \binom{v_i}{2}.$$

Supplementary Explanation

Answer. $\displaystyle\sum_{i=1}^{m} \binom{v_i}{2}$.

Derivation/proof. No edge joins distinct components. Within the simple component Ci, at most one edge can occur for each unordered pair of its vi vertices, giving at most C(vi,2) edges. Summing these disjoint edge sets bounds X by the displayed value. Attain it by making every component a complete graph K_(vi), which is connected for vi≥1 and creates no cross edges. Hence this is the exact maximum for the supplied component sizes, not a further optimization over those sizes.

The bound is $X \le \sum_{i=1}^{m} \binom{v_i}{2}$. Complete each component to K_{v_i}, including singletons, to attain it. The given sizes cannot be reallocated. The printed answer itself uses a summation despite the paper's general no-summation instruction; that source formatting inconsistency does not change the extremal value.

C136

Permutations and Combinations

2022 Final Part 2 | Question 6(c) | Source page 4 | Difficulty 3 | 4 marks
Q-2022-FINAL2-P06-c | Answer provenance: official

> There is a random simple undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the probability that G with $n = 4$ vertices has 3 edges and is not connected.

Official Answer

Source: physical PDF page 4 (2022-final-part-2.pdf).

> (c) The simple graph G with 4 vertices and 3 edges is not connected if and only if three edges form a triangle and the last vertex is disconnected, which happens with the probability $\binom{4}{3}p^3(1-p)^3$.

Supplementary Explanation

Answer. $4p^3(1-p)^3$.

Derivation/proof. A disconnected four-vertex simple graph with three edges cannot split2+2, since that allows at most two edges. Its nontrivial component must have three vertices, and all three possible edges must be present: a triangle. Choose the isolated vertex (or triangle vertices) in 4 ways. Each exact configuration has three present and three absent edges among K4's six possibilities, hence probability $p^3(1-p)^3$. These four configurations are disjoint, giving the result.

The structure is a triangle plus an isolated vertex among six potential edges. The four configurations are disjoint. This requires absence of all three edges to the fourth vertex, not merely the presence of a triangle.

C137

Permutations and Combinations

2021 Final Part 2 | Question 2 | Source page 1 | Difficulty 3 | 10 marks
Q-2021-FINAL2-P02 | Answer provenance: official

> How many positive integers less than 100,000,000 have the sum of their digits equal to 19?

Official Answer

Source: physical PDF page 1 (2021-final-part-2.pdf), physical PDF page 2 (2021-final-part-2.pdf).

> Let $d_1, d_2, d_3, d_4, d_5, d_6, d_7, d_8$ be the eight digits of the positive integer less than 100,000,000. Assuming each d_i can take any non-negative integers, there are $\binom{8+19-1}{19}$ ways to assign non-negative integers to d_i for every i so that their sum equals to 19. At most one d_i can be greater than 9, because $10 + 10 = 20 > 19$. The integer exceeding 9 can be any of the eight d_i. After selecting one, say d_1, to be exceeding 9, there are $\binom{8+9-1}{9}$ positive integers that the digits sum to 19.
>
> So, the total number of positive integers less than 100,000,000 and have the sum of their digits equal to 19 is $\binom{8+19-1}{19} - 8 \cdot \binom{8+9-1}{9} = 566280$.

Supplementary Explanation

Answer. $\binom{26}{7} - 8\binom{16}{7} = 566280$.

Derivation/proof. Pad every allowed integer to eight digits; digit sum 19 excludes the all-zero integer automatically. Ignoring the digit upper bound, nonnegative solutions of d1+...+d8=19 number C(26,7). If a specified digit is at least10, subtract10 from it, leaving sum9 and C(16,7) solutions. Two digits cannot both be at least10 because their sum would exceed19. The eight bad sets are therefore disjoint, so subtract 8C(16,7) and obtain 566280.

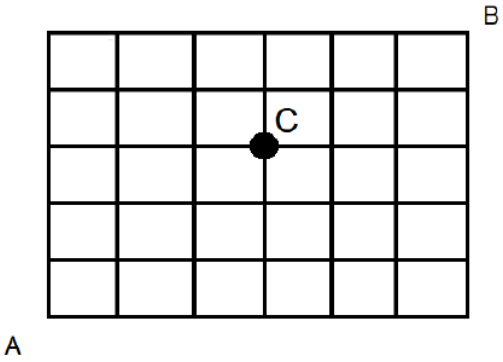
Numerically, $\binom{26}{7} - 8\binom{16}{7} = 657800 - 8 \cdot 11440 = 566280$. Leading zeros create no duplication because each integer has a unique representation padded to exactly eight positions.

The eight bad sets are pairwise disjoint.

C138

Permutations and Combinations

2020 Final Part 1 | Question 3(b) | Source page 2 | Difficulty 3 | 3 marks
Q-2020-FINAL1-P03-b | Answer provenance: official

> You are staying on point A and you are allowed to walk either up or to the right. Answer these questions:
>
> 
>
>
> In how many ways you can go to point B without passing through point C?

Official Answer

Source: physical PDF page 2 (2020-final-part-1.pdf).

$\binom{11}{5} - \binom{6}{3} \cdot \binom{5}{3}$.

Supplementary Explanation

Answer. $\binom{11}{5} - \binom{6}{3}\binom{5}{3} = 262$.

Derivation/proof. There are $\binom{11}{5} = 462$ unrestricted paths. From A to C the figure requires three right and three up moves, giving $\binom{6}{3} = 20$ prefixes. From C to B it requires three right and two up moves, giving $\binom{5}{3} = 10$ suffixes. A path through C uniquely decomposes into one prefix and one suffix, so exactly 200 pass through C. Subtracting leaves 262.

All paths number 462, and the forbidden-point paths number 200 by the unique prefix/suffix split. Their difference is $462 - 200 = 262$.

C139

Permutations and Combinations

2020 Final Part 2 | Question 5(d) | Source page 4 | Difficulty 3 | 3 marks
Q-2020-FINAL2-P05-d | Answer provenance: official

> There is a random undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the probability that G with $n = 4$ vertices is connected and X takes its minimal possible value?

Official Answer

Source: physical PDF page 4 (2020-final-part-2.pdf).

G is connected and X is minimal if and only if G is a tree. The probability of having 3 edges in G is $\binom{6}{3}p^3(1-p)^3$. However, if the 3 edges form a triangle, then the last vertex will be disconnected, which happens with probability $\binom{4}{3}p^3(1-p)^3$. So, the final probability is

$$\binom{6}{3}p^3(1-p)^3 - \binom{4}{3}p^3(1-p)^3 = 16p^3(1-p)^3.$$

Supplementary Explanation

Answer. $16p^3(1-p)^3$.

Derivation/proof. A connected four-vertex graph has at least three edges, achieved exactly by trees. There are six potential edges and $\binom{6}{3} = 20$ three-edge subsets. A disconnected three-edge simple graph on four vertices must be a triangle plus an isolate: a 2+2 split permits at most two edges. There are $\binom{4}{3} = 4$ such triangles, leaving 16 trees. Each exact edge set has probability $p^3(1-p)^3$, so multiply by 16.

To verify the disconnected exception is exhaustive, a partition into two pairs permits at most two edges; finer partitions permit no more. Thus a disconnected three-edge graph must consist of a triangle and an isolated vertex.

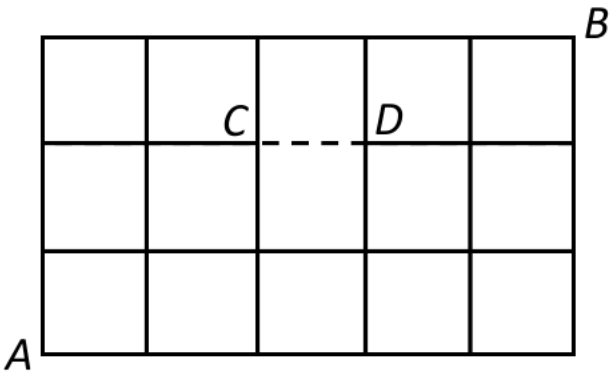
Of the twenty three-edge subsets, exactly four form a triangle plus an isolated vertex. Removing them leaves $20 - 4 = 16$ trees, each with the same probability $p^3(1-p)^3$.

C140

Permutations and Combinations

2018 Final | Question 11(b) | Source page 16 | Difficulty 3
Q-2018-FINAL-P11-b | Answer provenance: official

In the following shape, consider paths that start from A and finish at B following the lines. Every path may only have steps going up or right. E.g., one path would be $\{\uparrow, \uparrow, \uparrow, \rightarrow, \rightarrow, \rightarrow, \rightarrow, \rightarrow\}$. How many distinct such paths are there if:



b) The line from C to D (the dashed line in the figure below) is missing, so paths cannot pass between C and D?

Official Answer

Source: physical PDF page 16 (2018-final.pdf).

Having removed this edge, it is no longer possible to reach D from C, since down movements are not possible. Therefore, we need to subtract from the total number of paths, those that go through both C and D. There are $C(4,2) = 6$ paths from A to C, 1 path from C to D, and $C(3,2)$ from D to B, which means there are $6 \times 1 \times 3$ paths from A to B that pass through C and D. So there are $56 - 18 = 38$ from A to B that don't pass between use the edge between C and D.

Supplementary Explanation

Answer. $56 - 6 \cdot 1 \cdot 3 = 38$.

Derivation/proof. Count paths using the missing edge and subtract. Reaching C requires two right and two up steps, giving $\binom{4}{2} = 6$ prefixes. The step $C \to D$ is forced. From D to B, two right and one up step give $\binom{3}{2} = 3$ suffixes. Every forbidden path has this unique decomposition and cannot use the edge twice, so there are 18 forbidden paths. Subtracting from 56 gives 38; a grid dynamic-programming count independently agrees.

A monotone path cannot traverse the forbidden edge twice, so the prefix-edge-suffix count has no duplicate paths.

Every path through the forbidden point splits uniquely into a prefix reaching it and a suffix leaving it. Subtracting those eighteen paths from all fifty-six gives $56 - 18 = 38$ valid paths.

C141

Permutations and Combinations

2016 Final | Question 7(a) | Source page 13 | Difficulty 3
Q-2016-FINAL-P07-a-proof | Answer provenance: official

> Assume that a party of n people wants to select some of their members to form a committee. The committee must have at least one member. Let $T(n)$ be the total number of possible committees that can be formed from a party of n people.
>
> (a) Define $T(n)$ as a recurrence equation, and prove that your recurrence equation is correct by induction.
>
> Atomic focus [editorial]: prove the recurrence counts committees; defining it is a separate atom.

Official Answer

Source: PDF physical page 13 (2016-final.pdf).

(a) $T(1) = 1$, and $T(n) = 2T(n-1) + 1$ for $n > 1$.

The base case $T(1)$ is obviously true.

Assume that $T(n-1)$ counts the number of committees that are formed from $n-1$ people correctly for $n > 1$.

When there are n people, the number of possible committees that are formed without the n-th person is counted by $T(n-1)$. On the other hand, each committee without the n-th person union with the n-th person forms another possible committee. So, we have $2T(n-1)$ possible committees. There is only one case is missed, which is the committee that contains only the n-th person. Therefore, when there are n people, $T(n) = 2T(n-1) + 1$ counts all possible committees with at least one member from n people. By the principle of induction, we have proved that the recurrence equation is correct.

Supplementary Explanation

Answer. The recurrence $T(1) = 1$, $T(n) = 2T(n-1) + 1$ counts all allowed committees for every $n \geq 1$.

Derivation/proof. Base case: for one person, $\{1\}$ is the only nonempty committee, so $T(1) = 1$. Inductive hypothesis: $T(n-1)$ correctly counts nonempty committees on the first $n-1$ people. At size n, committees excluding n are counted by $T(n-1)$. Those containing n and at least one old member correspond bijectively to a nonempty old committee and give another $T(n-1)$. The sole remaining committee is $\{n\}$. The disjoint cases exhaust all possibilities, so $2T(n-1) + 1$ is correct. Induction completes the proof.

Removing the new person gives an old nonempty committee whenever the original committee also contains an old member; adding that person back is the inverse map.

C142

Permutations and Combinations

2016 Midterm | Question 2(b) | Source page 4 | Difficulty 3 | 6 marks
Q-2016-MID-P02-b | Answer provenance: official

> Consider a university with 300 faculties, 500 teaching assistants, 600 postgraduate students and 10,000 undergraduate students. A tennis club would like to invite a group of guests to join an activity. The group of guests must consist of 4 faculties, 10 teaching assistants, 10 postgraduate students and 40 undergraduate students. Assume that each person initially has only one role (i.e. no one person can be both a postgraduate student and a teaching assistant, etc).
>
> (b) Now, one postgraduate student has a special treatment. This postgraduate student could also be accepted to be a teaching assistant (i.e., this particular person has two roles, a postgraduate student and a teaching assistants). The tennis club can invite this guy as a role of either a teaching assistant or a postgraduate student (but not both), or does not invite this guy at all. How many possible groups of guests are there?

Official Answer

Source: PDF physical page 4 (2016-midterm.pdf).

(b) The special guy can be in the group as a postgraduate student or not in the group at all, this is the same as part (a). In addition, this special guy can be in the group as a teaching assistant. In this case, we only need 9 more teaching assistants, and the total number of possible postgraduate students becomes 599 because the special guy already in the group as teaching assistant. Therefore, we have

$$\binom{300}{4}\binom{500}{10}\binom{600}{10}\binom{10000}{40}+\binom{300}{4}\binom{500}{9}\binom{599}{10}\binom{10000}{40}$$

possible groups of guests.

Another possible solution is

$$\binom{300}{4}\binom{500}{10}\binom{599}{9}\binom{10000}{40}+\binom{300}{4}\binom{501}{10}\binom{599}{10}\binom{10000}{40}.$$

Supplementary Explanation

Answer. $\binom{300}{4}\binom{10000}{40}\left[\binom{500}{10}\binom{600}{10}+\binom{500}{9}\binom{599}{10}\right]$.

Derivation/proof. Partition by whether the special person serves as an assistant. If not, the original count allows that person either as a postgraduate or absent. If yes, choose 9 ordinary assistants from 500 and 10 other postgraduates from 599. Faculty and undergraduate selections are unchanged. These cases are disjoint, and the distinguished role is determined by the other attendees' role counts.

The original count already includes that person as a postgraduate or omits that person; those cases must not be added again.

C143

Permutations and Combinations

2015 Midterm | Question 6(d) | Source page 11 | Difficulty 3 | 5 marks
Q-2015-MID-P06-d | Answer provenance: official

> In a computer system, a password is a string of 8 characters, where each character is either a digit or a lowercase letter from the English alphabet. How many possible passwords are there under each of the following conditions?
>
> (d) The first character must be a letter, and there must be exactly two digits, which cannot be next to each other.
>
> Briefly explain how your answers are derived.

Official Answer

Source: PDF physical page 11 (2015-midterm.pdf).

(d) Let us create a password in two steps. (1) Choose 6 letters, and arrange them on a line such that there is a space between each pair and there is a space at the end:

$$L - L - L - L - L - L -$$

(2) Choose two digits, place them in two of the spaces, and remove the remaining spaces. There are 26^6 possibilities for step 1, and $10^2 \times \binom{6}{2}$ possibilities for step 2. Hence the total number of passwords is $26^6 \times 10^2 \times \binom{6}{2}$.

Supplementary Explanation

Answer. $26^6 \cdot 10^2 \binom{6}{2}$.

Derivation/proof. Choose the ordered sequence of six letters in 26^6 ways. Its five internal gaps and one trailing gap are eligible; the leading gap is forbidden. Choose two of these six gaps and put one digit in each, in 10^2 ways. Deleting the digits recovers the letter sequence and chosen gaps uniquely, so no password is double counted. As a check, choosing two nonconsecutive positions from 2–8 gives $\binom{7}{2} - 6 = 15 = \binom{6}{2}$.

C144

Permutations and Combinations

2015 Midterm | Question 8(b) | Source page 13 | Difficulty 3 | 5 marks
Q-2015-MID-P08-b | Answer provenance: official

> A shopping bag contains 12 pair of gloves. Each pair consists of two different gloves, one for the left hand and another for the right hand, and the two gloves are in the same color. Moreover, different pairs are in different colors. You blindly take 6 gloves (not 6 pairs of gloves) from the bag. How many possible ways are there if
>
> (b) the number of colors of these 6 gloves is 4?
>
> Briefly explain how your answers are derived.

Official Answer

Source: PDF physical page 13 (2015-midterm.pdf).

(b) The number of colors of these 6 gloves is 4 means firstly you pick 2 pairs of gloves out of 12 so there are $\binom{12}{2}$ possibilities. Then the remaining 2 gloves are from 2 different pairs out of the remaining 10 pairs. You choose only one glove from each pair. So there are $\binom{10}{2} \cdot 2 \cdot 2$ possibilities. Overall there are $\binom{12}{2}\binom{10}{2} \cdot 2 \cdot 2$ cases.

Supplementary Explanation

Answer. $\binom{12}{2}\binom{10}{2}2^2 = 11880$.

Derivation/proof. Let d be the number of doubled colors and s singleton colors. Then $d + s = 4$ and $2d + s = 6$, so $d = s = 2$. Choose doubled colors in $\binom{12}{2}$ ways, singleton colors from the rest in $\binom{10}{2}$ ways, then each singleton glove in two ways. This decomposition is unique.

Subtracting $d + s = 4$ from $2d + s = 6$ gives $d = 2, s = 2$.

C145

Permutations and Combinations

2014 Midterm | Question 1(c) | Source page 3 | Difficulty 3
Q-2014-MID-P01-c | Answer provenance: official

> A by-election is being held to fill 5 seats in a council that have recently become vacant. A total of 9 candidates have registered to run in the election. The candidates are divided into 3 groups based on their political affiliations with: 4 in Group A, 3 in Group B, and 2 in Group C.
>
> (c) In how many ways can a voter choose 5 candidates such that at least 1 candidate from each group is chosen?
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 3 (2014-midterm.pdf).

(c) In this case we need to partition into all the possible scenarios of choosing from (A, B, C), which are $(1,2,2), (1,3,1), (2,1,2), (2,2,1), (3,1,1)$. This gives

$$\binom{4}{1}\binom{3}{2}\binom{2}{2} + \binom{4}{1}\binom{3}{3}\binom{2}{1} + \binom{4}{2}\binom{3}{1}\binom{2}{2} + \binom{4}{2}\binom{3}{2}\binom{2}{1} + \binom{4}{3}\binom{3}{1}\binom{2}{1}$$

or

$$4 \cdot 3 \cdot 1 + 4 \cdot 1 \cdot 2 + 6 \cdot 3 \cdot 1 + 6 \cdot 3 \cdot 2 + 4 \cdot 3 \cdot 2 = 98$$

Supplementary Explanation

Answer. 98.

Derivation/proof. The positive triples (A, B, C) summing to five within capacities $(4, 3, 2)$ are $(1,2,2), (1,3,1), (2,1,2), (2,2,1), (3,1,1)$. Summing the corresponding products of binomial coefficients gives $12 + 8 + 18 + 36 + 24 = 98$.

For a triple (a, b, c), the count is $\binom{4}{a}\binom{3}{b}\binom{2}{c}$. The five cases are mutually exclusive and cover all choices.

C146

Permutations and Combinations

2014 Midterm | Question 3 | Source page 5 | Difficulty 3 | 11 marks
Q-2014-MID-P03 | Answer provenance: official

> Give a combinatorial proof of the identity, for all $n \geq 13$.
>
> $$\binom{n}{4}\binom{n-4}{6}\binom{n-10}{3} = \binom{n}{6}\binom{n-6}{3}\binom{n-9}{4}.$$
>
> Note: An algebraic proof of this identity will not be accepted as a solution.

Official Answer

Source: PDF physical page 5 (2014-midterm.pdf).

Consider the problem of "how to color n items so that 4 are red, 6 are green, 3 are blue and the remaining $n - 13$ are yellow.

The left hand side of the inequality obviously counts this. Notice that the problem is the same if we change the order in which we ask the question.

Consider the problem of how to color n items so that 6 are green, 3 are blue, 4 are red and the remaining $n - 13$ are yellow. This is what the right hand side of the equation is counting, so the two sides are the same.

Supplementary Explanation

Answer. Both sides count colourings with 4 red, 6 green, 3 blue, and $n - 13$ yellow items.

Derivation/proof. On the left choose red then green then blue; on the right choose green then blue then red. Each procedure selects exactly the same labelled colour assignment, so the counts agree.

The two orders give $\binom{n}{4}\binom{n-4}{6}\binom{n-10}{3}$ and $\binom{n}{6}\binom{n-6}{3}\binom{n-9}{4}$. Every final colouring occurs exactly once in each process; $n \geq 13$ ensures all selections are possible.

C147

Permutations and Combinations

2013 Midterm | Question 3 | Source page 5 | Difficulty 3 | 10 marks
Q-2013-MID-P03 | Answer provenance: official

> Let n be a positive integer. Give a combinatorial proof of the following equation:
>
> $$\binom{2n}{2} = 2\binom{n}{2} + n^2.$$
>
> Note: An algebraic proof will not be accepted as a correct solution.

Official Answer

Source: PDF physical page 5 (2013-midterm.pdf).

- Left side: select two items from $2n$ item.
- Right side: split $2n$ items into left half and right half each contain n items. We have $\binom{n}{2}$ ways to select two items both are in the left half, $\binom{n}{2}$ ways to select two items both are in the right half, and n^2 ways to select two items such that one in left half and the other in right half. That's all possible cases, so the number of ways to select two items from $2n$ items is $2\binom{n}{2} + n^2$.

Both left and right hand side are counting the number of ways to select two items from $2n$ items, so they are the same.

Supplementary Explanation

Answer. Both sides count 2-element selections from two labelled groups of n objects.

Derivation/proof. Partition $2n$ objects into left and right groups of size n. A pair lies wholly left ($\binom{n}{2}$ choices), wholly right ($\binom{n}{2}$ choices), or has one object in each group (n^2 choices). These cases are exhaustive and disjoint, proving the identity.

Direct counting gives $\binom{2n}{2}$, so $\binom{2n}{2} = 2\binom{n}{2} + n^2$.

C148

Permutations and Combinations

2013 Midterm | Question 8 | Source page 13 | Difficulty 3 | 12 marks
Q-2013-MID-P08 | Answer provenance: official

> In how many different ways can we write 100 as the sum of three non-negative integers? In other words, how many lists $x_1x_2x_3$ of three non-negative integers x_1, x_2 and x_3 are there such that $x_1 + x_2 + x_3 = 100$?
>
> You might need to use the Bijection principle for this question. Clearly state the two sets involved, and the function between the two sets.

Official Answer

Source: PDF physical page 13 (2013-midterm.pdf).

The collection that we need to count is:

$$S = \{x_1x_2x_3 \mid \text{the } x_i \geq 0\text{'s are integers such that } x_1 + x_2 + x_3 = 100\}.$$

This set is difficult to count directly. To get another set that is easy to count, consider arranging 100 balls on one line and dividing the line of balls into three segments by inserting two separators between the balls. Some of the segments might be empty. Let T be the set of all possible ways to place the two separators.

To define a function from T to S, consider one particular way to place the two separators. We can get three non-negative integers as follows: Let x_1 be the number of balls one the left side of the first separator, x_2 be the number of balls between the two separators, and x_3 be the number of balls on the right side of the second separator. It is clear that $x_1 + x_2 + x_3 = 100$. It is also obvious that the function is a bijection.

To count T, consider placing the 100 balls and the two separators on one line. There are totally 102 positions. The two separators can be placed at any two of those positions. Therefore, the size of T is:

$$\binom{102}{2}.$$

Because there is a bijection between S and T, the size of S is the same.

Supplementary Explanation

Answer. $\binom{102}{2} = 5151$.

Derivation/proof. Map a triple to 100 stars separated into blocks of sizes x_1, x_2, x_3 by two bars, allowing adjacent bars and end bars. This is a bijection with placements of two bars among 102 positions, hence $\binom{102}{2}$.

Explicitly, S is the set of ordered nonnegative triples of sum 100, and T is the set of length-102 strings with 100 identical stars and two identical bars. Counting stars in the three blocks inverts the map, proving bijectivity. Thus $|S| = |T| = \binom{102}{2} = 5151$.

C149

Permutations and Combinations

2025 Fall Final | Question 4(c) | Source page 8 | Difficulty 4 | 3 marks
Q-2025F-FINAL-P04-c | Answer provenance: official

> Assume there are 4 teachers (T), 5 engineers (E), 6 doctors (D) and 7 students (S). These $4 + 5 + 6 + 7 = 22$ people want to stand in a single queue (a line with a well-defined front/head and back/end). Every individual is a distinct person (i.e., they are all different people) Determine the number of ways to arrange them in the queue for each of the following conditions. No argument is needed. There is no need to calculate the exact numerical answer. You are not allowed write the answer in any form of summation.
>
> (c) [3 points] The first teacher and first engineer are standing next to each other (By first teacher and first engineer we mean the earliest in the queue toward its front).

Official Answer

Source: physical PDF page 8 (COMP2711_F2025_Final.pdf).

(c) $\binom{21}{13} \times 13! \times (4 \times 5 \times 2) \times 7!$

Supplementary Explanation

Answer. $\binom{21}{13}\, 13!\, (4 \times 5 \times 2)\, 7!$.

Derivation/proof. Choose the earliest teacher in four ways, the earliest engineer in five ways, and their internal adjacent order in two ways: $4 \cdot 5 \cdot 2 = 40$. Compression leaves 21 objects: 13 doctors and students, the block, and seven other teachers and engineers. Choose and fill the positions of the 13 unrestricted people in $\binom{21}{13} 13!$ ways. Of the eight remaining positions, the earliest must contain the block, or some teacher or engineer would appear before the selected earliest pair. Arrange the remaining seven people in 7! ways. Expanding the block uniquely restores a queue with the selected pair actually adjacent, so the product counts every valid queue exactly once.

C150

Binomial Coefficients

2017 Midterm | Question 8(b) | Source page 10 | Difficulty 1
Q-2017-MID-P08-b | Answer provenance: official

> (b) Find the number of terms in the expansion of $(5a + 8b)^{15}$.

Official Answer

Source: physical PDF page 10 (2017-midterm.pdf).

16.

Supplementary Explanation

Answer. 16.

Derivation/proof. The terms are $\binom{15}{k}(5a)^{15-k}(8b)^k$ for $k = 0, 1, \ldots, 15$. Every coefficient is nonzero and each exponent pair $(15-k, k)$ is different, so no terms combine. The inclusive range has sixteen integers.

In the binomial expansion the term indexed by k is $\binom{15}{k}5^{15-k}8^k a^{15-k}b^k$. Matching the requested exponent of b fixes k and therefore its coefficient uniquely.

C151

Binomial Coefficients

2024 Fall Midterm | Question 9(b) | Source page 14 | Difficulty 2 | 5 marks
Q-2024F-MID-P09-b | Answer provenance: official

> (b) Use combinatorial argument to prove:
>
> $$2^{2n-2} = \sum_{r=0}^{n-1} \binom{2n-1}{r}.$$
>
> Hint: You may use part (a), even if you do not manage to prove it.

Official Answer

Source: physical PDF page 14 (COMP2711_F2024_Midterm.pdf).

> For the second part, the LHS is the number of subsets of $\{1, 2, 3, \cdots, 2n-1\}$ with at most $n-1$ elements. The RHS counts the number of subsets that have exactly 0 elements, 1 element, 2 elements, $\cdots$, $n-1$ elements separately and adds them together.

Supplementary Explanation

Answer. $2^{2n-2} = \sum_{r=0}^{n-1} \binom{2n-1}{r}$.

Derivation/proof. Consider subsets of $U = \{1, \ldots, 2n-1\}$ with at most $n-1$ elements. Part (a), which may be used directly, counts them as $2^{2n-1}/2 = 2^{2n-2}$, the left side. For fixed size r, choosing r of the $2n-1$ elements gives $\binom{2n-1}{r}$. These size classes are disjoint, and $r = 0, \ldots, n-1$ covers all desired subsets. Adding gives the right side. Both methods count the same collection of subsets exactly once, proving the identity.

C152

Binomial Coefficients

2018 Final | Question 4(b) | Source page 6 | Difficulty 2
Q-2018-FINAL-P04-b | Answer provenance: official

> (b) Fill in the following recursive definition so that $f(n) = \binom{n}{m}$, the binomial coefficient for fixed m.
>
> $f(m) = 1$
>
> $f(k+1) = f(k)$ ____
>
> For this question, no proof is required.

Official Answer

Source: physical PDF page 6 (2018-final.pdf).

$$f(k+1) = f(k) \cdot \frac{k+1}{k+1-m}.$$

Supplementary Explanation

Answer. $f(k+1) = f(k)\frac{k+1}{k+1-m}$ for $k \geq m$.

Derivation/proof. Taking the ratio gives $\binom{k+1}{m}/\binom{k}{m} = [(k+1)!/(m!(k+1-m)!)]/[k!/(m!(k-m)!)] = (k+1)/(k+1-m)$. The denominator is positive for $k \geq m$, and the initial value $\binom{m}{m} = 1$ matches the given basis.

The recurrence multiplier is $\dfrac{k+1}{k+1-m}$, obtained by cancelling factorials in successive binomial coefficients. The boundary value is $f(m) = \binom{m}{m} = 1$, so the rule begins without a zero denominator.

C153

Binomial Coefficients

2017 Midterm | Question 8(a) | Source page 10 | Difficulty 2
Q-2017-MID-P08-a | Answer provenance: official

> (a) Find the coefficient of x^7y^5 in the expansion of $(3x - y)^{12}$.

Official Answer

Source: physical PDF page 10 (2017-midterm.pdf).

$-\binom{12}{7}3^7$.

Supplementary Explanation

Answer. $-\binom{12}{7}3^7$.

Derivation/proof. To obtain x^7y^5, choose $3x$ from seven of the twelve factors and $-y$ from the other five. There are $\binom{12}{7}$ choices of the seven positions. Their scalar contribution is $3^7(-1)^5 = -3^7$, giving the stated coefficient.

C154

Binomial Coefficients

2017 Midterm | Question 11 | Source page 13 | Difficulty 2 | 10 marks
Q-2017-MID-P11 | Answer provenance: official

> Give a combinatorial proof of $\binom{n}{k}\binom{k}{1} = \binom{n}{1}\binom{n-1}{k-1}$.

Official Answer

Source: physical PDF page 13 (2017-midterm.pdf).

$\binom{n}{k}\binom{k}{1}$: $\binom{n}{k}$ is the number of ways to select k people into a club from n people. $\binom{k}{1}$ is the number of ways to select a leader from the k people club. Therefore, this is the number of ways to form a k people club and assign one of them to be club leader. $\binom{n}{1}\binom{n-1}{k-1}$: $\binom{n}{1}$ is the number of ways to select a person into a club and acts as club leader from n people. $\binom{n-1}{k-1}$ is the number of ways to select $k-1$ people into the club. Therefore, both expression counts the number of ways to form a k people club and assign one of them to be club leader.

Supplementary Explanation

Answer. Both sides count a k-person club with one leader chosen from its members.

Derivation/proof. For $1 \le k \le n$, first choose the club in $\binom{n}{k}$ ways and then its leader in $\binom{k}{1}$ ways. Alternatively, choose the leader first in $\binom{n}{1}$ ways and the remaining $k-1$ members from the other $n-1$ people in $\binom{n-1}{k-1}$ ways. Both constructions give every club-leader pair exactly once, proving equality.

Count a committee with one distinguished member in two orders. Choosing the committee and then its distinguished member, or that member and then the remaining committee, gives $\binom{n}{k}\binom{k}{1} = \binom{n}{1}\binom{n-1}{k-1}$.

C155

Binomial Coefficients

2016 Midterm | Question 3(b) | Source page 5 | Difficulty 2 | 6 marks
Q-2016-MID-P03-b | Answer provenance: official

> Consider the following identity. For any two integers n and k where $0 < k \leq n$,
>
> $$\binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1}.$$
>
> (b) Prove the identity above by means of an algebraic proof.

Official Answer

Source: PDF physical page 6 (2016-midterm.pdf).

(b)

$$\begin{aligned} RHS &= \frac{n}{k}\binom{n-1}{k-1} \\ &= \frac{n}{k} \cdot \frac{(n-1)!}{(k-1)!((n-1)-(k-1))!} \\ &= \frac{n}{k} \cdot \frac{(n-1)!}{(k-1)!(n-k)!} \\ &= \frac{n!}{k!(n-k)!} \\ &= \binom{n}{k} \\ &= LHS \end{aligned}$$

Supplementary Explanation

Answer. $\frac{n}{k}\binom{n-1}{k-1} = \binom{n}{k}$.

Derivation/proof. $\frac{n}{k}\binom{n-1}{k-1} = \frac{n(n-1)!}{k(k-1)!((n-1)-(k-1))!} = \frac{n!}{k!(n-k)!} = \binom{n}{k}$. The specified $0 < k \leq n$ ensures division by k and all factorials are valid.

The final factorial ratio is $n!/[k!(n-k)!]$.

C156

Binomial Coefficients

2025 Fall Final | Question 2(c) | Source page 5 | Difficulty 3 | 4 marks
Q-2025F-FINAL-P02-c | Answer provenance: official

> (c) [4 points] Use combinatorial argument to prove:
>
> $$\binom{n}{k}P(k,r) = \binom{n-r}{k-r}P(n,r)$$

Official Answer

Source: physical PDF page 5 (COMP2711_F2025_Final.pdf), physical PDF page 6 (COMP2711_F2025_Final.pdf).

(c) From a group of n distinct people, choose an ordered committee of r people (positions matter) and also choose an unordered group of k people who will be the full selection committee (ther ordered members are part of these k).

Left-hand side:

- First choose the k members of the full selection committee: $\binom{n}{k}$ ways.
- Then choose r ordered positions (e.g., chairperson, vice-chair, etc.) from these k people: $P(k,r)$ ways. Total: $\binom{n}{k} \cdot P(k,r)$.

Right-hand side:

- First choose the r people who will hold the r ordered positions, in order: $P(n,r)$ ways.
- Now we still need to choose the remaining $k-r$ members of the full (unordered) committee. These must come from the remaining $n-r$ people (excluding the r already chosen for the ordered roles): $\binom{n-r}{k-r}$ ways. Total: $\binom{n-r}{k-r} \cdot P(n,r)$.

Both sides count the same thing, so

$$\binom{n}{k} \cdot P(k,r) = \binom{n-r}{k-r} \cdot P(n,r).$$

Supplementary Explanation

Answer. Both sides count a k-person committee with r ordered offices assigned to distinct committee members.

Derivation/proof. In the usual defined range $0 \le r \le k \le n$, first choose the full committee from n people in $\binom{n}{k}$ ways, and then assign the offices in $P(k,r)$ ways. This gives the left side. Alternatively, assign the r offices from all n people in $P(n,r)$ ways, then choose $k-r$ additional members from the remaining $n-r$ people in $\binom{n-r}{k-r}$ ways. Both constructions are in one-to-one correspondence with the same objects, so their counts agree.

C157

Binomial Coefficients

2025 Fall Midterm | Question 5 | Source page 14 | Difficulty 3 | 10 marks
Q-2025F-MID-P05 | Answer provenance: official

Use combinatorial reasoning to prove the identity.

$$\binom{n}{k} - \binom{n-3}{k} = \binom{n-1}{k-1} + \binom{n-2}{k-1} + \binom{n-3}{k-1}.$$

Official Answer

Source: physical PDF page 14 (COMP2711_F2025_Midterm.pdf).

Let S be a set of n elements, and let a, b, c be distinct elements of S. The number of k-subsets of S is $\binom{n}{k}$, and the number of k-subsets of $S - \{a, b, c\}$ is $\binom{n-3}{k}$. Then the LHS is the number of k-subsets of S that contain at least one of the elements of $\{a, b, c\}$. Such k-subsets can be divided into 3 types: (1) the k-subsets that contain the element a; (2) the k-subsets that do not contain a but contain b; and (3) the k-subsets that do not contain a, b but contain c. The numbers k-subsets of type (1), type (2), type (3) are

$$\binom{n-1}{k-1}, \quad \binom{n-2}{k-1}, \quad \binom{n-3}{k-1}$$

respectively. The sum of these numbers is exactly the RHS.

Supplementary Explanation

Answer. Both sides count the k-element subsets containing at least one of three designated elements.

Derivation/proof. In the range where the binomial coefficients have their usual combinatorial meaning, take an n-element set S with distinct elements a,b,c. From all $\binom{n}{k}$ k-subsets, remove the $\binom{n-3}{k}$ containing none of these elements. Partition the remainder into subsets containing a; subsets not containing a but containing b; and subsets not containing a, b but containing c. The counts are respectively $\binom{n-1}{k-1}$, $\binom{n-2}{k-1}$, and $\binom{n-3}{k-1}$, obtained by choosing the remaining k−1 elements from the allowed pool. The three classes are disjoint and exhaustive, so their sum is the right side and proves the identity. Impossible boundary selections count as zero.

C158

Binomial Coefficients

2024 Spring Final | Question 9(a) | Source page 11 | Difficulty 3 | 5 marks
Q-2024S-FINAL-P09-a | Answer provenance: official

> (a) Let p be a prime number and k be an integer such that $1 \leq k \leq p-1$. Prove that the binomial coefficient $\binom{p}{k}$ is divisible by p.

Official Answer

Source: physical PDF page 11 (COMP2711_S2024_Final.pdf).

> Solution: (a) Consider the binomial coefficient: $\binom{p}{k} = \frac{p!}{k!(p-k)!}$. The numerator $p!$ can be written as $p \cdot (p-1) \cdots 2 \cdot 1$, which is clearly divisible by p. The denominator is the product of two factorials, $k!$ and $(p-k)!$. Since $1 \leq k \leq p-1$, both k and $p-k$ are integers strictly less than p. Thus, the denominator $k!(p-k)!$ is the product of integers all of which are non-zero and strictly less than p. This means that neither $k!$ nor $(p-k)!$ is divisible by p. Since $\binom{p}{k}$ is an integer, and it is expressed as a fraction where the numerator is divisible by p but the denominator is not, we can conclude that $\binom{p}{k}$ itself must be divisible by p.

Supplementary Explanation

Answer. $p \mid \binom{p}{k}$ for $1 \leq k \leq p-1$.

Derivation/proof. Let $C = \binom{p}{k} \in \mathbb{Z}$. Then $C\,k!(p-k)! = p!$, whose right side is divisible by p. Since $k, p-k < p$, every positive denominator factor lies between 1 and $p-1$, so none is divisible by p. The prime product property gives $p \nmid k!(p-k)!$. Applying the same property to $C\,k!(p-k)!$ gives $p \mid C$. At $k = 0, p$ the coefficient is one, showing why these endpoints must be excluded.

C159

Binomial Coefficients

2024 Spring Final | Question 9(b) | Source page 11 | Difficulty 3 | 10 marks
Q-2024S-FINAL-P09-b | Answer provenance: official

(b) Let p be a prime number. Use mathematical induction to prove that for any positive integer a, the number $a^p - a$ is divisible by p. In other words, you need to prove Fermat's Little Theorem using induction on the integer a. Note: If you give the number-theoretic proof of Fermat's Little Theorem presented in class, you will receive 0 marks. Hint: In your proof of part (b), you may find the claim given in part (a) useful. Note that in part (b), you are permitted to use this claim from part (a), even if you were unable to prove it.

Official Answer

Source: physical PDF page 11 (COMP2711_S2024_Final.pdf), physical PDF page 12 (COMP2711_S2024_Final.pdf).

(b) The base case is when $a = 1$, in which case we have $a^p - a = 1^p - 1 = 0$, which is divisible by p, and hence the base case holds. Now supposing that the claim holds for an arbitrary a, i.e., $p \mid a^p - a$, we wish to show that it holds for $a + 1$, i.e., $p \mid (a+1)^p - (a+1)$. Expanding $(a+1)^p - (a+1)$ using the binomial formula, we get

$$\begin{aligned}
(a+1)^p - (a+1) &= \left(\sum_{i=0}^{p} \binom{p}{i} \cdot a^i \cdot 1^{p-i}\right) - (a+1) \\
&= \left(\sum_{i=0}^{p} \binom{p}{i} \cdot a^i\right) - (a+1) \\
&= \left(\sum_{i=1}^{p-1} \binom{p}{i} \cdot a^i\right) + \binom{p}{0} \cdot a^0 + \binom{p}{p} \cdot a^p - (a+1) \\
&= \left(\sum_{i=1}^{p-1} \binom{p}{i} \cdot a^i\right) + 1 + a^p - (a+1) \\
&= \left(\sum_{i=1}^{p-1} \underbrace{\binom{p}{i}}_{\text{divisible by } p \text{ from part (a)}} \cdot a^i\right) + \underbrace{a^p - a}_{\text{divisible by } p \text{ by IH}}.
\end{aligned}$$

Thus, $(a+1)^p - (a+1)$ is equal to the sum of terms each of which is divisible by p, and therefore we have $p \mid (a+1)^p - (a+1)$, which completes our inductive proof.

Supplementary Explanation

Answer. $p \mid a^p - a$ for every positive integer a.

Derivation/proof. Fix the prime p. Base case: $a = 1$ gives $1^p - 1 = 0$, divisible by p. The inductive hypothesis is $p \mid a^p - a$. By the binomial theorem, $(a+1)^p - (a+1) = (a^p - a) + \sum_{i=1}^{p-1} \binom{p}{i} a^i$: the constant one cancels the subtracted one, and $a^p - a$ forms the first term. Part (a) makes every internal coefficient, $1 \le i \le p-1$, divisible by p, so the entire sum is divisible by p. The first term is divisible by p by the inductive hypothesis. Thus $p \mid (a+1)^p - (a+1)$, completing induction without assuming Fermat's theorem.

C160

Binomial Coefficients

2023 Fall Midterm | Question 5 | Source page 12 | Difficulty 3 | 7 marks
Q-2023F-MID-P05 | Answer provenance: official

> Prove the following identity using combinatorial arguments:
>
> $$\sum_{i=0}^{n} \binom{n}{i}^2 = \binom{2n}{n}.$$

Official Answer

Source: physical PDF page 12 (COMP2711_F2023_Midterm.pdf).

> It is equivalent to proving the identity $\sum_{i=0}^{n} \binom{n}{i} \cdot \binom{n}{n-i} = \binom{2n}{n}$. Let us consider a set of $2n$ elements, and we want to select n elements from this set. The right-hand side (RHS) of the equation directly gives us the total number of ways to do this, which is $\binom{2n}{n}$. Now, let us focus on the left-hand side (LHS). We divide the set into two parts, each containing n items. We then select items for each part. If we select i items for the first part, we must select $n - i$ items for the second part. Therefore, there are $\binom{n}{i} \cdot \binom{n}{n-i}$ ways to make these selections. To consider all possible values of i, we sum up these quantities for i ranging from 0 to n. This gives us the sum $\sum_{i=0}^{n} \binom{n}{i} \cdot \binom{n}{n-i}$. Hence, both the RHS and LHS represent the number of ways to select n elements from a set of $2n$ elements, and thus, the equation holds true.

Supplementary Explanation

Answer. Both sides count the same n-element subsets of a fixed $2n$-element set.

Derivation/proof. Split a fixed set of $2n$ distinct elements into disjoint labeled groups A and B, each of size n. Directly choosing an n-element subset gives $\binom{2n}{n}$. Alternatively, classify a chosen subset by $i = |S \cap A|$. There are $\binom{n}{i}$ choices from A and $\binom{n}{n-i}$ from B. The classes $i = 0, \ldots, n$ are disjoint and exhaustive, and $\binom{n}{n-i} = \binom{n}{i}$. Summing yields the left side. Every subset appears once in each counting method, proving the identity.

For fixed i, multiply the two choices. The classes are disjoint and cover every selection.

C161

Binomial Coefficients

2020 Final Part 2 | Question 1 | Source page 1 | Difficulty 3 | 8 marks
Q-2020-FINAL2-P01 | Answer provenance: official

> Prove that $\binom{n}{1} + 2^2\binom{n}{2} + \ldots + n^2\binom{n}{n} = n(n+1)\cdot 2^{n-2}$, where n is a positive integer. [Hint: $i^2 = i(i-1)+i$.]

Official Answer

Source: physical PDF page 1 (2020-final-part-2.pdf); physical PDF page 2 (2020-final-part-2.pdf).

$$\binom{n}{1} + 2^2 \cdot \binom{n}{2} + \cdots + n^2 \cdot \binom{n}{n} = \sum_{i=1}^{n} i^2 \cdot \binom{n}{i} = \sum_{i=1}^{n} i(i-1) \cdot \binom{n}{i} + \sum_{i=1}^{n} i \cdot \binom{n}{i}.$$

We have proved $\sum_{i=1}^{n} i \cdot \binom{n}{i} = n \cdot 2^{n-1}$ in tutorial 9,

$$\begin{aligned}
\sum_{i=1}^{n} i \cdot \binom{n}{i} &= \sum_{i=1}^{n} i \cdot \frac{n!}{i!(n-i)!} \\
&= \sum_{i=1}^{n} \frac{n!}{(i-1)!(n-i)!} \\
&= \sum_{i=1}^{n} n \cdot \frac{(n-1)!}{(i-1)!((n-1)-(i-1))!} \\
&= n \cdot \sum_{i=0}^{n-1} \binom{n-1}{i} \\
&= n \cdot 2^{n-1}.
\end{aligned}$$

And we use the same method to calculate $\sum_{i=1}^{n} i(i-1) \cdot \binom{n}{i}$,

$$\begin{aligned}
\sum_{i=1}^{n} i(i-1) \cdot \binom{n}{i} &= \sum_{i=1}^{n} i(i-1) \cdot \frac{n!}{i!(n-i)!} \\
&= \sum_{i=1}^{n} \frac{n!}{(i-2)!(n-i)!} \\
&= \sum_{i=1}^{n} n(n-1) \cdot \frac{(n-2)!}{(i-2)!((n-2)-(i-2))!} \\
&= n(n-1) \cdot 2^{n-2}.
\end{aligned}$$

Combine the two terms together, we have,

$$\begin{aligned}
\binom{n}{1} + 2^2 \cdot \binom{n}{2} + \cdots + n^2 \cdot \binom{n}{n} &= \sum_{i=1}^{n} i(i-1) \cdot \binom{n}{i} + \sum_{i=1}^{n} i \cdot \binom{n}{i} \\
&= n \cdot 2^{n-1} + n(n-1) \cdot 2^{n-2} \\
&= n(n+1) \cdot 2^{n-2}.
\end{aligned}$$

Supplementary Explanation

Answer. The identity holds for every positive n.

Derivation/proof. For n≥2, use $i\binom{n}{i} = n\binom{n-1}{i-1}$ and $i(i-1)\binom{n}{i} = n(n-1)\binom{n-2}{i-2}$, starting the second sum at i=2. The binomial theorem gives $\sum_{i=1}^{n} i\binom{n}{i} = n2^{n-1}$ and $\sum_{i=2}^{n} i(i-1)\binom{n}{i} = n(n-1)2^{n-2}$. Add them using the hint to obtain $n[2+(n-1)]2^{n-2} = n(n+1)2^{n-2}$. At n=1 both sides equal 1, checked separately so no negative factorial occurs.

C162

Binomial Coefficients

2019 Final | Question 5 | Source page 7 | Difficulty 3 | 8 marks
Q-2019-FINAL-P05 | Answer provenance: official

> Give a combinatorial proof to the following identity:
>
> $$\sum_{i=0}^{n} \binom{n}{i} \sum_{j=0}^{n-i} \binom{n-i}{j} = 3^n.$$

Official Answer

Source: physical PDF page 7 (2019-final.pdf).

The right side is easier to interpret, so we start with that. We take a set of size n, and partition them into 3 sets A, B and C. The RHS is the number of ways to split. For the left side, suppose we already know that we want i elements in A and j elements in B. Then, we have $\binom{n}{i}$ ways to choose the elements of A, and $\binom{n-i}{j}$ ways to choose the elements of B once we have already chosen the elements of A. After we have chosen the elements of A and B, the rest of the elements go to set C. Finally, we have to sum over all of the possible values of i and j.

Supplementary Explanation

Answer. Both sides count assignments of n distinct elements to three labeled groups A, B and C.

Derivation/proof. Each element independently chooses one of A,B,C, giving 3^n. Alternatively fix i elements for A in $\binom{n}{i}$ ways. From the remaining n-i elements choose j for B in $\binom{n-i}{j}$ ways, and assign every remaining element to C. Sum over $0 \le j \le n-i$ and then $0 \le i \le n$. These possibilities are disjoint by the actual groups and exhaust every assignment, proving the identity. Empty groups are allowed on both sides.

The construction allows a group to have size zero: empty groups are allowed. Consequently the count concerns all functions to the labeled groups, rather than only surjective functions.

C163

Binomial Coefficients

2016 Midterm | Question 3(a) | Source page 5 | Difficulty 3 | 9 marks
Q-2016-MID-P03-a | Answer provenance: official

> Consider the following identity. For any two integers n and k where $0 < k \leq n$,
>
> $$\binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1}.$$
>
> (a) Prove the identity above by means of a combinatorial proof.

Official Answer

Source: PDF physical page 5 (2016-midterm.pdf).

(a) $\binom{n}{k} = \frac{n}{k}\binom{n-1}{k-1}$ could be re-written as

$$k\binom{n}{k} = n\binom{n-1}{k-1}$$

That is,

$$\binom{n}{k}\binom{k}{1} = \binom{n}{1}\binom{n-1}{k-1}$$

Consider n distinct objects.

Suppose that we want to distribute these n distinct objets into 3 distinguishable buckets. The first bucket contains 1 object, the second bucket contains $k-1$ objects, and the third bucket contains the remaining objects (i.e., $n-k$ objects).

The total number of ways of distributing the objects into these 3 buckets can be interpreted in the following two methods.

- The first method is to select k objects out of n objects first and to keep the remaining objects (i.e., $n-k$ objects) in the third bucket. Then, select 1 object out of the selected k objects and place it to the first bucket and keep the remaining objects (i.e., $k-1$ objects) in the second bucket. Here, the total number of ways in this method is $\binom{n}{k}\binom{k}{1}$.
- The second method is to select 1 object out of n objects first and put it to the first bucket. Then, we select $k-1$ objects out of the remaining objects (i.e., $n-1$ objects) and put it to the second bucket. All remaining objects (i.e., $n-k$ objects) are put to the third bucket. Here, the total number of ways in the second method is $\binom{n}{1}\binom{n-1}{k-1}$.

Since the first method has the same physical meaning as the second method, we have

$$\binom{n}{k}\binom{k}{1} = \binom{n}{1}\binom{n-1}{k-1}$$

Supplementary Explanation

Answer. $k\binom{n}{k} = n\binom{n-1}{k-1}$, hence the required identity.

Derivation/proof. Count pairs (S, s) with $S \subseteq \{1, \ldots, n\}$, $|S| = k$, and $s \in S$. Choosing S then s gives $\binom{n}{k}k$. Choosing s first gives n choices and then $\binom{n-1}{k-1}$ choices for the other members. Equate the counts and divide by the positive integer k.

The distinguished member, the other selected members, and the unselected objects correspond exactly to the source's three buckets of sizes $1, k-1, n-k$. Since $k > 0$, division by k is legitimate.

C164

Binomial Coefficients

2025 Spring Final | Question 9 | Source page 12 | Difficulty 4 | 7 marks
Q-2025S-FINAL-P09 | Answer provenance: official

> Consider all 1000-element subsets of the set $\{1, 2, 3, \ldots, 2022\}$. From each such subset, choose the least element. Find the sum of all these least elements. Hint: You can use the Hockey Stick Identity introduced in class without proving it:
>
> $$\binom{n+1}{r+1} = \sum_{j=r}^{n} \binom{j}{r}$$

Official Answer

Source: physical PDF page 12 (COMP2711_S2025_Final.pdf).

- Number of subsets with 1 as the least element: $C(2021, 999)$
- Number of subsets with 2 as the least element: $C(2020, 999)$
- ...
- Number of subsets with 1023 as the least element: $C(999, 999)$

Sum of all least elements equals:

$$C(2021, 999) + 2 \cdot C(2020, 999) + \ldots + 1023 \cdot C(999, 999).$$

We can group the sum into columns, where each column can be computed using the Hockey Stick Identity:

$$\begin{array}{llllll} C(2021,999)+ & & & & & \\ C(2020,999)+ & C(2020,999)+ & & & & \\ C(2019,999)+ & C(2019,999)+ & C(2019,999)+ & & & \\ \ldots & & & & & \\ C(999,999)+ & C(999,999)+ & C(999,999)+ & C(999,999)+ & \ldots \end{array}$$

- 1st column sums to $C(2022, 1000)$
- 2nd column sums to $C(2021, 1000)$
- ...
- 1023rd column sums to $C(1000, 1000)$

We can again apply the Hockey Stick identity to the sum of all columns. This gives us $C(2023, 1001)$.

Supplementary Explanation

Answer. $\binom{2023}{1001}$.

Derivation/proof. If the minimum is m, select the other 999 elements from $m+1, \ldots, 2022$, in $\binom{2022-m}{999}$ ways. The possible minimum satisfies $1 \leq m \leq 1023$. The required weighted sum is $\sum_{m=1}^{1023} m\binom{2022-m}{999}$. Substituting $m = \sum_{t=1}^{m} 1$ and interchanging finite sums gives $\sum_{t=1}^{1023} \sum_{m=t}^{1023} \binom{2022-m}{999}$. The inner hockey-stick sum is $\binom{2023-t}{1000}$. Summing the outer terms, whose upper indices run from 1000 to 2022, applies the identity again and yields $\binom{2023}{1001}$. Every subset contributes once in each of its first m layers, retaining its full weight.

C165

Binomial Coefficients

2021 Final Part 2 | Question 1 | Source page 1 | Difficulty 4 | 10 marks
Q-2021-FINAL2-P01 | Answer provenance: official

> Recall the definition of Fibonacci number:
>
> $F(1) = 1, F(2) = 1, F(n) = F(n-1) + F(n-2)$ for $n \geq 2$.
>
> Give a combinatorial proof to the following identity:
>
> $$F(n+2) = \sum_{i=0}^{(n+1)/2} \binom{n+1-i}{i}.$$
>
> You can assume $n \geq 1$ and n is odd.
>
> [Hint: Consider bit strings with no consecutive 0s.]

Official Answer

Source: physical PDF page 1 (2021-final-part-2.pdf).

> Let S_n be the set of bit strings of length n with no consecutive 0s. We will show that both sides are equal to $|S_n|$. Let s_i denote any bit string in S_i.
>
> Left-hand side: For $n = 0$, the empty string is the only string and it has no consecutive 0s. So, $F(0 + 2) = 1 = |S_0|$ bit string that have no consecutive 0s. For $n = 1$, there are $F(1 + 2) = F(3) = F(2) + F(1) = 2 = |S_1|$ bit strings of length 1 that have no consecutive 0s. For $n > 1$, a bit string of length n that has no consecutive 0s can be obtained by appending a "1" to s_{n-1} and appending a "10" to s_{n-2}. The two strings must be different and these two are the only possible cases. By induction, we have $|S_n| = F(n + 2)$.
>
> Right-hand side: $\binom{n+1-i}{i}$ is counting the number of bit strings of length n that has exactly i 0s and no consecutive 0s. When a bit string of length n that has more than $(n + 1)/2$ 0s, there must be at least two consecutive 0s. So, the expression sums over all mutually disjoint cases, and thus it equals to $|S_n|$.

Supplementary Explanation

Answer. Both sides count length-n bit strings without consecutive zeros.

Derivation/proof. Let s_n count these strings. At length zero s0=1 and length one s1=2. For n≥2, a valid string ends either in 1 after a valid length-(n−1) prefix, or in 10 after a valid length-(n−2) prefix; these disjoint exhaustive cases give s_n=s_(n−1)+s_(n−2). Hence s_n=F(n+2). Independently fix i zeros. Arrange n−i ones; their n−i+1 gaps (including endpoints) can each contain at most one zero, giving C(n−i+1,i). Feasibility requires i≤(n+1)/2; summing disjoint zero counts proves the formula. The forward recurrence only needs Fibonacci indices ≥3; the printed boundary n≥2 is not used to evaluate an unspecified F(0).

The gap condition is $i \leq n - i + 1$, equivalent to $i \leq (n + 1)/2$. The stipulated odd n makes the upper limit integral. Both decompositions are unique and the final-block cases are disjoint and exhaustive.

The recurrence is $|S_n| = |S_{n-1}| + |S_{n-2}|$, and the fixed-zero count is $\binom{n-i+1}{i}$.

C166

Inclusion-Exclusion

2020 Final Part 2 | Question 6(a) | Source page 4 | Difficulty 2 | 2 marks
Q-2020-FINAL2-P06-a | Answer provenance: official

> Consider the permutations of the 10 digits (0–9). We denote a permutation as A, whose elements are $A[0], A[1], \ldots, A[9]$. Calculate the number of permutations that satisfy each of the following conditions:
>
> (a) There is exactly one i such that $A[i] = i$. (b) There are exactly nine i's such that $A[i] = i$. (c) $A[0] = 1, A[1] = 0$. (d) $A[4] = 4, A[5] = 5$.
>
> You can express your answers using binomial coefficients, factorials, and D_n (derangement number). However, you should not have summation $\sum$ in your final answers.
>
> There is exactly one i such that $A[i] = i$.

Official Answer

Source: physical PDF page 4 (2020-final-part-2.pdf); physical PDF page 5 (2020-final-part-2.pdf).

D_n is the number of derangements of a set with n elements,

$$D_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + (-1)^n \frac{1}{n!}\right].$$

Exactly 1 digit appears in its index position means that, we randomly select 1 digit and put it in its index position and dearrange the remaining 9 digits. So the answer is,

$$\binom{10}{1} D_9 = 10 \cdot D_9.$$

Supplementary Explanation

Answer. $10D_9$.

Derivation/proof. Choose the unique fixed index in 10 ways and put that digit there. The remaining nine digits must avoid their own nine positions, otherwise there would be more than one fixed point. They form a derangement counted by D9. Each permutation has one unique chosen fixed point, so the product 10D9 has no duplication.

The final expression uses the explicitly allowed derangement number and contains no summation sign.

The condition $A[i] = i$ is a fixed point. After choosing its unique index, the remaining positions must be deranged to exclude a second fixed point, which explains the factor D_9.

C167

Inclusion-Exclusion

2019 Final | Question 8(b) | Source page 10 | Difficulty 2 | 3 marks
Q-2019-FINAL-P08-b | Answer provenance: official

> How many strings of six lowercase letters from the English alphabet contain
>
> (b) the letters a and b?

Official Answer

Source: physical PDF page 10 (2019-final.pdf).

We use W_a to denote the set of words containing a, and similarly $W_b, W_{a\wedge b}, W_{a\vee b}$. By the inclusion-exclusion rule, we have $|W_{a\vee b}| = |W_a| + |W_b| - |W_{a\wedge b}|$. From (a), we know $|W_a| = |W_b| = 26^6 - 25^6$. Similarly, $|W_{a\vee b}| = 26^6 - 24^6$. So

$$|W_{a\wedge b}| = |W_a| + |W_b| - |W_{a\vee b}| = 2(26^6 - 25^6) - (26^6 - 24^6) = 26^6 - 2\cdot 25^6 + 24^6.$$

Supplementary Explanation

Answer. $26^6 - 2\cdot 25^6 + 24^6$.

Derivation/proof. Let A be strings missing a and B strings missing b. Their sizes are 25^6 each, and their intersection misses both, with 24^6 strings. The strings containing both are outside $A \cup B$. Inclusion-exclusion gives $26^6 - (25^6 + 25^6 - 24^6)$, which simplifies to the answer.

The intersection omits both named letters, leaving twenty-four choices at each position, so $|A \cap B| = 24^6$. Inclusion-exclusion adds this intersection back after the two subtractions.

C168

Inclusion-Exclusion

2025 Fall Midterm | Question 6 | Source page 15 | Difficulty 3 | 9 marks
Q-2025F-MID-P06 | Answer provenance: official

> Use inclusion-exclusion principle to solve the following problem. No other methods are allowed. You do not need to calculate the final value. In how many 6-digit numbers do all three digits 1, 2, and 3 appear at least once? Note that the leftmost digit can not be equal to 0.

Official Answer

Source: physical PDF page 15 (COMP2711_F2025_Midterm.pdf).

> Assume S to be the set of all 6-digit numbers.
>
> $$|S| = 9 \times 10^5.$$
>
> Let A_1 be the set of 6-digit numbers missing digit 1, A_2 be the set of 6-digit numbers missing digit 2 and A_3 be the set of 6-digit numbers missing digit 3. We need the number of 6-digit numbers where digits $1, 2$ and 3 appear, which is the total minus the numbers missing at least one of 1 or 2 or 3:
>
> $$S - (A_1 \cup A_2 \cup A_3).$$
>
> By inclusion-exclusion:
>
> $$|A_1 \cup A_2 \cup A_3| = |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap A_3|.$$
>
> $-|A_1|$: Numbers with no 1. First digit has 8 choices (2 to 9), other digits have 9 choices ($0, 2$ to 9):
>
> $$|A_1| = 8 \times 9^5.$$
>
> Similarly, $|A_1| = |A_2| = |A_3| = 8 \times 9^5$. $-|A_1 \cap A_2|$: Numbers with no 1 and no 2. First digit has 7 choices (3 to 9), other digits have 8 choices ($0, 3$ to 9):
>
> $$|A_1 \cap A_2| = 7 \times 8^5.$$
>
> Similarly, $|A_2 \cap A_3| = |A_1 \cap A_3| = |A_1 \cap A_2| = 7 \times 8^5$. $-|A_1 \cap A_2 \cap A_3|$: Numbers with no 1 and no 2 and no 3. First digit has 6 choices (4 to 9), other digits have 7 choices ($0, 4$ to 9):
>
> $$|A_1 \cap A_2 \cap A_3| = 6 \times 7^5.$$
>
> Now compute:
>
> $$|A_1 \cup A_2 \cup A_3| = 3 \times (8 \times 9^5) - 3 \times (7 \times 8^5) + 6 \times 7^5.$$
>
> So the answer is $|S| - |A_1 \cup A_2 \cup A_3| = 9 \times 10^5 - 3 \times (8 \times 9^5) + 3 \times (7 \times 8^5) - 6 \times 7^5$.

Supplementary Explanation

Answer. $9 \cdot 10^5 - 3 \cdot 8 \cdot 9^5 + 3 \cdot 7 \cdot 8^5 - 6 \cdot 7^5$.

Derivation/proof. The universe of six-digit numbers has $|S| = 9 \cdot 10^5$ elements. Let A_i be the numbers missing digit i, for i=1,2,3. Missing one specified nonzero digit leaves eight leading choices and nine for each other position, so $|A_i| = 8 \cdot 9^5$. Missing two leaves seven leading choices and eight later choices, giving $7 \cdot 8^5$ per pairwise intersection. Missing all three gives $6 \cdot 7^5$. Inclusion-exclusion subtracts the three single-event counts, adds the three pairwise intersections, and subtracts the one triple intersection from the universe. This gives the stated expression.

C169

Inclusion-Exclusion

2024 Fall Midterm | Question 7 | Source page 12 | Difficulty 3 | 5 marks
Q-2024F-MID-P07 | Answer provenance: official

> Assume that there are 5 processing jobs that must be assigned to 3 servers. Use Inclusion-Exclusion to calculate how many assignments exist such that each server is assigned to at least one job? Briefly explain why.

Official Answer

Source: physical PDF page 12 (COMP2711_F2024_Midterm.pdf).

> Answer: Solution 1: This is the number of onto functions from a set of 5 elements to a set of 3 elements: $3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5$ Solution 2:

Supplementary Explanation

Answer. $3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5 = 150$.

Derivation/proof. Each of five jobs has three choices, giving $3^5 = 243$ functions. Let A_i consist of assignments leaving server i empty; then $|A_i| = 2^5$. If two specified servers are empty, all jobs go to the remaining server, so the intersection has size $1^5 = 1$. The triple intersection is empty because five jobs cannot use no server. Inclusion-exclusion gives $243 - 3 \cdot 32 + 3 \cdot 1 = 150$. Jobs and servers have identities, so these are labelled assignments, not unordered partitions.

C170

Inclusion-Exclusion

2024 Spring Final | Question 8 | Source page 9 | Difficulty 3 | 10 marks
Q-2024S-FINAL-P08 | Answer provenance: official

> Let n be a positive integer such that $n = p^2q^4r^2$, where p, q, and r are three distinct prime numbers. Prove that the number of integers from 1 to n that are relatively prime with n is given by the expression:
>
> $$n\left(1-\frac{1}{p}\right)\left(1-\frac{1}{q}\right)\left(1-\frac{1}{r}\right).$$

Official Answer

Source: physical PDF page 9 (COMP2711_S2024_Final.pdf), physical PDF page 10 (COMP2711_S2024_Final.pdf).

> Solution: To find the number of integers from 1 to n that are relatively prime to n, we need to count the number of integers from 1 to n that are not divisible by p, q, or r. Let $\phi(n)$ denote this quantity. Let $S = \{1, 2, \ldots, n\}$. Define: A_p as the set of integers in S divisible by p, A_q as the set of integers in S divisible by q, A_r as the set of integers in S divisible by r. Using the principle of inclusion-exclusion, the number of integers in S that are divisible by p, q, or r is given by:
>
> $$|A_p \cup A_q \cup A_r| = |A_p| + |A_q| + |A_r| - |A_p \cap A_q| - |A_p \cap A_r| - |A_q \cap A_r| + |A_p \cap A_q \cap A_r|.$$
>
> We need to determine the size of each of these sets:
>
> $$|A_p| = \frac{n}{p}, \quad |A_q| = \frac{n}{q}, \quad |A_r| = \frac{n}{r}, \quad |A_p \cap A_q| = \frac{n}{pq}, \quad |A_p \cap A_r| = \frac{n}{pr}, \quad |A_q \cap A_r| = \frac{n}{qr}, \quad |A_p \cap A_q \cap A_r| = \frac{n}{pqr}.$$
>
> Substituting these values, we get:
>
> $$|A_p \cup A_q \cup A_r| = \frac{n}{p} + \frac{n}{q} + \frac{n}{r} - \frac{n}{pq} - \frac{n}{pr} - \frac{n}{qr} + \frac{n}{pqr}.$$
>
> The number of integers in S that are relatively prime to n is: $\phi(n) = n - |A_p \cup A_q \cup A_r|$. Substituting the expression for $|A_p \cup A_q \cup A_r|$, and simplifying, we get:
>
> $$\begin{aligned}\phi(n) &= n - \left(\frac{n}{p} + \frac{n}{q} + \frac{n}{r} - \frac{n}{pq} - \frac{n}{pr} - \frac{n}{qr} + \frac{n}{pqr}\right) \\ &= n\left(1 - \left(\frac{1}{p} + \frac{1}{q} + \frac{1}{r} - \frac{1}{pq} - \frac{1}{pr} - \frac{1}{qr} + \frac{1}{pqr}\right)\right) \\ &= n\left(1 - \frac{1}{p} - \frac{1}{q} - \frac{1}{r} + \frac{1}{pq} + \frac{1}{pr} + \frac{1}{qr} - \frac{1}{pqr}\right) \\ &= n\left(1-\frac{1}{p}\right)\left(1-\frac{1}{q}\right)\left(1-\frac{1}{r}\right).\end{aligned}$$

Supplementary Explanation

Answer. $n(1-1/p)(1-1/q)(1-1/r)$.

Derivation/proof. Within $S = \{1, \ldots, n\}$, let A_p, A_q, A_r be the multiples of the three respective primes. Only these primes can divide both an integer and n, so the target is $n - |A_p \cup A_q \cup A_r|$. The individual sizes are $n/p, n/q, n/r$; pairwise intersections have sizes $n/(pq), n/(pr), n/(qr)$ and the triple intersection has size $n/(pqr)$, since each product divides n. Inclusion-exclusion gives $n - n/p - n/q - n/r + n/(pq) + n/(pr) + n/(qr) - n/(pqr)$. Factor out n to obtain exactly the expansion of $(1-1/p)(1-1/q)(1-1/r)$.

C171

Inclusion-Exclusion

2023 Fall Midterm | Question 6 | Source page 13 | Difficulty 3 | 12 marks
Q-2023F-MID-P06 | Answer provenance: official

> How many solutions are there for the equation $a+b+c=13$, where a, b, c are nonnegative integers and $a \leq 4$, $b \leq 5$, $c \leq 6$. Use the inclusion-exclusion principle to calculate the result instead of enumerating the values of possible solutions.

Official Answer

Source: physical PDF page 13 (COMP2711_F2023_Midterm.pdf), physical PDF page 14 (COMP2711_F2023_Midterm.pdf).

$$U = \{(a,b,c) \mid a+b+c=13, a,b,c \geq 0\}$$

$$A = \{(a,b,c) \mid a+b+c=13, a>4, b,c \geq 0\}$$

$$B = \{(a,b,c) \mid a+b+c=13, b>5, a,c \geq 0\}$$

$$C = \{(a,b,c) \mid a+b+c=13, c>6, a,b \geq 0\}$$

So the result of our question is $|\overline{A} \cap \overline{B} \cap \overline{C}| = |U| - |A \cup B \cup C|$. And we use inclusion-exclusion to count $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$. We know that the number of r-combinations from a set with n elements when repetition of elements is allowed is $\binom{n+r-1}{r}$, which means that the number of solutions of $a+b+c=n$, where $a,b,c \geq 0$, is $\binom{n+2}{2}$.

$$|U| = \binom{15}{2} = 105$$

Since we can rewrite the set A to be

$$A = \{(a,b,c) \mid a+b+c=13, a>4, b,c \geq 0\} = \{(a'+5,b,c) \mid a'+b+c=8, a',b,c \geq 0\}$$

$$|A| = \binom{10}{2} = 45$$

In the same way, we can calculate the sizes of B, C as well as their intersections.

$$|B| = \binom{9}{2} = 36, \quad |C| = \binom{8}{2} = 28$$

$$|A \cap B| = \binom{4}{2} = 6, \quad |A \cap C| = \binom{3}{2} = 3, \quad |B \cap C| = \binom{2}{2} = 1$$

While the set $A \cap B \cap C$ should satisfy $a+b+c=13$ where $a>4, b>5, c>6$, there is no solution for this equation. So we have,

$$|A \cap B \cap C| = 0$$

In all, the result is,

$$\begin{aligned} |\overline{A} \cap \overline{B} \cap \overline{C}| &= |U| - (|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|) \\ &= 105 - (45 + 36 + 28 - 6 - 3 - 1 + 0) \\ &= 6. \end{aligned}$$

Supplementary Explanation

Answer. 6.

Derivation/proof. Let U be all nonnegative solutions and A, B, C the violations $a \geq 5$, $b \geq 6$, $c \geq 7$. Stars and bars gives $|U| = \binom{15}{2}$. Shifting the violating variables gives $|A| = \binom{10}{2}$, $|B| = \binom{9}{2}$, $|C| = \binom{8}{2}$. The double shifts give $|A \cap B| = \binom{4}{2}$, $|A \cap C| = \binom{3}{2}$, and $|B \cap C| = \binom{2}{2}$. The triple shift would leave total $13 - 5 - 6 - 7 < 0$, so its count is zero. Inclusion-exclusion therefore gives $105 - 45 - 36 - 28 + 6 + 3 + 1 = 6$, with all forbidden overlaps accounted for.

The thresholds are $A : a \geq 5$, $B : b \geq 6$, and $C : c \geq 7$. The triple intersection is empty, and explicitly $105 - 45 - 36 - 28 + 6 + 3 + 1 - 0 = 6$. Legal boundary values remain included.

C172

Inclusion-Exclusion

2020 Final Part 1 | Question 4(a) | Source page 3 | Difficulty 3 | 3 marks
Q-2020-FINAL1-P04-a | Answer provenance: official

> How many integers $0 \leq n < 1000$ are there such that
>
> n is divisible by 2, 3 or 5?

Official Answer

Source: physical PDF page 3 (2020-final-part-1.pdf).

Let $f(k)$ be the number of integers $0 \leq n < 1000$ that are divided by k, then $f(k) = 1 + \lfloor 999/k \rfloor$. We have $f(2) = 500$, $f(3) = 334$, $f(5) = 200$, $f(6) = 167$, $f(10) = 100$, $f(15) = 67$ and $f(30) = 34$. By inclusion-exclusion principle, the answer is $f(2) + f(3) + f(5) - f(6) - f(10) - f(15) + f(30) = 734$.

Supplementary Explanation

Answer. 734.

Derivation/proof. For d>0 the number of multiples in [0,999] is $f(d) = 1 + \lfloor 999/d \rfloor$, including zero. The counts for 2,3,5 are 500,334,200; pairwise intersections use 6,10,15 and give 167,100,67; the triple uses 30 and gives 34. Inclusion-exclusion yields $500 + 334 + 200 - 167 - 100 - 67 + 34 = 734$. Zero is counted once after cancellation.

The coefficient of zero in inclusion-exclusion is $3 - 3 + 1 = 1$.

C173

Inclusion-Exclusion

2019 Final | Question 10(d) | Source page 12 | Difficulty 3 | 3 marks
Q-2019-FINAL-P10-d | Answer provenance: official

> Consider the equation $x_1 + x_2 + x_3 = 20$ where x_1, x_2, and x_3 are nonnegative integers.
>
> For this question, give exact numbers as the solution.
>
> (d) How many solutions are there so that x_1, x_2 and x_3 are less than 9?

Official Answer

Source: physical PDF page 12 (2019-final.pdf).

We use S_ϕ to denote the set of solutions subject to condition ϕ. By the inclusion-exclusion principle, we have

$$\begin{aligned}|S_{x_1\geq 9\vee x_2\geq 9\vee x_3\geq 9}| &= 3\cdot|S_{x_1\geq 9}| - 3\cdot|S_{x_1\geq 9\wedge x_2\geq 9}| + |S_{x_1\geq 9\wedge x_2\geq 9\wedge x_3\geq 9}| \\ &= 3\cdot 78 - 3\cdot 6 + 0 = 216.\end{aligned}$$

So, $|S_{x_1<9\wedge x_2<9\wedge x_3<9}| = |S| - |S_{x_1\geq 9\vee x_2\geq 9\vee x_3\geq 9}| = 231 - 216 = 15$. Alternative solution: Write $y_1 = 8 - x_1, y_2 = 8 - x_2, y_3 = 8 - x_3$. Then the equation becomes $24 - y_1 - y_2 - y_3 = 20$, or $y_1 + y_2 + y_3 = 4$, which has $C(4+3-1, 3-1) = C(6,2) = 15$ solutions.

Supplementary Explanation

Answer. 15.

Derivation/proof. Let A_i be solutions with $x_i \geq 9$. The unrestricted count is 231; each A_i has 78 solutions and each pair intersection has 6. The triple intersection is empty because $27 > 20$. Thus the desired count is $231 - 3\cdot 78 + 3\cdot 6 = 15$. Independently set $y_i = 8 - x_i \geq 0$: then $y_1 + y_2 + y_3 = 4$, whose $\binom{6}{2} = 15$ solutions automatically have $y_i \leq 4 \leq 8$, giving a bijection back.

In the complementary-variable check, each $y_i \leq 4 \leq 8$, so the inverse $x_i = 8 - y_i$ is automatically nonnegative and the transformation is a bijection.

C174

Inclusion-Exclusion

2018 Final | Question 9(a) | Source page 13 | Difficulty 3
Q-2018-FINAL-P09-a | Answer provenance: official

> (a) How many elements in $\mathbb{Z}_{70}$ have multiplicative inverses? Explain your answer.

Official Answer

Source: physical PDF page 13 (2018-final.pdf).

We can write $70 = 2 \cdot 5 \cdot 7$.

$$M_7 = \{0, 7, 14, 21, 28, 35, 42, 49, 56, 63\}$$

$$M_5 = \{0, 5, 10, 15, 20, \ldots, 65\}$$

$$M_2 = \{0, 2, 4, 6, \ldots, 68\}.$$

The integers that do not have an inverse are those that are not relatively prime with 70, i.e., all the multiples of $7, 5, 2$, i.e., $M_7 \cup M_5 \cup M_2$. Note that $|M_7| = 10$, $|M_5| = 14$, $|M_2| = 35$. Moreover, $|M_7 \cap M_5| = 2$, $|M_7 \cap M_2| = 5$, and $|M_5 \cap M_2| = 7$. Finally, $|M_7 \cap M_5 \cap M_2| = 1$. By the inclusion exclusion principle, $|M_7 \cup M_5 \cup M_2| = 10 + 14 + 35 - 2 - 5 - 7 + 1 = 36$. Therefore, the number of elements that have an inverse is $70 - 36 = 24$.

Supplementary Explanation

Answer. 24.

Derivation/proof. An element is invertible exactly when coprime to $70 = 2 \cdot 5 \cdot 7$. Among residues 0 through 69, counts of multiples of 2,5,7 are 35,14,10; pairwise intersections have sizes 7,5,2 and the triple intersection has size 1. Thus nonunits number $35 + 14 + 10 - 7 - 5 - 2 + 1 = 46$, and units number $70 - 46 = 24$. The independent totient product $70(1 - 1/2)(1 - 1/5)(1 - 1/7) = 24$ agrees.

As an independent arithmetic check, $70(1 - 1/2)(1 - 1/5)(1 - 1/7) = 24$.

C175

Inclusion-Exclusion

2017 Final | Question 1 | Source page 3 | Difficulty 3 | 4 marks
Q-2017-FINAL-P01 | Answer provenance: official

> An office manager has four employees and nine reports to be done. In how many ways can the reports be assigned to the employees so that each employee has at least one report to do?

Official Answer

Source: physical PDF page 3 (2017-final.pdf).

$4^9 - \binom{4}{1}3^9 + \binom{4}{2}2^9 - \binom{4}{3}1^9$.

Supplementary Explanation

Answer. $4^9 - \binom{4}{1}3^9 + \binom{4}{2}2^9 - \binom{4}{3}1^9 = 186480$.

Derivation/proof. There are 4^9 unrestricted assignments. For each employee, define the bad event that they receive no report. If a specified set of j employees is omitted, each report has $4 - j$ choices, giving $(4 - j)^9$ assignments. Inclusion-exclusion gives $\sum_{j=0}^{4}(-1)^j\binom{4}{j}(4 - j)^9$, whose last term is zero. Evaluating yields 186480.

Evaluation gives $262144 - 78732 + 3072 - 4 = 186480$. Each nonempty assignment survives inclusion-exclusion exactly once.

C176

Inclusion-Exclusion

2017 Final | Question 4(a) | Source page 6 | Difficulty 3 | 5 marks
Q-2017-FINAL-P04-a | Answer provenance: official

> Recall that in RSA, we need to choose two primes p and q, and the public key is (n, e) where $n = pq$. Let $T = (p-1)(q-1)$. The private key is d, where $ed \equiv 1 \pmod{T}$. The encryption of x is $x^e \bmod n$. This means that d must be an integer in $\mathbb{Z}_T$ that is relatively prime with T. We can find d using the following algorithm:
>
> 1. Randomly pick $d \in \mathbb{Z}_T$.
> 2. If d and T are relatively prime, then return d. If they are not, then pick a new d and loop from step 1.
>
> Assume that we will never pick the same d twice.
>
> (a) Suppose that $p = 13, q = 61$. In the worst case, what is the number of times we will execute step 1 (that is, the number of times we have to pick a random d)?

Official Answer

Source: physical PDF page 6 (2017-final.pdf); physical PDF page 7 (2017-final.pdf).

$p - 1 = 2^2 \cdot 3, q - 1 = 2^2 \cdot 3 \cdot 5$. So this is equivalent to finding the number of integers from $\{0, 1, \cdots, (p-1)(q-1) - 1 = 719\}$ that are relatively prime to $2, 3$, and 5. Using the inclusion exclusion principle, we first find the number of integers that are multiples of $2 : 360, 3 : 240, 5 : 144, 6 : 120, 10 : 72, 15 : 48, 30 : 24$. The number of integers that are not relatively prime to $2, 3$, and 5 would then be $360 + 240 + 144 - 120 - 72 - 48 + 24 = 528$. By the pigeonhole principle, in the worst case it will take 529 tries to find a proper d.

Supplementary Explanation

Answer. 529 picks.

Derivation/proof. Here $T = 12 \cdot 60 = 720 = 2^4 3^2 5$. Bad candidates in $\{0, \ldots, 719\}$ are multiples of 2, 3 or 5. Their union has size $360 + 240 + 144 - 120 - 72 - 48 + 24 = 528$ by inclusion-exclusion (zero is included in each set). All 528 may be picked first without replacement; the next pick is then good. This ordering also proves the bound is attainable.

C177

Inclusion-Exclusion

2025 Spring Final | Question 10 | Source page 13 | Difficulty 4 | 7 marks
Q-2025S-FINAL-P10 | Answer provenance: official

> A 3-by-3 unit-square grid is to be colored such that each unit square is independently colored red or blue with equal probability. What is the probability that the grid does not contain any 2-by-2 red square? Justify your answer.

Official Answer

Source: physical PDF page 13 (COMP2711_S2025_Final.pdf).

Since each of the 9 unit squares of the 3-by-3 square can take one of two colors, in total there are 2^9 different ways of coloring the 3-by-3 square.

Let us count now the colorings that form a 2-by-2 red square. This red square can be formed in the upper left, upper right, bottom left or bottom right corner of the 3-by-3 square. Let A, B, C, D be, respectively, the sets of the four types of colorings listed above that produce a 2-by-2 red square. Since the other 5 squares of the 3-by-3 square can have any color, we have $|A| = |B| = |C| = |D| = 2^5$. These sets are however not disjoint, so we must apply inclusion-exclusion:

$$\begin{aligned}|A \cup B \cup C \cup D| = {} & |A| + |B| + |C| + |D| \\ & - |A \cap B| - |A \cap C| - |A \cap D| - |B \cap C| - |B \cap D| - |C \cap D| \\ & + |A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| + |B \cap C \cap D| \\ & - |A \cap B \cap C \cap D|.\end{aligned}$$

As $A \cap B$ represents the set of the colorings having the first two rows red (and the 3 squares of the lowest row free to take any of the two colors), we have $|A \cap B| = 2^3$. Similarly, $|A \cap C| = 2^3$, $|B \cap D| = 2^3$ and $|C \cap D| = 2^3$. We have a different situation for $A \cap D$ and $B \cap C$: if two opposing 2-by-2 squares are red, this leaves uncertain only the colors of the two remaining 1-by-1 corners, therefore $|A \cap D| = |B \cap C| = 2^2$. The intersections of 3 of the sets leaves undecided only the color of one square, therefore each of these intersections has 2 elements. Finally, the intersection of all the four sets contains only the square completely colored in red, that is, one element. We obtain:

$$|A \cup B \cup C \cup D| = 4 \cdot 2^5 - (4 \cdot 2^3 + 2 \cdot 2^2) + 4 \cdot 2 - 1 = 95.$$

So from the 512 possible colorings, $512 - 95 = 417$ do not contain any 2-by-2 red square, so the probability of not having a 2-by-2 red square is $\frac{417}{512}$.

Supplementary Explanation

Answer. $417/512$.

Derivation/proof. The nine independent fair colors give $2^9 = 512$ equally likely outcomes. Events A,B,C,D require the top-left, top-right, bottom-left and bottom-right 2×2 squares to be all red. Each fixes four cells and permits $2^5 = 32$ colorings. Of the six pair intersections, four adjacent pairs fix six cells and permit eight colorings each; two diagonal pairs fix seven and permit four each. Any three events fix eight cells, yielding two colorings for each of four triple intersections. All four events require the unique all-red grid. Inclusion-exclusion gives $4 \cdot 32 - (4 \cdot 8 + 2 \cdot 4) + 4 \cdot 2 - 1 = 95$ colorings with a red square. The complement has $512 - 95 = 417$, giving probability $417/512$.

C178

Inclusion-Exclusion

2020 Final Part 2 | Question 6(e) | Source page 4 | Difficulty 4 | 5 marks
Q-2020-FINAL2-P06-e | Answer provenance: official

> Consider the permutations of the 10 digits (0–9). We denote a permutation as A, whose elements are $A[0], A[1], \ldots, A[9]$. Calculate the number of permutations that satisfy each of the following conditions:
>
> (a) There is exactly one i such that $A[i] = i$. (b) There are exactly nine i's such that $A[i] = i$. (c) $A[0] = 1, A[1] = 0$. (d) $A[4] = 4, A[5] = 5$.
>
> You can express your answers using binomial coefficients, factorials, and D_n (derangement number). However, you should not have summation $\sum$ in your final answers.
>
> Satisfying any of the above conditions.

Official Answer

Source: physical PDF page 5 (2020-final-part-2.pdf).

We let A, B, C, D denote the sets of permutations that satisfy the above conditions (a), (b), (c), (d), respectively. Obviously, $B = \varnothing$, and we want to calculate $|A \cup B \cup C \cup D| = |A \cup C \cup D|$. Considering set $A \cap C$, fixed the first two digits, and pick 1 digit from $\{2, 3, \ldots, 9\}$ to put in the corresponding index position and then dearrange the remaining 7 digits. Thus,

$$|A \cap C| = 8 \cdot D_7.$$

Considering set $C \cap D$, fix 4 digits and randomly permute the remaining, thus,

$$|C \cap D| = 6!.$$

Considering set $A \cap D$, middle two index is 4 and 5, so $A \cap D = \varnothing$. And $A \cap C \cap D = C \cap \varnothing = \varnothing$. So,

$$|A \cap D| = |A \cap C \cap D| = 0.$$

By using inclusion-exclusion theorem,

$$\begin{aligned} |A \cup C \cup D| &= |A| + |C| + |D| - |A \cap C| - |A \cap D| - |C \cap D| + |A \cap C \cap D| \\ &= 10 \cdot D_9 + 2 \cdot 8! - 8 \cdot D_7 - 6!. \end{aligned}$$

Supplementary Explanation

Answer. $10D_9 + 2 \cdot 8! - 8D_7 - 6!$.

Derivation/proof. Call the four condition sets A,B,C,D. B is empty because nine fixed points force ten. The single counts are |A|=10D9, |C|=|D|=8!. In A∩C, the first two digits are swapped; choose one fixed point among the other eight and derange the remaining seven, giving 8D7. C∩D fixes four named positions, giving 6!. A∩D and the triple intersection are empty because D forces at least two fixed points. Inclusion-exclusion now gives the displayed expression.

Equivalently name the four condition sets S_1, S_2, S_3, S_4. In $S_1 \cap S_3$, choose the unique fixed point from the eight positions outside the prescribed swap and derange the remaining seven, giving $8D_7$. In $S_3 \cap S_4$, four positions are prescribed, leaving 6! possibilities. The other intersections involving S_2 are empty, and $S_1 \cap S_4$ is empty because it forces two fixed points. Inclusion-exclusion therefore subtracts exactly these two nonzero intersections.

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